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Dipartimento di Fisica e Astronomia G.Galilei

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Information transduction with multiple states

Internal supervisor
Prof. Sandro Azaele

Candidate
William Conte

External supervisor
Dr. Daniel Maria Busiello

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Abstract

This thesis investigates information transduction in biological systems, focusing on the ability of an entity to extract information from a noisy environment using limited degrees of freedom, while minimizing the associated relative thermodynamic cost. Starting from a foundational three-component model, composed of a probe (y) that extracts information, a membrane (x) representing the accessible interface, and an active bath (η) acting as the information source, we generalize the system architecture to an arbitrary number of membrane states and active baths. To analyze the composition of the information driving the optimization of the sensory couplings, we employ Partial Information Decomposition (PID) on more complex architectures, considering systems with two membranes and one bath ($M = 2, N = 1$), and vice versa ($M = 1, N = 2$). By doing so, we are able to see whether synergy or redundancy in the accessible degrees of freedom is associated with a higher information transfer from the environment. We develop a framework that optimizes internal sensory parameters to maximize information extraction under a constraint of equal relative dissipation (the relative entropy production rate induced by the coupling parameters), ensuring a fair comparison with the baseline model.

We find that the prevalence of synergistic or redundant information is closely associated with the system architecture. For the $M = 2$, $N = 1$ topology, synergistic information is generally favored, whereas for the $M = 1$, $N = 2$ case redundant information generally plays a more prominent role. Within both architectures, the optimal allocation of sensory resources is not uniform across parameter space: as the system transitions between different regimes, individual sensory couplings are observed to switch on or off, coinciding with structural transitions in the optimal configuration. These transitions are associated with the interplay between the environmental drive (σ) and the internal structural couplings (k) of the system.

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Chapter 1

Introduction

1.1 Information Transduction

Before introducing the problem quantitatively and mathematically, we start by providing a physical intuition of the system we are going to study.

Biological systems cannot typically guide their behavior by having access to all the degrees of freedom characterizing the external world [1]. Rather, they must rely on a restricted subset of stochastic observables transmitting such external information to their internal processes. This is precisely the mechanism captured by the term transduction: the process through which a system captures information about an unobservable external world and adapts its internal dynamics accordingly.

Cells, for instance, infer environmental conditions and act accordingly by monitoring the stochastic activity of molecular receptors, whose responses are inherently limited by finite dynamic range, biochemical noise, and energetic constraints [2, 3, 4, 5]. Similar challenges arise in metabolic pathways modulating their propensity to varying nutrient concentrations, and in adaptive materials in which only a distorted trace of the underlying nonequilibrium dynamics is accessible to engineered sensors [5].

However, very little is known about how much information on inaccessible observables is actually contained in the accessible ones [1]. This is especially relevant since all biological systems operate with a limited energy budget. As a consequence, they have to tune their transduction strategies accordingly to achieve the maximal amount of information on the hidden degrees of freedom. Moreover, the presence of multiple timescales, the nonreciprocity of the interactions, and the intrinsic activity of biological systems make the identification of optimal strategies a complex task.

In the following, we formalize an exemplary model of such a system by focusing on a chemical network [5]. A chemical network is a complex dynamical system composed of multiple interacting species (e.g., molecules, proteins, or enzymes) that undergo continuous stochastic reactions. These microscopic interactions coordinate to perform macroscopic biological functions, such as signal processing, metabolic regulation, or, precisely, information transduction from a hidden signal to an internal readout molecule.

1.1.1 Example of a Chemical Transduction Network

We focus on a chemical signaling process, sketched in Figure 1.1. A hidden chemical network stimulates the production of an intermediate signaling molecule X, which in turn binds to a receptor to produce an internal readout Y. This model can be described by the following set of general Langevin equations for the concentrations [6]:

$$\begin{aligned} \dot{H}_i &= A_i^{(H)}(H) + b_i^{(H)}(H) \cdot \xi_H \\ \dot{X} &= A^{(X)}(H, X, Y) + b^{(X)}(H, X, Y)\xi_X \\ \dot{Y} &= A^{(Y)}(X, Y) + b^{(Y)}(X, Y)\xi_Y \end{aligned} \tag{1.1}$$

where H_i are the concentrations of the hidden molecules. They interact both deterministically, via a nonlinear force field with components $A_i^{(H)}$, and through their fluctuations, as $b_i^{(H)}$ is a concentration-dependent vector mixing uncorrelated white noises ξ_H coming from different hidden reactions. Conversely, X and Y are the species accessible to the sensor: each features a single noise source (ξ_X, ξ_Y) and deterministic interactions ($A^{(X)}$, $A^{(Y)}$).

To ground this general formulation in a concrete case, we now introduce a specific chemical network that serves as an elementary example of the class of systems studied in this work, i.e. a particular realization of Eq. (1.1) once the reaction scheme is fixed. The internal dynamics producing the signaling molecule X involves an enzyme E that catalyzes the conversion of internal molecules through an intermediate complex C_E:

$$\begin{aligned} \sum_j \alpha_j H_j &\rightleftharpoons \sum_j \beta_j H_j \\ \sum_j \gamma_j H_j + E &\rightleftharpoons C_E \to E + X \ . \end{aligned} \tag{1.2}$$

Here, α_j, β_j, and γ_j are arbitrary stoichiometric coefficients: they do not represent specific numerical values, but stand for an unspecified, general combination of the hidden species H_j. The first reaction is a generic reversible transformation among the H_j, representing the internal dynamics of the hidden network; the second states that some combination of hidden species, weighted by γ_j, acts as the substrate through which the enzyme E produces X. This notation lets us keep full generality for the internal dynamics of the H_i species while fixing a concrete production mechanism for X.

The molecule X can then interact with a receptor R to generate the readout Y via the complex C_R, followed by a degradation step:

$$\begin{aligned} X + R &\rightleftharpoons C_R \to R + Y \\ Y &\to \emptyset \ . \end{aligned} \tag{1.3}$$

Starting from these reactions, the associated stochastic evolution for the concentrations can be obtained via a Kramers-Moyal expansion of the corresponding chemical master equation. Specifically, in the macroscopic limit of large copy numbers, truncating this expansion to the second order produces a chemical Langevin equation

whose deterministic drift follows the mass-action rate laws of the reaction scheme, and whose (multiplicative) noise amplitude follows from the same expansion. The resulting dynamics features non-linear reaction terms and non-diagonal multiplicative noise, coupling H, X, C_E, C_R, and Y through the intermediate complexes.

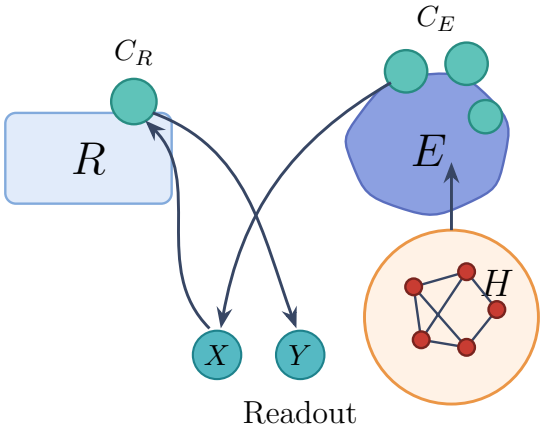


Figure 1.1: Schematic of the chemical transduction cascade: the hidden network H drives the intermediate complexes C_E, which synthesize the signaling molecule X. The signal X then forms the complex C_R upon binding with the receptor, ultimately triggering the production of the readout Y.

We first incorporate the conservation laws $R(t) = R_{tot} - C_R(t)$ and $E(t) = E_{tot} - C_E(t)$ [5]. Since intermediate complexes are typically fast variables, as in nucleotide-exchange and several enzyme kinetics [7, 8], and since all couplings between H, X, and Y are mediated by these fast species, we can integrate out C_R and C_E from the corresponding Fokker-Planck equation [5, 9]. This timescale separation yields an effective dynamics that only features diagonal multiplicative noise in X and Y. The physical reason is that off-diagonal noise terms, which represent instantaneous cross-correlations in the diffusion matrix, only arise from elementary reactions that simultaneously change the state of multiple species. By integrating out the fast intermediate complexes, we effectively separate the reaction steps: the functional dependencies between H, X, and Y are preserved in their respective deterministic drifts and diagonal noise amplitudes, but no direct simultaneous reactions remain to generate off-diagonal noise. A negative feedback from the readout Y onto the signal X can also be implemented within this scheme, through a reaction such as $Y + R \rightleftharpoons C_R$: the additional fast complex formation effectively modifies the concentration of X once integrated out.

We are ultimately interested in how the sensor reads and processes the fluctuations of concentrations around their stationary values. Denoting by $H_i^{(0)}$, $X^{(0)}$, $Y^{(0)}$ the deterministic fixed point of the effective dynamics obtained above, we define the fluctuations $\eta_i = H_i - H_i^{(0)}$, $x = X - X^{(0)}$, $y = Y - Y^{(0)}$ (lowercase letters). Linearizing the dynamics around this fixed point via a standard Linear Noise Approximation (LNA) [10, 11] yields a purely linear effective model for the coupled fluctuations. However, to realistically capture the non-linearities induced by the limited dynamic resolution of the sensor, a non-linear term $g(x)$ is often added phenomenologically a

posteriori. This mathematically effective procedure leads to the generalized model:

$$\begin{aligned}\dot{\eta} &= -\tau_\eta^{-1} W_\eta \eta + \sqrt{2D_\eta}\xi_\eta \\ \dot{x} &= -\tau_x^{-1} x + \sigma \cdot \eta + \alpha a y + \sqrt{2D_x}\xi_x \\ \dot{y} &= -\tau_y^{-1} y + a g(x) + \sqrt{2D_y}\xi_y\end{aligned} \tag{1.4}$$

where a is the tunable coupling, α encodes a possible inhibitory regulatory mechanism ($\alpha < 0$), and W_η accounts for the internal interactions of the hidden degrees of freedom. The parameters τ_η, τ_x, and τ_y denote the characteristic timescales of each degree of freedom.

Nomenclature and Physical Interpretation

While Eq. (1.4) has been derived starting from a specific chemical signaling network, its mathematical structure can be used to describe a broad class of hierarchical, nonreciprocal systems: the dynamics of each variable is driven by the one immediately above it in the hierarchy, without a direct feedback in the opposite direction (unless explicitly introduced, as with the inhibitory term α). Concretely, the hidden dynamics η drives the intermediate variable x, which in turn drives the accessible variable y, but neither x nor y influences the dynamics upstream of it.

To maintain clarity and provide a consistent physical intuition, throughout the rest of this thesis we adopt the biophysical nomenclature of a mechanically coupled hierarchical model, abstracting away the chemical details of the specific example discussed above [1]:

- η, the active bath: the inaccessible degrees of freedom that drive the system out of equilibrium from behind the scenes. This is the hidden network fluctuations H_i in the chemical picture, or, physically, the active fluctuations of a cellular cytoskeleton;
- x, the membrane: the intermediate variable that relays information from the active bath to the outside world. This is the signaling molecule X in the chemical picture, or, physically, a fluctuating lipid bilayer;
- y, the probe: the only variable actually accessible to an external observer, used to infer the state of the active bath. This is the internal readout Y in the chemical picture, or, physically, an external sensor tracking the membrane's undulations.

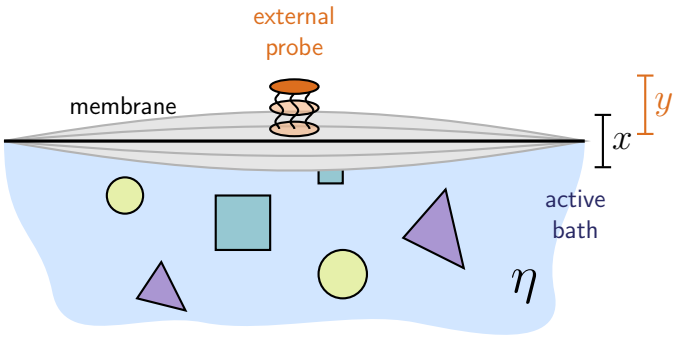


Figure 1.2: Schematic of the mechanical transduction model: an external probe y is coupled to a fluctuating membrane x, which is in turn driven out of equilibrium by an underlying active bath η.

With this terminology, the problem of *information transduction* translates into asking how the probe y can tune its coupling a to the membrane x to maximize the information gathered on the active bath η, while minimizing the thermodynamic cost (dissipation) of the sensing process [1].

1.2 Information Theory

A key measure in our model is the amount of information that can be extracted from a noisy system. To do this, we leverage the Information Theory framework: after a brief introduction in Sec. 1.2.1, we will explain the measures of information used in this work (Sec. 1.2.2 and 1.2.3).

1.2.1 Shannon entropy

In this section, we introduce some basic definitions on which we will base the concepts used in our model. We follow the approach proposed in [12].

First of all, we consider a discrete random variable X, whose alphabet (i.e., the set of possible values it can assume) is denoted as $\mathcal{X}$. Over this space, it is possible to define a probability mass function $p(x)$, where $x \in \mathcal{X}$ is a possible outcome of X.

A natural question that arises in this context is how to quantify the amount of uncertainty associated with the outcome of X. The first to measure this degree of uncertainty was Shannon [13], who defined the Shannon entropy for a discrete system as:

$$H(X) = -\sum_{x\in\mathcal{X}} p(x)\log p(x)\ . \tag{1.5}$$

From the form of Eq. (1.5), we can identify the similarities with thermodynamic entropy. In this context, each term $-\log p(x)$ (with $x \in \mathcal{X}$) represents the amount of uncertainty associated with a single element of the alphabet. Thus, the Shannon entropy represents the average uncertainty of the system.

Usually, the logarithm is in base 2, and the entropy is expressed in *bits*. For example, the entropy of a fair coin toss is 1 bit. If the base of the logarithm is

a generic b, we denote the entropy as $H_b(X)$. In particular, if the base is e, the entropy is measured in *nats*. Moreover, we will use the convention that $0 \log 0 = 0$. This is mathematically justified by continuity, since $x \log x \to 0$ as $x \to 0$. As a consequence, adding terms with zero probability does not change the total entropy of the system.

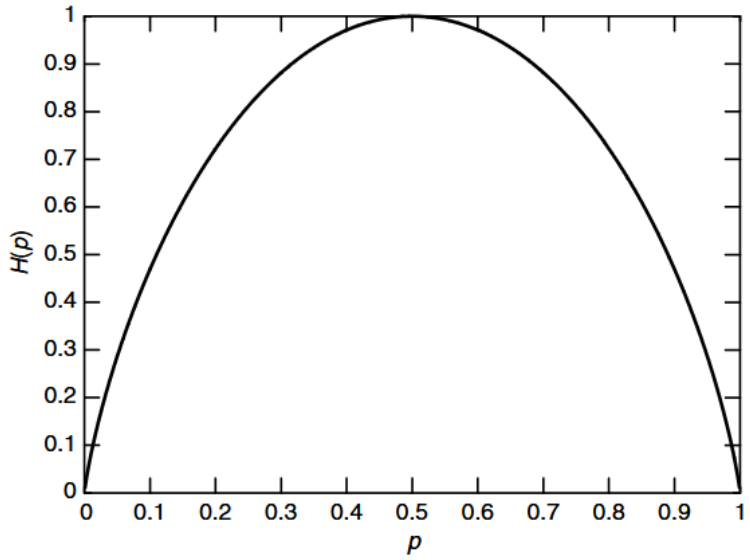


Figure 1.3: Shannon entropy $H(p)$ vs p (bits).

Joint and Conditional entropy

We can easily extend this concept to a pair of discrete random variables, X and Y, characterized by a joint probability mass function $p(x, y)$. The joint entropy $H(X, Y)$ measures the total uncertainty associated with the combined system. It is defined as [12]:

$$H(X, Y) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y) \ . \tag{1.6}$$

Another fundamental quantity is the conditional entropy $H(Y|X)$, which quantifies the remaining uncertainty about the variable Y when the value of X is already known. It is defined as the expected value of the entropies of Y conditioned on the different values of X:

$$H(Y|X) = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(y|x) \ , \tag{1.7}$$

where $p(y|x)$ is the conditional probability. A very useful property that links these quantities is the chain rule for entropy, which states that the total uncertainty of a joint system is the uncertainty of the first variable plus the remaining uncertainty of the second variable given the first one:

$$H(X, Y) = H(X) + H(Y|X) \ . \tag{1.8}$$

Shannon entropy in continuous systems

Up to this point, we have considered discrete random variables. However, our physical model involves continuous state variables (such as the position in a Langevin system). Therefore, we need to extend these definitions to the continuous case.

For a continuous random variable X with a probability density function $p(x)$ defined over a support S, the equivalent of the Shannon entropy is called *differential entropy*, usually denoted as $h(X)$ [12]:

$$h(X) = -\int_S p(x) \log p(x)\ dx\ . \tag{1.9}$$

Similarly, joint and conditional differential entropies are defined simply by replacing the summations with multiple integrals over the respective continuous probability density functions.

It is important to point out a fundamental conceptual difference between the discrete and continuous cases. If we quantize a continuous variable into bins of size Δ, its discrete entropy behaves as $H(X^\Delta) \approx h(X) - \log \Delta$. As the resolution becomes infinitely fine ($\Delta \to 0$), the discrete entropy diverges to infinity, reflecting the fact that infinite information is required to specify a continuous real number with absolute precision. The differential entropy $h(X)$ simply drops this infinite offset. As a consequence, unlike discrete entropy, $h(X)$ can take negative values and is not invariant under coordinate transformations.

To overcome this resolution problem and extract physically meaningful quantities, we must rely on relative measures of information where the diverging terms exactly cancel out. This naturally leads us to the concept of Mutual Information.

1.2.2 Mutual information

Before introducing the concept of mutual information (MI) between a pair of discrete random variables, we first need to define the so-called Kullback-Leibler (KL) divergence, also known as relative entropy [12].

Given two probability mass functions associated with a random variable X, say $p(x)$ and $q(x)$, the KL divergence is defined as:

$$D_{KL}(p||q) = \sum_{x\in\mathcal{X}} p(x) \log \frac{p(x)}{q(x)}\ . \tag{1.10}$$

We can intuitively understand the meaning of this definition: it acts as a measure of the difference between two probability distributions. Note that it is not strictly a distance metric, since it is not symmetric under the exchange of arguments ($D_{KL}(p||q) \neq D_{KL}(q||p)$) and it does not satisfy the triangle inequality.

Given two random variables X and Y with a joint probability mass function $p(x, y)$ and marginal probability mass functions $p(x)$ and $p(y)$, the mutual information $I(X;Y)$ is defined as the KL divergence between the joint distribution and the product of the marginal distributions:

$$I(X;Y) = D_{KL}(p(x,y)||p(x)p(y)) = \sum_{x\in\mathcal{X}}\sum_{y\in\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}\ . \tag{1.11}$$

This expression has an immediate physical intuition: $I(X;Y)$ quantifies the amount of information that the knowledge of one random variable provides about the other. By rearranging the terms in Eq. (1.11) and using the definitions from the previous section, the mutual information can be elegantly rewritten in terms of Shannon entropies [12]:

$$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X) \ . \tag{1.12}$$

This formulation clearly shows that MI measures the reduction of uncertainty in X given the knowledge of Y (and vice versa, as MI is perfectly symmetric: $I(X;Y) = I(Y;X)$).

We can better understand this by looking at two extreme limit cases:

- Complete independence: if X and Y are completely independent, their joint probability factorizes perfectly as $p(x,y) = p(x)p(y)$. In this case, $H(X|Y) = H(X)$ (knowing Y tells us absolutely nothing about X), and therefore $I(X;Y) = 0$.

- Complete correlation: if X and Y are perfectly correlated (or deterministically linked, meaning one uniquely defines the other), knowing Y completely removes any uncertainty about X. In this case, the conditional entropy vanishes ($H(X|Y) = 0$), and the mutual information reaches its maximum possible value: $I(X;Y) = H(X) = H(Y)$.

The generalization of the mutual information to continuous random variables is straightforward. If we express the discrete entropy as the sum of the differential entropy and the discretization bias, and substitute it into Eq. (1.12), the diverging bias terms exactly cancel out. As a result, the continuous mutual information maintains the exact same mathematical structure, where each discrete term is simply replaced by the corresponding differential entropy:

$$I(X;Y) = h(X) - h(X|Y) = h(Y) - h(Y|X) \ . \tag{1.13}$$

Alternatively, using the continuous probability density functions, $I(X;Y)$ can be explicitly written as the Kullback-Leibler divergence between the joint density and the product of the marginals:

$$I(X;Y) = \int_{\mathcal{X}} \int_{\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \ dx \ dy \ . \tag{1.14}$$

Unlike the differential entropy, this quantity is always non-negative, making it a robust and physically meaningful measure of information for continuous systems.

1.2.3 Partial Information Decomposition (PID)

The mutual information $I(X;Y)$ is a powerful tool to quantify the information shared between two random variables. However, physical systems often involve multiple interacting components. Suppose we have a target variable Y and two source variables, X_1 and X_2. The standard mutual information $I(X_1, X_2; Y)$ quantifies the

total amount of information that the joint state of the sources provides about the target.

However, this macroscopic quantity leaves a fundamental question unanswered: how is this information distributed among the sources? For instance, X_1 and X_2 might provide the exact same information about Y (like two identical sensors measuring the same temperature), or they might provide independent, unique pieces of information. Furthermore, they could interact to reveal information about Y that is not accessible from either source alone.

To solve this limitation, Williams and Beer [14] introduced the Partial Information Decomposition (PID) framework. The core idea is to decompose the total joint mutual information into four distinct, non-negative components:

$$I(X_1, X_2; Y) = U(X_1 \backslash X_2; Y) + U(X_2 \backslash X_1; Y) + R(X_1, X_2; Y) + S(X_1, X_2; Y) \ , \quad (1.15)$$

where the four terms represent intuitively distinct ways the sources can inform the target:

- Redundancy (R): The overlapping information about Y that can be extracted equivalently from either X_1 or X_2 alone.
- Unique Information (U): The information about Y provided solely by one source and not by the other (e.g., $U(X_1 \setminus X_2; Y)$ is the information unique to X_1).
- Synergy (S): The extra information about Y that is only accessible when knowing the joint state of X_1 and X_2 simultaneously, and cannot be deduced by observing them separately.

A key realization of the PID framework is that standard pairwise mutual information can also be expressed in terms of these components. The information provided by a single source is simply the sum of its unique information and the redundant information it shares with the other source:

$$I(X_1; Y) = U(X_1 \setminus X_2; Y) + R(X_1, X_2; Y) \ , \quad (1.16)$$
$$I(X_2; Y) = U(X_2 \setminus X_1; Y) + R(X_1, X_2; Y) \ . \quad (1.17)$$

However, Williams and Beer demonstrated that standard Shannon information theory does not uniquely define these four quantities. Eqs. (1.15), (1.16), and (1.17) form a system of three equations with four unknowns (U_1, U_2, R, S). Therefore, to completely define the decomposition, one must introduce an explicit definition for at least one of the components (typically the redundancy, R), from which the others can be derived [15].

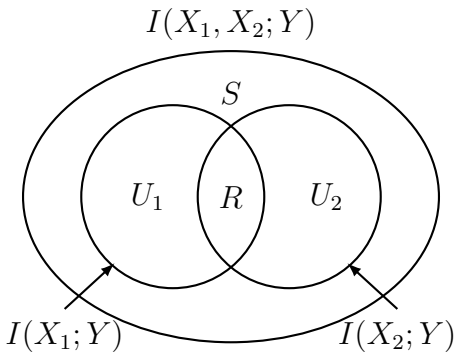


Figure 1.4: Venn diagram representation of the Partial Information Decomposition (PID). The total joint mutual information $I(X_1, X_2; Y)$ is decomposed into redundant (R), synergistic (S), and unique (U) contributions.

It is worth noting that while the decomposition in Eq. (1.15) is mathematically well-posed for two source variables, extending the PID framework to an arbitrary number of sources introduces deep theoretical complexities. As the number of sources increases, the decomposition does not merely add new unique terms, but rather explodes into a set known as the Dedekind lattice [14]. For instance, with three sources, the framework requires identifying 18 distinct components, including partial redundancies (e.g., information shared by sources 1 and 2, but not 3) and partial synergies (see Fig. 1.5 for a visual representation of these terms). Because analytically isolating these sub-components remains a challenge in Information Theory, especially for continuous Gaussian variables, the explicit PID application in this work will be focused on architectures involving only two sources of information.

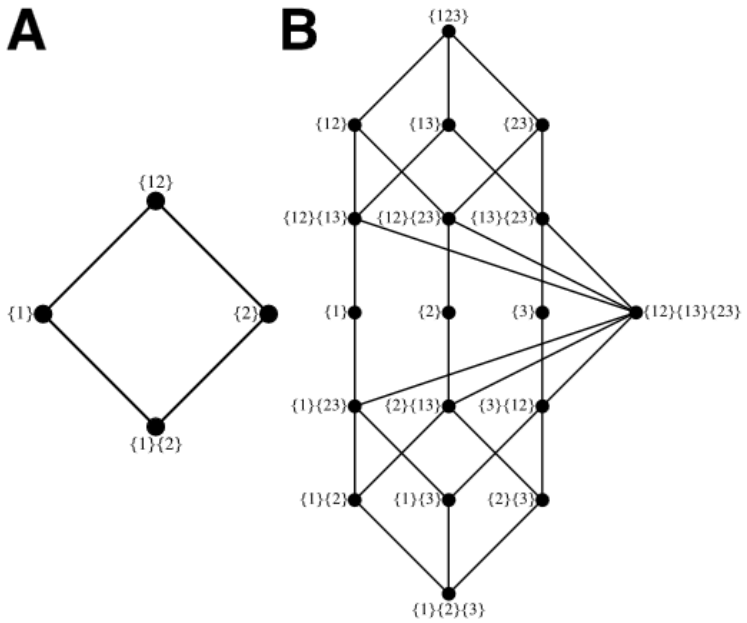


Figure 1.5: Combinatorial explosion of the PID, image taken from [14]. The structures are visualized as Hasse diagrams, whose node count is strictly governed by the Dedekind numbers. The bracket notation specifies how information is distributed among source combinations: {1} and {2} denote information available from source 1 or 2 alone (unique information); {12} represents synergistic information exclusively available by observing sources 1 and 2 jointly; and {1}{2} denotes redundant information between source 1 and 2. **(A)** For two sources, the Dedekind lattice contains exactly 4 tractable nodes bounded by pure Redundancy ({1}{2}) at the bottom and Synergy ({12}) at the top. **(B)** For three sources, the lattice explodes to 18 nodes. It introduces complex hybrid atoms, such as {1}{23} (redundant information shared between source 1 and the synergy of 2 and 3). It is important also to point out that this scaling may also lead to negative components of information.

Minimal Mutual Information PID (MMI PID)

To tackle the underdetermined system presented in Sec. 1.2.3, we adopt the strategy analyzed by A. Barrett in [15]. This approach is known as the Minimal Mutual Information Partial Information Decomposition (MMI PID).

The core idea is to define the redundant component of the joint information as the minimum of the two marginal mutual information provided by the individual sources:

$$R(X_1, X_2; Y) = \min\{I(X_1; Y), I(X_2; Y)\} \ . \tag{1.18}$$

By plugging Eq. (1.18) into Eqs. (1.16) and (1.17), the system is completely determined. Specifically, the source with the smallest mutual information will have zero unique information, meaning all its predictive power about the target is entirely shared with (and redundant to) the other source.

A key metric that naturally emerges from this framework is the Whole-Minus-Sum (WMS) information, defined as the difference between the joint mutual information and the sum of the individual marginal informations:

$$\mathrm{WMS}(X_1, X_2; Y) = I(X_1, X_2; Y) - (I(X_1; Y) + I(X_2; Y)) \ . \tag{1.19}$$

By substituting the PID equations (1.15), (1.16), and (1.17) into Eq. (1.19), we can express the WMS in terms of the fundamental PID components for our two-source system:

$$\begin{aligned} \mathrm{WMS} &= \big(U_1 + U_2 + R + S\big) - \big(U_1 + R\big) - \big(U_2 + R\big) \\ &= S(X_1, X_2; Y) - R(X_1, X_2; Y) \ . \end{aligned} \tag{1.20}$$

This result shows that the WMS acts as a macroscopic indicator of the dominant interaction regime: a positive WMS implies a synergy-dominated system ($S > R$), whereas a negative WMS indicates a redundancy-dominated one ($R > S$).

It is important to state that Eq. (1.18) is not the unique way to define redundancy in the literature. In fact, formulating a universally accepted measure for R remains an open challenge in Information Theory. A known limitation of the MMI approach is its tendency to overestimate redundancy, as it can sometimes confuse the correlation between the sources themselves with the actual information they redundantly share about the target. To address these theoretical paradoxes, several alternative metrics have been proposed over the last decade. For instance, Bertschinger *et al.* [16] introduced a geometric measure (often denoted as I_{broja}) that extracts the unique information by optimizing over a specific space of fictitious probability distributions. Another notable alternative is the Common Change in Surprisal (I_{ccs}) proposed by Ince [17], which computes redundancy at the local level of individual events rather than using global averages. Despite these refined alternatives, the MMI formulation remains highly regarded for its computational tractability and its direct applicability to continuous and Gaussian variables, making it an ideal choice for the Langevin dynamics analyzed in this work.

1.3 Stochastic Thermodynamics

This chapter introduces the conceptual and mathematical tools of Stochastic Thermodynamics, which are needed to extend the macroscopic laws of classical thermodynamics to microscopic or mesoscopic systems, where thermal fluctuations can no longer be neglected. We start with a brief recap of classical thermodynamics (Sec. 1.3.1), before introducing the stochastic description of the dynamics through the Langevin equation and the Fokker-Planck equation (Sec. 1.3.2). Building on these tools, we develop stochastic energetics, which defines heat and work along single trajectories, and stochastic entropy, which allows us to formulate a microscopic version of the Second Law. Finally, we focus on Non-Equilibrium Steady States (NESS), for which we derive the decomposition of the entropy production rate into traffic $\mathcal{T}$ and inflow rate $\mathcal{G}$. This relation will play a central role in the following, where it will be used to estimate the thermodynamic cost of information harvesting.

1.3.1 Recap of Classical Thermodynamics

In this work, an important part deals with Stochastic Thermodynamics. However, before introducing it, it can be useful to perform a brief recap of classical Thermodynamics.

Classical Thermodynamics studies, from a macroscopic perspective, the transitions from one equilibrium state to another [18]. In this framework, a set of extensive (or intensive, depending on the considered ensemble) variables, for instance energy E, volume V or number of particles N, is used to define the state of the system. The system can interact with its surroundings in the form of exchanged heat or work done by external forces. This energy balance is governed by the first law of thermodynamics, which is essentially an extension of the principle of conservation of energy, stating that energy can neither be created nor destroyed, but only transformed from one form to another. For a closed macroscopic system undergoing a transformation from an initial equilibrium state to a final one, the change in its internal energy ΔU must balance the net amounts of heat Q and work W exchanged with the environment (referred also as reservoir). Mathematically, considering the convention where both heat and work are considered positive when energy enters the system, the first law is expressed as:

$$\Delta U = Q + W \ ,$$

where:

- U is the internal energy of the system, which is a state function (meaning its variation depends only on the initial and final states, not on the specific path followed);
- Q is the heat absorbed by the system from the reservoir ($Q > 0$ if heat flows in, $Q < 0$ if heat flows out);
- W is the work performed on the system by external forces ($W > 0$ if work is done on the system, compressing or driving it, $W < 0$ if the system performs work on the surroundings).

When dealing with infinitesimal transformations, which are crucial for the development of stochastic thermodynamics, the first law is written in differential form:

$$dU = \delta Q + \delta W \ . \tag{1.21}$$

Here, dU is an exact differential because U is a state function. Conversely, δQ and δW are inexact differentials (denoted by the symbol δ), highlighting that heat and work are path-dependent quantities; they represent methods of energy transfer rather than properties stored within the system. In the microscopic realm, this exact balance will be applied to individual stochastic trajectories, where dU, δQ, and δW become fluctuating, trajectory-dependent variables.

A fundamental concept in classical thermodynamics, encapsulated by the Second Law of Thermodynamics, is that the relaxation to a new equilibrium is governed by the "maximum entropy principle". This principle states that an isolated system will naturally evolve towards the macroscopic state that maximizes its entropy. Furthermore, for a system in contact with a thermal bath at temperature T, the change in entropy during a reversible process is intimately connected to the exchanged heat via the Clausius relation, $dS = \delta Q/T$.

To better understand this concept, it is important to clarify the difference between a macrostate and a microstate. A macrostate is defined by the macroscopic, measurable properties of the system, such as temperature, pressure, or volume. On the other hand, a microstate represents a specific, detailed configuration of all the microscopic components of the system (for example, the exact position and momentum of every single particle). Usually, a massive number of different microstates can correspond to the exact same macrostate.

According to the Gibbs formulation (which generalizes Boltzmann's approach), entropy can be defined as a state function that measures the degree of disorder in a physical system [19]. It quantifies this disorder in relation to the probability distribution of its underlying microscopic states.

More precisely, the Gibbs entropy S is defined as [20]:

$$S = -k_B \sum_i p_i \ln p_i \ ,$$

where k_B is the Boltzmann constant and p_i represents the probability of finding the system in a particular microstate i corresponding to the observed macrostate. Crucially, classical thermodynamics operates in the so-called thermodynamic limit, where the number of components is extremely large (e.g., Avogadro's number $N \sim 10^{23}$). In this macroscopic regime, statistical fluctuations scale as $1/\sqrt{N}$ and are therefore completely negligible; the observable macroscopic quantities perfectly match their deterministic average values.

However, when dealing with microscopic or mesoscopic systems - such as individual molecules or biological networks - thermal fluctuations become comparable in magnitude to deterministic forces. Consequently, they can no longer be ignored and must be explicitly taken into account in the modeling process. The necessity to properly describe these fluctuating dynamics and extend the classical macroscopic laws to individual trajectories determined the development of Stochastic Thermodynamics.

1.3.2 Stochastic Dynamics

As stated in the previous paragraph, for microscopic or mesoscopic systems it is no longer possible to rely only on a description of macroscopic average behaviors, due to the importance of fluctuations in this regime. In fact, at this scale, the energies involved are of the order of the thermal energy k_BT. This continuous interaction with the environment leads to a highly stochastic dynamical process. To properly model this stochasticity, the theoretical framework can be developed along two complementary paths: either considering the single trajectory dynamics, which leads to the Langevin equation, or by describing the time evolution of the system's probability distribution, which is governed by the Fokker-Planck equation [21]. We now briefly recap these two theoretical tools: the Langevin equation is a stochastic differential equation used to model the trajectory of a system which undergoes a non-negligible thermal noise. Its general overdamped form is given by:

$$\gamma \dot{x}(t) = F(x,t) + \sqrt{2\gamma k_B T}\, \xi(t)\,, \tag{1.22}$$

where:

- γ is the friction coefficient, representing the viscous drag exerted by the surrounding environment;
- $x(t)$ is the state of the system (for instance, the position of a particle), and $\dot{x}(t)$ is its time derivative;
- $F(x,t)$ represents the deterministic forces acting on the system, which can arise from external potentials or non-conservative driving forces;
- k_BT is the thermal energy, with k_B being the Boltzmann constant and T the absolute temperature of the bath;
- $\xi(t)$ is a stochastic term modeling the thermal noise. It is typically assumed to be a standard Gaussian white noise, characterized by a zero mean, $\langle \xi(t) \rangle = 0$, and a delta-correlated variance, $\langle \xi(t)\xi(t') \rangle = \delta(t-t')$.

The specific amplitude of the noise, $\sqrt{2\gamma k_B T}$, is a direct consequence of the Fluctuation-Dissipation Theorem. This relation implies a physical connection between the energy dissipated through friction (γ) and the energy injected by the thermal fluctuations, which is necessary to reach a proper thermal equilibrium when no external driving is applied. Thanks to this relation, we can introduce the diffusion coefficient D, defined by the Einstein relation as [22]:

$$D = \frac{k_BT}{\gamma}\,. \tag{1.23}$$

As we have stated before, while the Langevin equation describes the system focusing on a single stochastic trajectory, we can also adopt a statistical ensemble perspective. In this case, we study the time evolution of the probability density function $p(x,t)$, which represents the probability of finding the system in the state

x at time t. The continuous dynamics of this probability distribution is governed by the Fokker-Planck equation:

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x}\left(\frac{F(x,t)}{\gamma}p(x,t)\right) + D\frac{\partial^2}{\partial x^2}p(x,t)\,. \tag{1.24}$$

The latter equation can be rewritten by making the probability current $J(x,t)$ explicit. By collecting the spatial derivative, the Fokker-Planck equation assumes the form of a classic continuity equation:

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial J(x,t)}{\partial x}\,, \tag{1.25}$$

where the probability current $J(x,t)$ is defined as:

$$J(x,t) = \frac{F(x,t)}{\gamma}p(x,t) - D\frac{\partial p(x,t)}{\partial x}\,. \tag{1.26}$$

Furthermore, a fundamental concept directly connected to the Fokker-Planck equation and the probability current is the Non-Equilibrium Steady State (NESS). A system reaches a NESS when its probability distribution relaxes to a stationary profile, meaning that it solves the Fokker-Planck equation with a zero time derivative, $\frac{\partial p(x,t)}{\partial t} = 0$.

Mathematically and physically, this regime is completely different from a true thermodynamic equilibrium. While both conditions share a time-independent probability distribution, a thermal equilibrium satisfies the detailed balance principle, which strictly requires the probability current to vanish everywhere ($J(x) = 0$). In contrast, a NESS is characterized by a non-zero probability current continuously flowing through the system. In general multidimensional spaces, this steady flow is defined by a divergence-free current ($\nabla \cdot \mathbf{J} = 0$), which specifically reduces to a spatially constant current ($J(x) = \text{const} \neq 0$) in one-dimensional systems. This steady flow is usually maintained by the continuous action of external non-conservative forces, keeping the system away from equilibrium.

1.3.3 Stochastic Energetics

It is possible to introduce a principle of energy conservation, according to the assumption that if a Langevin equation represents the balance of forces on a system, then the dynamics conserves the energy of the system plus the reservoir [23]. This is essentially the first law of thermodynamics applied to an individual realization of the stochastic process.

Let's start with an overdamped Langevin equation. To discuss thermodynamic processes, we assume that the potential energy U depends not only on the system's state x, but also on a variable a which represents the effect of an external agent. Accordingly, the deterministic force appearing in Eq. (1.22) is here taken to be purely conservative, $F(x,a) = -\partial U(x,a)/\partial x$, so that the equation reads:

$$\gamma\dot{x}(t) = -\frac{\partial U(x,a)}{\partial x} + \sqrt{2\gamma k_B T}\;\xi(t)\;,$$

with the meanings of each symbol explained in Section 1.3.2.

We can rearrange the terms to express the instantaneous balance of forces, which sum up to zero at every moment:

$$0 = -\gamma \dot{x}(t) + \sqrt{2\gamma k_B T}\ \xi(t) - \frac{\partial U(x,a)}{\partial x}\ .$$

The first two terms on the right side represent the interaction between the system and the heat bath (the friction and the thermal noise). The last term is the systematic force due to the potential.

Suppose now that the state of the system changes by an infinitesimal amount dx: if we multiply the forces by $(-dx)$, we obtain the energy balance. This multiplication of fluctuating quantities must be understood in the sense of Stratonovich calculus (denoted by $\circ$):

$$0 = -\left(-\gamma \dot{x}(t) + \sqrt{2\gamma k_B T}\ \xi(t)\right) \circ dx + \frac{\partial U(x,a)}{\partial x} \circ dx\ .$$

The first term represents the work done by the reaction force exerted by the system on the heat bath. We identify it as the heat discarded by the system into the bath, which we denote by δQ:

$$\delta Q = -\left(-\gamma \dot{x}(t) + \sqrt{2\gamma k_B T}\ \xi(t)\right) \circ dx\ . \tag{1.27}$$

Notice that the sign convention here is opposite to usual macroscopic thermodynamics (here $\delta Q > 0$ means heat is released).

The second term, $\frac{\partial U}{\partial x} \circ dx$, is not the total differential of U because of the variable a. To complete the differential form, we must add the quantity $\frac{\partial U}{\partial a} da$ to both sides of the energy balance equation. Knowing that the total differential is $dU = \frac{\partial U}{\partial x} \circ dx + \frac{\partial U}{\partial a} da$, we obtain:

$$\frac{\partial U(x,a)}{\partial a} da = \delta Q + dU\ .$$

The left side of this equation is the work done by the external system through the change of the parameter a. Therefore, we define the stochastic work δW as:

$$\delta W = \frac{\partial U(x,a)}{\partial a} da\ . \tag{1.28}$$

Finally, we find the general expression of the energy conservation law for a single stochastic trajectory:

$$\delta W = \delta Q + dU\ .$$

It is important to note that this expression is perfectly consistent with the first law of thermodynamics introduced in Eq. (1.21). The apparent difference in the signs is simply due to the chosen convention. Here, as done in Eq. (1.27), δQ is defined as the heat dissipated into the bath, meaning it is the exact opposite of the absorbed heat (which we can denote as δQ_{in}) used in classical thermodynamics ($\delta Q = -\delta Q_{in}$). If we substitute this relation into the equation, we immediately recover the standard form $dU = \delta Q_{in} + \delta W$.

1.3.4 Stochastic Entropy

Drawing a connection with information theory (as discussed in Section 1.2), we can quantify the uncertainty regarding the system's state. Following the stochastic thermodynamics framework established by Peliti and Pigolotti [21], the macroscopic non-equilibrium entropy of a system, based on the probability density function $p(x,t)$ obtained from the Fokker-Planck equation, is defined as:

$$S_{sys}(t) = -k_B \int p(x,t) \ln p(x,t) \; dx \; . \tag{1.29}$$

This macroscopic quantity describes the behavior of a whole statistical ensemble. However, physical systems at the mesoscale are heavily influenced by thermal noise. As a result, even if we start with the exact same initial conditions, each system will follow a unique and independent trajectory $x(t)$.

To properly study the thermodynamics of a single realization, we introduce the trajectory-dependent stochastic entropy:

$$s_{sys}(t) = -k_B \ln p(x,t) \; . \tag{1.30}$$

It is important to point out a conceptual detail: while $s_{sys}(t)$ evaluates the entropy for a specific individual path, its computation still relies on the macroscopic probability distribution $p(x,t)$. By taking the average over all possible stochastic realizations, we naturally recover the ensemble entropy:

$$S_{sys}(t) = \langle s_{sys}(t) \rangle \; .$$

When the process evolves from an initial time t_0 to a final time t_f, according to Eq. (1.30) the stochastic entropy of the system changes by:

$$\Delta s_{sys} = s_{sys}(t_f) - s_{sys}(t_0) \; .$$

To formulate a microscopic equivalent of the Second Law of thermodynamics, this internal variation must be combined with the entropy change of the surrounding environment. Therefore, the total stochastic entropy production along a single trajectory is:

$$\Delta s_{tot} = \Delta s_{sys} + \Delta s_{res} \; , \tag{1.31}$$

where Δs_{res} represents the entropy variation of the thermal bath.

Since the environment is considered large enough to always remain in thermodynamic equilibrium at a constant temperature T, we can evaluate its entropy change using the classical Clausius relation:

$$\Delta s_{res} = \frac{Q}{T} \; . \tag{1.32}$$

Here, Q is the total heat dissipated into the environment, which directly links this entropy balance to the stochastic energetics framework established in the previous section.

1.3.5 Entropy Production Rate

While the integrated variations Δs are useful to evaluate a complete process between an initial and a final state, it is often more insightful to look at the continuous dynamics. For this reason, we define the stochastic entropy production rates by taking the time derivative of the previous quantities [21].

The rate of change of the reservoir entropy is simply related to the instantaneous heat flux:

$$\dot{s}_{res}(t) = \frac{1}{T}\frac{\delta Q(t)}{dt} \ ,$$

where, according to stochastic energetics, Eq. (1.27) and Eq. (1.28), the heat rate must be mathematically interpreted using the Stratonovich product between the forces of the bath and the velocity $\dot{x}(t)$:

$$\frac{1}{T}\frac{\delta Q(t)}{dt} = \frac{1}{T}\ F(x,t) \circ \dot{x}(t) \ . \tag{1.33}$$

For the system, we can calculate the stochastic entropy production rate by applying the chain rule to Eq. (1.30). Since the probability $p(x,t)$ depends on both position and time, its total time derivative evaluated along the trajectory $x(t)$ gives two terms:

$$\dot{s}_{sys}(t) = -k_B \frac{d}{dt}\ln p(x,t) = -k_B \frac{\partial_t p(x,t)}{p(x,t)} - k_B \frac{\partial_x p(x,t)}{p(x,t)}\dot{x}(t) \ .$$

The total stochastic entropy production rate along a single trajectory is then the sum of these two contributions:

$$\dot{s}_{tot}(t) = \dot{s}_{sys}(t) + \dot{s}_{res}(t) \ .$$

Because of the thermal noise, $\dot{s}_{tot}(t)$ is a highly fluctuating variable that can frequently take negative values, meaning that the Second Law can be transiently violated at the microscopic scale. However, we can demonstrate that its statistical ensemble average is always strictly non-negative.

To understand where this result comes from, we calculate the ensemble averages of the system and reservoir entropy rates separately. Starting with the system, the average entropy rate is the time derivative of the macroscopic Shannon entropy (Eq. (1.29)):

$$\langle \dot{s}_{sys}(t) \rangle = \frac{dS_{sys}(t)}{dt} = -k_B \int \partial_t p(x,t) \ln p(x,t)\ dx \ .$$

By using the Fokker-Planck continuity equation ($\partial_t p(x,t) = -\partial_x J(x,t)$) and integrating by parts (assuming the probability current $J(x,t)$ vanishes at the boundaries), we find:

$$\langle \dot{s}_{sys}(t) \rangle = k_B \int (\partial_x J(x,t)) \ln p(x,t)\ dx = -k_B \int J(x,t)\frac{\partial_x p(x,t)}{p(x,t)}\ dx \ . \tag{1.34}$$

For the reservoir, the average entropy rate is given by the average heat dissipation rate divided by temperature. The heat dissipated is the work done by the systematic

force $F(x,t)$ against the viscous drag (Eq. (1.33)), which in terms of probability current is [24]:

$$\langle \dot{s}_{res}(t) \rangle = \frac{\langle \dot{Q}(t) \rangle}{T} = \frac{\langle F(x,t) \circ \dot{x}(t) \rangle}{T} = \frac{1}{T} \int F(x,t) J(x,t) \, dx \ . \tag{1.35}$$

From the definition of the probability current $J(x,t) = \frac{F(x,t)}{\gamma} p(x,t) - D\partial_x p(x,t)$, we can express the force as $F(x,t) = \gamma \frac{J(x,t)}{p(x,t)} + \gamma D \frac{\partial_x p(x,t)}{p(x,t)}$. Recalling the Einstein relation $\gamma = \frac{k_B T}{D}$, this becomes $F(x,t) = \frac{k_B T}{D} \frac{J(x,t)}{p(x,t)} + k_B T \frac{\partial_x p(x,t)}{p(x,t)}$. Substituting this into the reservoir entropy rate yields:

$$\langle \dot{s}_{res}(t) \rangle = k_B \int \left(\frac{J(x,t)^2}{D p(x,t)} + J(x,t) \frac{\partial_x p(x,t)}{p(x,t)} \right) \, dx \ .$$

When we sum the two contributions given by Eq. (1.34) and Eq. (1.35) to obtain the total average entropy production rate, the terms involving the gradient of the probability density exactly cancel out, leaving only the quadratic term:

$$\langle \dot{s}_{tot}(t) \rangle = k_B \int \frac{J(x,t)^2}{D \ p(x,t)} \, dx \ .$$

Since the diffusion coefficient D and the probability density $p(x,t)$ are intrinsically positive quantities, the integral of $J(x,t)^2$ mathematically ensures that $\langle \dot{s}_{tot}(t) \rangle \geq 0$. This provides a rigorous microscopic proof of the Second Law of Thermodynamics: while individual trajectories can experience negative entropy fluctuations, the macroscopic ensemble perfectly obeys the thermodynamic irreversibility.

Finally, an important reduction of this framework occurs when analyzing a Non-Equilibrium Steady State (NESS). A system reaches a NESS when it is continuously driven by external non-conservative forces, but its macroscopic probability distribution has relaxed to a time-independent profile, meaning $\partial_t p(x,t) = 0$.

In this specific regime, the macroscopic entropy of the system is constant in time, which immediately implies that its average rate of change is zero:

$$\langle \dot{s}_{sys} \rangle = -k_B \int \partial_t p(x,t) \ln p(x,t) \, dx = 0 \ .$$

As a direct consequence, the total entropy production rate in a steady state is entirely determined by the continuous dissipation of heat into the environment:

$$\langle \dot{s}_{tot} \rangle = \langle \dot{s}_{res} \rangle = \frac{\langle \dot{Q} \rangle}{T} \geq 0 \ .$$

Entropy Production Rate for Multidimensional Langevin Systems

The results derived above for a single degree of freedom $x(t)$ extend directly to a general multidimensional Langevin system. Such a system is described by a state vector $\mathbf{z}(t)$, whose continuous dynamics are governed by the multidimensional overdamped Langevin equation:

$$\dot{\mathbf{z}}(t) = \boldsymbol{\gamma}^{-1} \mathbf{F}(\mathbf{z}, t) + \sqrt{2\mathbf{D}} \boldsymbol{\xi}(t) \ , \tag{1.36}$$

where $\boldsymbol{\xi}(t)$ is a vector of independent standard Gaussian white noises. Here, $\mathbf{F}(\mathbf{z}, t)$ represents the deterministic physical force field, and $\boldsymbol{\gamma}$ is the friction tensor. The matrix $\mathbf{D}$ is the diffusion tensor, which is a symmetric, positive-definite matrix that characterizes the amplitude and the spatial coupling of the thermal fluctuations induced by the environment.

By virtue of the Einstein relation [22], the friction tensor and diffusion matrices are strictly linked by $\mathbf{D} = k_B T \boldsymbol{\gamma}^{-1}$. The time evolution of the probability density $p(\mathbf{z}, t)$ for this system obeys the corresponding Fokker-Planck equation, $\partial_t p(\mathbf{z}, t) + \nabla_{\mathbf{z}} \cdot \mathbf{J}(\mathbf{z}, t) = 0$. The probability current is defined using the deterministic drift vector $(\boldsymbol{\gamma}^{-1}\mathbf{F}(\mathbf{z}, t))$:

$$\mathbf{J}(\mathbf{z}, t) = \boldsymbol{\gamma}^{-1}\mathbf{F}(\mathbf{z}, t)p(\mathbf{z}, t) - \mathbf{D}\nabla_{\mathbf{z}} p(\mathbf{z}, t) \ .$$

Following the same steps as in the scalar case, the average entropy production rate splits into a system and a reservoir contribution, both of which are generally time-dependent:

$$\langle \dot{s}_{tot}(t) \rangle = \langle \dot{s}_{sys}(t) \rangle + \langle \dot{s}_{res}(t) \rangle \ ,$$

where the system term is again Eq. (1.29). For the reservoir term, the heat dissipated into the environment is calculated directly as the power exerted by the physical force field $\mathbf{F}(\mathbf{z}, t)$. According to the rules of stochastic calculus, the ensemble average of a Stratonovich product between a state-dependent vector field and the stochastic velocity $\dot{\mathbf{z}}(t)$ rigorously translates into a spatial integral of that same vector field dotted with the probability current $\mathbf{J}(\mathbf{z}, t)$ [24]. As a result, the reservoir entropy rate is expressed as (Eq. (1.35)):

$$\langle \dot{s}_{res}(t) \rangle = \frac{1}{T} \langle \mathbf{F}(\mathbf{z}, t) \circ \dot{\mathbf{z}}(t) \rangle = \frac{1}{T} \int \mathbf{F}(\mathbf{z}, t) \cdot \mathbf{J}(\mathbf{z}, t) \, d\mathbf{z} \ .$$

A NESS can only exist if the force field and the diffusion tensor are autonomous, i.e., time-independent, $\mathbf{F}(\mathbf{z}, t) \to \mathbf{F}(\mathbf{z})$ and $\mathbf{D}(t) \to \mathbf{D}$; a Fokker-Planck equation with explicit time-dependent coefficients could not, in general, admit a stationary solution. Under this assumption, the system reaches a Non-Equilibrium Steady State (NESS) when its probability distribution becomes time-independent as well, $\partial_t p(\mathbf{z}, t) = 0 \implies p(\mathbf{z}, t) \to p(\mathbf{z})$, yielding a purely spatial probability current $\mathbf{J}(\mathbf{z})$. Consequently, the macroscopic entropy of the system is strictly constant $(\langle \dot{s}_{sys} \rangle = 0)$.

For the remainder of this section we adopt natural dimensionless units, setting $k_B T = 1$; under this convention the Einstein relation simplifies to $\mathbf{D} = \boldsymbol{\gamma}^{-1}$, which we use throughout the following derivation. It is important to clarify that, as a consequence of this convention, the entropy production is effectively being measured in units of $k_B T$. The total average entropy production rate becomes a time-independent constant and reduces to the dissipative reservoir contribution:

$$\langle \dot{s}_{tot} \rangle = \langle \dot{s}_{res} \rangle = \langle \mathbf{F}(\mathbf{z}) \circ \dot{\mathbf{z}}(t) \rangle = \int \mathbf{F}(\mathbf{z}) \cdot \mathbf{J}(\mathbf{z}) \, d\mathbf{z} \ . \tag{1.37}$$

This exact steady-state expression serves as the mathematical starting point for the derivation of the traffic-inflow rate decomposition presented next.

1.3.6 Traffic $\mathcal{T}$

Several works [25, 26, 27] have demonstrated that, for a system in a NESS, the total average entropy production rate can be exactly decomposed into two distinct contributions:

$$\langle \dot{s}_{tot} \rangle = 4\mathcal{T} + \mathcal{G} \ ,$$

where $\mathcal{T}$ denotes the traffic (or frenesy), a time-symmetric kinetic observable that characterizes the dynamical activity of the system, whereas $\mathcal{G}$ represents the inflow rate [28], an observable acting as the continuous-state equivalent of the phase-space contraction rate.

While the existence of this decomposition is established in [25, 28], the explicit derivation presented below for the multidimensional overdamped Langevin system was carried out independently as part of this thesis. We operate under the assumptions of a NESS with autonomous fields $\mathbf{F}(\mathbf{z})$ and $\mathbf{D}$. For the following mathematical derivation, we assume also that the diffusion tensor $\mathbf{D}$ is space-independent. Furthermore, in order to physically decouple the dynamical activities of the individual degrees of freedom, which is a property that will be used in the following chapters, we assume $\mathbf{D}$ to be diagonal, representing uncorrelated additive white noise. Using $k_B T = 1$, implying $\boldsymbol{\gamma}^{-1} = \mathbf{D}$, the stationary probability current simplifies to $\mathbf{J}(\mathbf{z}) = \mathbf{D}\mathbf{F}(\mathbf{z})p(\mathbf{z}) - \mathbf{D}\nabla_{\mathbf{z}} p(\mathbf{z})$.

From $\mathbf{J}(\mathbf{z})$, we define the mean local velocity $\boldsymbol{\nu}(\mathbf{z})$ as the probability current divided by the probability density, which can be interpreted from a physical perspective as the average drift experienced by a trajectory passing through point $\mathbf{z}$:

$$\begin{aligned} \boldsymbol{\nu}(\mathbf{z}) &= \frac{\mathbf{J}(\mathbf{z})}{p(\mathbf{z})} \\ &= \frac{\mathbf{D}\mathbf{F}(\mathbf{z})p(\mathbf{z}) - \mathbf{D}\nabla_{\mathbf{z}} p(\mathbf{z})}{p(\mathbf{z})} \\ &= \mathbf{D}\left(\mathbf{F}(\mathbf{z}) - \frac{\nabla_{\mathbf{z}} p(\mathbf{z})}{p(\mathbf{z})}\right) \ . \end{aligned}$$

Using the vector logarithmic identity $\nabla_{\mathbf{z}} \log p(\mathbf{z}) = \frac{\nabla_{\mathbf{z}} p(\mathbf{z})}{p(\mathbf{z})}$, we arrive at the expression for the mean local velocity:

$$\boldsymbol{\nu}(\mathbf{z}) = \mathbf{D}\left(\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z})\right) \ . \tag{1.38}$$

Inverting this relation, the physical force field can be written as $\mathbf{F}(\mathbf{z}) = \mathbf{D}^{-1}\boldsymbol{\nu}(\mathbf{z}) + \nabla_{\mathbf{z}} \log p(\mathbf{z})$. This decomposition already anticipates the physical picture we are aiming for: the force splits into a part directly tied to the observable mean velocity, and a part tied to the local curvature of the stationary distribution.

We now decompose the total entropy production rate, which we denote as $\dot{S} \equiv \langle \dot{s}_{tot} \rangle$. Substituting the definition from Eq. (1.37):

$$\dot{S} = \int \mathbf{F}(\mathbf{z}) \cdot \left(p(\mathbf{z})\boldsymbol{\nu}(\mathbf{z})\right) \, d\mathbf{z} = \langle \mathbf{F}(\mathbf{z})^T \boldsymbol{\nu}(\mathbf{z}) \rangle \ .$$

Substituting our expression for the force into the expected value yields:

$$\begin{aligned} \dot{S} &= \langle \left(\mathbf{D}^{-1}\boldsymbol{\nu}(\mathbf{z}) + \nabla_{\mathbf{z}} \log p(\mathbf{z})\right)^T \boldsymbol{\nu}(\mathbf{z}) \rangle \\ &= \langle \boldsymbol{\nu}(\mathbf{z})^T \mathbf{D}^{-1} \boldsymbol{\nu}(\mathbf{z}) \rangle + \langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \boldsymbol{\nu}(\mathbf{z}) \rangle \ . \end{aligned}$$

Under the NESS hypothesis, the contribution from the term $\langle(\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \boldsymbol{\nu}(\mathbf{z})\rangle$ vanishes, leaving the standard positive-definite quadratic form. This can be understood intuitively: the vanishing term couples the gradient of the stationary density to the mean local velocity, and it is shown below (via the divergence-free property of $\mathbf{J}(\mathbf{z})$ in a NESS, Eq. 1.41) that this coupling integrates to zero over the whole space. What remains is a purely kinetic, non-negative quantity:

$$\dot{S} = \langle \boldsymbol{\nu}(\mathbf{z})^T \mathbf{D}^{-1} \boldsymbol{\nu}(\mathbf{z})\rangle \ . \tag{1.39}$$

We expand this quadratic form in Eq. (1.39) by substituting the direct definition of $\boldsymbol{\nu}(\mathbf{z})$ and exploiting the symmetry of the diffusion matrix ($\mathbf{D}^T = \mathbf{D}$):

$$\begin{aligned}
\dot{S} &= \langle (\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \, \mathbf{D}^T \mathbf{D}^{-1} \mathbf{D} \, (\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z}))\rangle \\
&= \langle (\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \, \mathbf{D} \, (\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z}))\rangle \\
&= \langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \mathbf{F}(\mathbf{z})\rangle - 2\langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle + \langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle \ .
\end{aligned}$$

The first term above depends purely on the force field and will turn out to belong to the traffic contribution; the last term depends purely on the geometry of the stationary distribution and will turn out to belong to the inflow rate. The cross-term in the middle mixes the two and must be resolved before a clean separation is possible.

We evaluate the cross-term separately using integration by parts (assuming the probability density vanishes at infinity):

$$\begin{aligned}
\langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle &= \int (\mathbf{D}\mathbf{F}(\mathbf{z})) \cdot \nabla_{\mathbf{z}} p(\mathbf{z}) \, d\mathbf{z} \\
&= -\int \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z})) \, p(\mathbf{z}) \, d\mathbf{z} \\
&= -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z}))\rangle \ .
\end{aligned}$$

Substituting this result back yields the expanded form:

$$\dot{S} = \langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \mathbf{F}(\mathbf{z})\rangle + 2\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z}))\rangle + \langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle \ . \tag{1.40}$$

At this stage the three terms are separated, but not yet in the physically meaningful traffic/inflow grouping: as we show next, the last term can itself be rewritten purely in terms of the force field, which will let us regroup the whole expression differently.

To simplify the final term, we recall that in a NESS the probability current is divergence-free ($\nabla_{\mathbf{z}} \cdot \mathbf{J}(\mathbf{z}) = 0$). This implies:

$$\int \nabla_{\mathbf{z}} \log p(\mathbf{z}) \cdot \mathbf{J}(\mathbf{z}) \, d\mathbf{z} = \int (\nabla_{\mathbf{z}} p(\mathbf{z})) \cdot \boldsymbol{\nu}(\mathbf{z}) \, d\mathbf{z} = 0 \ . \tag{1.41}$$

This is precisely the vanishing coupling term anticipated after Eq. (1.39).

Integrating by parts, we find that the expected divergence of the mean local velocity is zero ($\langle \nabla_{\mathbf{z}} \cdot \boldsymbol{\nu}(\mathbf{z})\rangle = 0$). Substituting $\boldsymbol{\nu}(\mathbf{z})$ with its definition in Eq. (1.38) we get:

$$\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z}) - \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z}))\rangle = 0 \implies \langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z}))\rangle = \langle \nabla_{\mathbf{z}} \cdot (\mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z}))\rangle \ .$$

Resolving the right-hand side via integration by parts:

$$\begin{aligned}\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\nabla_{\mathbf{z}} \log p(\mathbf{z}))\rangle &= -\int \nabla_{\mathbf{z}} p(\mathbf{z}) \cdot (\mathbf{D}\nabla_{\mathbf{z}} \log p(\mathbf{z}))\, d\mathbf{z} \\ &= -\langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D}\nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle \ .\end{aligned}$$

This gives the fundamental stationary-state relation, which is the key identity enabling the traffic/inflow separation:

$$\langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D}\nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle \ . \tag{1.42}$$

Substituting this back into the expanded entropy formula in Eq. (1.40), exactly cancels half of the cross-term:

$$\begin{aligned}\dot{S} &= \langle \mathbf{F}(\mathbf{z})^T \mathbf{DF}(\mathbf{z})\rangle + 2\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle - \langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle \\ &= \langle \mathbf{F}(\mathbf{z})^T \mathbf{DF}(\mathbf{z})\rangle + \langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle \ .\end{aligned} \tag{1.43}$$

This compact, two-term expression is entirely equivalent to Eq. (1.40) and depends only on the force field $\mathbf{F}(\mathbf{z})$, no longer explicitly on $\nabla_{\mathbf{z}} \log p(\mathbf{z})$. It will serve as a useful cross-check for the final decomposition derived below.

Connection to Traffic and Inflow Rate

Let us now compare the three-term expression obtained in Eq. (1.40) with the stationary-state relation of Eq. (1.42). The latter tells us that the probability-gradient term and the divergence of the drift coincide, up to a sign. This suggests a natural regrouping of the three terms in Eq. (1.40): the first two, which depend explicitly on the physical force field $\mathbf{F}(\mathbf{z})$, define the time-symmetric dynamical activity of the system, while the third, thanks to Eq. (1.42), can be entirely rewritten as (minus) the divergence of the drift. This observation motivates the following definitions.

First, the terms depending purely on the physical force field $\mathbf{F}(\mathbf{z})$ define the expected value of the time-symmetric dynamical activity ($\mathcal{T}$):

$$4\mathcal{T} = \langle \mathbf{F}(\mathbf{z})^T \mathbf{DF}(\mathbf{z})\rangle + 2\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle \ .$$

Physically, $\mathcal{T}$ measures how "active" the dynamics is, independently of its direction in time: it is insensitive to a time-reversal of the trajectories, which is why it is called time-symmetric.

Second, we isolate the term depending on the probability density gradient, which we demonstrated to be equivalent to the negative divergence of the drift vector ($\mathbf{DF}(\mathbf{z})$). This exactly matches the expected inflow rate $\mathcal{G}$:

$$\mathcal{G} = \langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D}\nabla_{\mathbf{z}} \log p(\mathbf{z})\rangle = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z}))\rangle \ .$$

Physically, $\mathcal{G}$ quantifies how strongly probability is being locally compressed or expanded by the deterministic drift.

Substituting these definitions back into the quadratic form, we obtain the fundamental decomposition:

$$\dot{S} = 4\mathcal{T} + \mathcal{G} \ .$$

This confirms that the thermodynamic cost of maintaining a steady state is driven by a fundamental competition between the system's time-symmetric dynamical activity ($4\mathcal{T}$) and the phase-space contraction induced by the environment ($\mathcal{G}$).

Chapter 2

Three-state model ($M = 1,\ N = 1$)

Following the notation introduced in Section 1.1.1, we consider a model with a single active bath η and a single membrane state x, coupled to a single probe y, without any feedback loop from downstream variables, and we take every coupling between states to be linear (see figure 2.1). Under these assumptions, the general model of Eq. (1.4) reduces to:

$$\begin{aligned} \tau_\eta \dot{\eta} &= -\eta + \sqrt{2D_\eta \tau_\eta} \cdot \xi_\eta(t) \\ \tau_x \dot{x} &= -x + \sigma\eta + \sqrt{2D_x \tau_x} \cdot \xi_x(t) \\ \tau_y \dot{y} &= -y + ax + \sqrt{2D_y \tau_y} \cdot \xi_y(t) \end{aligned} \tag{2.1}$$

These are essentially three coupled Ornstein-Uhlenbeck processes, with two asymmetric (non-reciprocal) couplings between the variables, given by the constants σ and a (both assumed to be greater than or equal to zero). The remaining symbols retain the meaning already introduced in Chapter 1.

To keep track of the model's complexity as we extend it in the following sections, we classify this model as a *three-state model*. Equivalently, we introduce two counting variables, M and N, denoting respectively the number of membrane states and the number of active bath states; the model above corresponds to the particular case $M = 1$, $N = 1$.

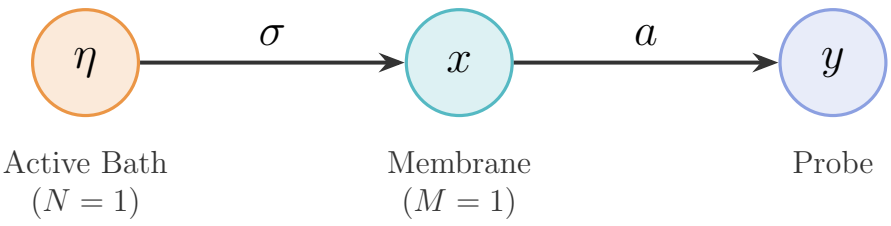


Figure 2.1: Network representation of the three-state model ($N = 1, M = 1$). The system consists of three coupled Ornstein-Uhlenbeck processes with purely feed-forward, non-reciprocal interactions. The active bath η drives the membrane state x with coupling strength σ, which in turn drives the external probe y with coupling strength a.

To solve the system in Eq. (2.1) analytically, we define the state vector $\mathbf{z}(t) = (y, x, \eta)^T$. The set of coupled Langevin equations can then be rewritten in the

compact matrix form:

$$\frac{d}{dt}\begin{pmatrix} y \\ x \\ \eta \end{pmatrix} = -\mathbf{A}\begin{pmatrix} y \\ x \\ \eta \end{pmatrix} + \tilde{\mathbf{D}}\begin{pmatrix} \xi_y(t) \\ \xi_x(t) \\ \xi_\eta(t) \end{pmatrix} \tag{2.2}$$

where $\mathbf{A}$ is the drift matrix and $\tilde{\mathbf{D}}$ is the noise coupling matrix, defined as:

$$\mathbf{A} = \begin{pmatrix} 1/\tau_y & -a/\tau_y & 0 \\ 0 & 1/\tau_x & -\sigma/\tau_x \\ 0 & 0 & 1/\tau_\eta \end{pmatrix}, \quad \tilde{\mathbf{D}} = \sqrt{2}\begin{pmatrix} \sqrt{D_y/\tau_y} & 0 & 0 \\ 0 & \sqrt{D_x/\tau_x} & 0 \\ 0 & 0 & \sqrt{D_\eta/\tau_\eta} \end{pmatrix} \tag{2.3}$$

The primary objective of this section is to optimize the probe-membrane coupling parameter a in order to maximize their mutual information, while strictly minimizing the associated thermodynamic dissipation. Once this optimal coupling is established, we will evaluate the information gained regarding the hidden active bath η through the observation of x. First, however, it is necessary to explicitly derive the analytical quantities underpinning this optimization procedure. Before starting the new section, it is important to highlight that all subsequent calculations are performed under the fundamental assumption that the system has relaxed to a NESS (Sec. 1.3.2).

2.1 Stationarity and Covariance Matrix

To calculate the mutual information between the variables in our system, we must first determine their joint stationary distribution. The time evolution of the probability density function $p(\mathbf{z}, t)$ for the state vector $\mathbf{z} = (y, x, \eta)^T$ is governed by the Fokker-Planck equation, already introduced in Sec. 1.3.2, corresponding to the Langevin dynamics in Eq. (2.2):

$$\frac{\partial p(\mathbf{z}, t)}{\partial t} = \sum_{i,j} \frac{\partial}{\partial z_i}\left(A_{ij} z_j p(\mathbf{z}, t)\right) + \sum_{i,j} D_{ij} \frac{\partial^2}{\partial z_i \partial z_j} p(\mathbf{z}, t) \tag{2.4}$$

where $\mathbf{A}$ is the drift matrix defined in Eq. (2.3), and $\mathbf{D}$ is the diffusion matrix, which is related to the noise coupling matrix by $\mathbf{D} = \frac{1}{2}\tilde{\mathbf{D}}\tilde{\mathbf{D}}^T$.

Because our system consists of linear coupled equations subject to additive Gaussian noise, the stationary solution (where $\frac{\partial p(\mathbf{z},t)}{\partial t} = 0$) is a multivariate Gaussian distribution [6]:

$$p(\mathbf{z}) = \frac{1}{\sqrt{(2\pi)^3 \det(\mathbf{\Sigma})}} \exp\left(-\frac{1}{2}(\mathbf{z} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1} (\mathbf{z} - \boldsymbol{\mu})\right) \tag{2.5}$$

where $\boldsymbol{\mu}$ denotes the mean vector and $\mathbf{\Sigma}$ is the covariance matrix. Given that the noise processes have zero mean, the stationary ensemble average of the state variables is also zero ($\boldsymbol{\mu} = \mathbf{0}$). Due to this, characterizing the probability distribution simply reduces to determining the covariance matrix $\mathbf{\Sigma}$, allowing us to bypass the need for repeated marginalizations of the joint probability density function.

Since the elements of $\mathbf{\Sigma}$ are defined by the second moments of the Gaussian distribution ($\Sigma_{nm} = \langle z_n z_m \rangle$), we can derive an equation for the covariance matrix by multiplying the stationary Fokker-Planck equation by $z_n z_m$ and integrating over the entire state space $d\mathbf{z}$:

$$-\int_{-\infty}^{+\infty} d\mathbf{z} \left(\sum_{i,j} \frac{\partial}{\partial z_i} (A_{ij} z_j p(\mathbf{z})) z_n z_m \right) = \int_{-\infty}^{+\infty} d\mathbf{z} \left(\sum_{i,j} D_{ij} \frac{\partial^2}{\partial z_i \partial z_j} p(\mathbf{z}) z_n z_m \right) \quad (2.6)$$

We solve this by applying integration by parts. For the left-hand side, assuming the probability density and its derivatives vanish at the boundaries ($\pm\infty$), we obtain:

$$\begin{aligned}
-\sum_{i,j} A_{ij} \int_{-\infty}^{+\infty} \frac{\partial}{\partial z_i} (z_j p)\, z_n z_m \, d\mathbf{z} &= \sum_{i,j} A_{ij} \int_{-\infty}^{+\infty} z_j p \left(\frac{\partial z_n}{\partial z_i} z_m + z_n \frac{\partial z_m}{\partial z_i} \right) d\mathbf{z} \\
&= \sum_{i,j} A_{ij} \int_{-\infty}^{+\infty} z_j p \left(\delta_{ni} z_m + \delta_{mi} z_n \right) d\mathbf{z} \\
&= \sum_{j} A_{nj} \langle z_j z_m \rangle + \sum_{j} A_{mj} \langle z_j z_n \rangle \\
&= (\mathbf{A\Sigma})_{nm} + (\mathbf{\Sigma A}^T)_{nm}
\end{aligned}$$

Applying a similar procedure to the right-hand side yields:

$$\begin{aligned}
\sum_{i,j} D_{ij} \int_{-\infty}^{+\infty} \frac{\partial^2 p}{\partial z_i \partial z_j} z_n z_m \, d\mathbf{z} &= \sum_{i,j} D_{ij} \int_{-\infty}^{+\infty} p \, \frac{\partial^2 (z_n z_m)}{\partial z_i \partial z_j} \, d\mathbf{z} \\
&= \sum_{i,j} D_{ij} \int_{-\infty}^{+\infty} p \left(\delta_{nj} \delta_{mi} + \delta_{mj} \delta_{ni} \right) d\mathbf{z} \\
&= D_{mn} + D_{nm} = 2D_{nm}
\end{aligned}$$

Equating the two sides produces the continuous-time Lyapunov equation for the covariance matrix:

$$\mathbf{A\Sigma} + \mathbf{\Sigma A}^T = 2\mathbf{D} \quad (2.7)$$

Solving this system yields the complete 3×3 symmetric covariance matrix $\mathbf{\Sigma}$:

$$\mathbf{\Sigma} = \begin{pmatrix} \Sigma_{yy} & \Sigma_{yx} & \Sigma_{y\eta} \\ \Sigma_{xy} & \Sigma_{xx} & \Sigma_{x\eta} \\ \Sigma_{\eta y} & \Sigma_{\eta x} & \Sigma_{\eta\eta} \end{pmatrix} \quad (2.8)$$

2.2 Mutual Information

Having established the covariance matrix, we can now derive the mutual information between any two variables in our network. As an illustrative example, we evaluate the mutual information between the membrane state x and the probe y (recalling that this procedure is equally valid for the target mutual information $I(y;\eta)$). By

combining the chain rule for entropy (Eq. 1.8) with the definition of mutual information for continuous variables (Eq. 1.13), we can express $I(x;y)$ in terms of marginal and joint differential entropies:

$$I(x;y) = h(x) + h(y) - h(x,y) \ . \tag{2.9}$$

Since the stationary distribution of our system is a multivariate Gaussian with zero mean, we can compute these quantities analytically. We begin by evaluating the marginal differential entropy (defined in Eq. 1.9) for the variable x, which has variance Σ_{xx}:

$$\begin{aligned} h(x) &= -\int_{-\infty}^{+\infty} p(x) \ln p(x)\, dx \\ &= -\int_{-\infty}^{+\infty} p(x) \left[-\frac{1}{2}\ln(2\pi\Sigma_{xx}) - \frac{x^2}{2\Sigma_{xx}} \right] dx \\ &= \frac{1}{2}\ln(2\pi\Sigma_{xx}) \int_{-\infty}^{+\infty} p(x)\, dx + \frac{1}{2\Sigma_{xx}} \int_{-\infty}^{+\infty} x^2 p(x)\, dx \ . \end{aligned} \tag{2.10}$$

Recognizing the normalization condition and the definition of the variance ($\int x^2 p(x) dx = \Sigma_{xx}$), the expression simplifies to the standard entropy of a 1D Gaussian:

$$h(x) = \frac{1}{2}\ln(2\pi\Sigma_{xx}) + \frac{1}{2} = \frac{1}{2}\ln(2\pi e \Sigma_{xx}) \ . \tag{2.11}$$

Next, let $\mathbf{w} = (x,y)^T$ be the bivariate state vector. The joint differential entropy is $h(x,y) = -\mathbb{E}[\ln p(\mathbf{w})]$. Substituting the bivariate Gaussian probability density function gives:

$$\begin{aligned} h(x,y) &= -\mathbb{E}\left[\ln\left(\frac{1}{2\pi\sqrt{\det(\mathbf{\Sigma}_{xy})}} \right) - \frac{1}{2}\mathbf{w}^T \mathbf{\Sigma}_{xy}^{-1} \mathbf{w} \right] \\ &= \ln(2\pi) + \frac{1}{2}\ln(\det(\mathbf{\Sigma}_{xy})) + \frac{1}{2}\mathbb{E}\left[\mathbf{w}^T \mathbf{\Sigma}_{xy}^{-1} \mathbf{w} \right] \ , \end{aligned} \tag{2.12}$$

where $\mathbf{\Sigma}_{xy}$ is the 2×2 submatrix containing the covariances of x and y. To evaluate the expectation of the quadratic form, we exploit the cyclical property of the trace operator ($\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$):

$$\mathbb{E}\left[\mathbf{w}^T \mathbf{\Sigma}_{xy}^{-1} \mathbf{w} \right] = \mathbb{E}\left[\mathrm{Tr}\left(\mathbf{\Sigma}_{xy}^{-1} \mathbf{w}\mathbf{w}^T \right) \right] = \mathrm{Tr}\left(\mathbf{\Sigma}_{xy}^{-1} \mathbb{E}\left[\mathbf{w}\mathbf{w}^T \right] \right) \ . \tag{2.13}$$

Because $\mathbb{E}[\mathbf{w}\mathbf{w}^T]$ is exactly the covariance matrix $\mathbf{\Sigma}_{xy}$, this term reduces to $\mathrm{Tr}(\mathbf{I}_{2\times 2}) = 2$. Plugging this result back into Eq. (2.12) yields the joint entropy:

$$h(x,y) = \frac{1}{2}\ln\left((2\pi e)^2 \det(\mathbf{\Sigma}_{xy}) \right) \ . \tag{2.14}$$

Finally, we substitute the marginal and joint entropies back into the mutual information equation:

$$\begin{aligned} I(x;y) &= \frac{1}{2}\ln(2\pi e\Sigma_{xx}) + \frac{1}{2}\ln(2\pi e\Sigma_{yy}) - \frac{1}{2}\ln\left((2\pi e)^2 \det(\mathbf{\Sigma}_{xy}) \right) \\ &= \frac{1}{2}\left[\ln(\Sigma_{xx}) + \ln(\Sigma_{yy}) - \ln(\det(\mathbf{\Sigma}_{xy})) \right] \ . \end{aligned} \tag{2.15}$$

By applying the properties of logarithms and expanding the determinant of the 2×2 submatrix ($\det(\mathbf{\Sigma}_{xy}) = \Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2$), we arrive at the exact analytical expression for the mutual information of our system:

$$I(x;y) = \frac{1}{2}\ln\left[\frac{\Sigma_{xx}\Sigma_{yy}}{\Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2}\right] \ . \tag{2.16}$$

2.3 Relative Entropy Rate and Traffic

Activating the probe-membrane coupling to extract information incurs an unavoidable thermodynamic cost. In this section, we quantify this energetic price through the *relative entropy production rate*, which we define as:

$$\dot{S}_{rel} = \dot{S}_a - \dot{S}_{a=0} \ . \tag{2.17}$$

Here, $\dot{S}_a$ represents the total steady-state entropy production rate of the fully coupled system, while $\dot{S}_{a=0}$ denotes the baseline dissipation of the uncoupled network.

By computing this difference, $\dot{S}_{rel}$ isolates the specific thermodynamic cost required to maintain the active sensory coupling. In doing so, it effectively factors out the continuous environmental dissipation generated by the non-tunable components of the hierarchy, which would occur regardless of whether the probe is monitoring the system or not. Following this thermodynamic assessment, we will shift our focus to the traffic $\mathcal{T}$ derived in the previous chapter, and we will explain why it serves as a more robust and physically insightful objective function than the standard entropy production rate for optimizing information transduction in such systems.

For a multidimensional Langevin system in a NESS, we derived earlier in Eq. (1.37) that the total average entropy production rate, denoted as $\dot{S}$, is strictly equivalent to the average heat rate dissipated into the environment:

$$\dot{S} = \langle \dot{s}_{res} \rangle = \langle \mathbf{F}(\mathbf{z}) \circ \dot{\mathbf{z}}(t) \rangle \ .$$

As established by the rules of stochastic thermodynamics [24] and as we have pointed out in Eq. (1.35), this expected value of the Stratonovich product can be equivalently computed leveraging the macroscopic probabilistic approach. This turns the ensemble average into a spatial integral involving the deterministic force field $\mathbf{F}(\mathbf{z})$ and the stationary probability current $\mathbf{J}(\mathbf{z})$:

$$\dot{S} = \int \mathbf{F}(\mathbf{z}) \cdot \mathbf{J}(\mathbf{z}) \, d\mathbf{z} \ . \tag{2.18}$$

To explicitly compute the entropy production rate for our linear model, we evaluate the integral by expressing both the thermodynamic force $\mathbf{F}(\mathbf{z})$ and the probability current $\mathbf{J}(\mathbf{z})$ in terms of the model's matrices.

Recalling our Langevin equation in matrix form (Eq. (2.2)), the deterministic drift is given by $-\mathbf{A}\mathbf{z}$. Following the standard relation $\dot{\mathbf{z}} = \mathbf{D}\mathbf{F}(\mathbf{z}) + \sqrt{2\mathbf{D}}\boldsymbol{\xi}(t)$, already expressed in Eq. (1.36), we identify the physical force field as:

$$\mathbf{F}(\mathbf{z}) = -\mathbf{D}^{-1}\mathbf{A}\mathbf{z} \ . \tag{2.19}$$

Next, we expand the stationary probability current $\mathbf{J}(\mathbf{z}) = \mathbf{D}\mathbf{F}(\mathbf{z})p(\mathbf{z}) - \mathbf{D}\nabla p(\mathbf{z})$. Because the stationary distribution is a zero-mean multivariate Gaussian, its gradient can be analytically evaluated as:

$$\nabla p(\mathbf{z}) = -\mathbf{\Sigma}^{-1}\mathbf{z}\, p(\mathbf{z}) \ . \tag{2.20}$$

Substituting this back into the current, we can factor out the state vector $\mathbf{z}$ and the probability density:

$$\begin{aligned} \mathbf{J}(\mathbf{z}) &= -\mathbf{A}\mathbf{z}\, p(\mathbf{z}) + \mathbf{D}\mathbf{\Sigma}^{-1}\mathbf{z}\, p(\mathbf{z}) \\ &= \left(\mathbf{D}\mathbf{\Sigma}^{-1} - \mathbf{A}\right)\mathbf{z}\, p(\mathbf{z}) \ . \end{aligned} \tag{2.21}$$

We can now plug the expressions for $\mathbf{F}(\mathbf{z})$ and $\mathbf{J}(\mathbf{z})$ into the integral for the entropy production rate:

$$\begin{aligned} \dot{S} &= \int \mathbf{F}(\mathbf{z}) \cdot \mathbf{J}(\mathbf{z})\, d\mathbf{z} \\ &= \int \left(-\mathbf{D}^{-1}\mathbf{A}\mathbf{z}\right)^T \left(\mathbf{D}\mathbf{\Sigma}^{-1} - \mathbf{A}\right)\mathbf{z}\, p(\mathbf{z})\, d\mathbf{z} \ . \end{aligned} \tag{2.22}$$

Recognizing that the integral over $p(\mathbf{z})\, d\mathbf{z}$ is simply the ensemble average of a quadratic form ($\mathbb{E}[\mathbf{z}^T\mathbf{M}\mathbf{z}]$), we can once again exploit the cyclical property of the trace ($\mathrm{Tr}(\mathbf{M}\mathbf{z}\mathbf{z}^T) = \mathrm{Tr}(\mathbf{M}\mathbf{\Sigma})$) to find an exact algebraic solution:

$$\begin{aligned} \dot{S} &= -\mathbb{E}\left[\mathbf{z}^T\mathbf{A}^T\mathbf{D}^{-1}\left(\mathbf{D}\mathbf{\Sigma}^{-1} - \mathbf{A}\right)\mathbf{z}\right] \\ &= -\mathrm{Tr}\left[\mathbf{A}^T\mathbf{D}^{-1}\left(\mathbf{D}\mathbf{\Sigma}^{-1} - \mathbf{A}\right)\mathbf{\Sigma}\right] \\ &= -\mathrm{Tr}\left[\mathbf{A}^T\mathbf{D}^{-1}\mathbf{D}\mathbf{\Sigma}^{-1}\mathbf{\Sigma} - \mathbf{A}^T\mathbf{D}^{-1}\mathbf{A}\mathbf{\Sigma}\right] \\ &= -\mathrm{Tr}\left[\mathbf{A}^T - \mathbf{A}^T\mathbf{D}^{-1}\mathbf{A}\mathbf{\Sigma}\right] \ . \end{aligned} \tag{2.23}$$

Finally, using the linearity of the trace and the property $\mathrm{Tr}(\mathbf{A}^T) = \mathrm{Tr}(\mathbf{A})$, we reverse the signs to obtain the standard compact formula for the entropy production rate of a multivariate linear Langevin system:

$$\dot{S} = \mathrm{Tr}\left[\mathbf{A}^T\mathbf{D}^{-1}\mathbf{A}\mathbf{\Sigma}\right] - \mathrm{Tr}\left[\mathbf{A}\right]. \tag{2.24}$$

Having established this result, the relative entropy production rate introduced in Eq. (2.17) can be explicitly computed as:

$$\dot{S}_{rel} = \mathrm{Tr}\left[\mathbf{A}_a^T\mathbf{D}^{-1}\mathbf{A}_a\mathbf{\Sigma}_a\right] - \mathrm{Tr}\left[\mathbf{A}_0^T\mathbf{D}^{-1}\mathbf{A}_0\mathbf{\Sigma}_0\right] \ , \tag{2.25}$$

where the subscripts denote the matrices evaluated with the coupling active ($a \neq 0$) and turned off ($a = 0$), respectively (note that the $-\mathrm{Tr}[\mathbf{A}]$ term cancels out perfectly in the subtraction).

According to our previous section 1.3.6, we can also express the traffic $\mathcal{T}$ for our linear Langevin system. Recalling its macroscopic definition:

$$4\mathcal{T} = \langle \mathbf{F}(\mathbf{z})^T\mathbf{D}\mathbf{F}(\mathbf{z})\rangle + 2\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z}))\rangle \ . \tag{2.26}$$

By substituting the explicit expression for the physical force field $\mathbf{F}(\mathbf{z}) = -\mathbf{D}^{-1}\mathbf{A}\mathbf{z}$, we can evaluate the two terms separately. For the first term, we exploit again the

cyclical property of the trace on the quadratic form, keeping in mind that the diffusion matrix is symmetric ($\mathbf{D}^T = \mathbf{D}$):

$$\begin{aligned}\langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \mathbf{F}(\mathbf{z}) \rangle &= \mathbb{E}\left[\left(-\mathbf{D}^{-1}\mathbf{A}\mathbf{z}\right)^T \mathbf{D} \left(-\mathbf{D}^{-1}\mathbf{A}\mathbf{z}\right) \right] \\ &= \mathbb{E}\left[\mathbf{z}^T \mathbf{A}^T \mathbf{D}^{-1} \mathbf{D} \mathbf{D}^{-1} \mathbf{A}\mathbf{z} \right] \\ &= \mathbb{E}\left[\mathbf{z}^T \mathbf{A}^T \mathbf{D}^{-1} \mathbf{A}\mathbf{z} \right] \\ &= \mathrm{Tr}\left[\mathbf{A}^T \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma} \right] \ . \end{aligned} \tag{2.27}$$

For the second term, we compute the expected divergence of the deterministic drift, $\mathbf{DF}(\mathbf{z}) = -\mathbf{A}\mathbf{z}$. Since the divergence of a linear vector field corresponds simply to the trace of its matrix ($\nabla_{\mathbf{z}} \cdot (\mathbf{M}\mathbf{z}) = \mathrm{Tr}(\mathbf{M})$), and this trace is a constant scalar, its ensemble average is trivially itself:

$$\langle \nabla_{\mathbf{z}} \cdot (-\mathbf{A}\mathbf{z}) \rangle = \langle -\mathrm{Tr}(\mathbf{A}) \rangle = -\mathrm{Tr}(\mathbf{A}) \ . \tag{2.28}$$

Combining these two results, we arrive at the exact analytical expression for the traffic in a multivariate linear NESS:

$$\mathcal{T} = \frac{1}{4}\left(\mathrm{Tr}\left[\mathbf{A}^T \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma} \right] - 2\mathrm{Tr}\left[\mathbf{A} \right] \right) \ . \tag{2.29}$$

As a consistency check, we can now explicitly verify the stationary relation $\dot{S} = 4\mathcal{T} + \mathcal{G}$ derived in the previous chapter. The inflow rate $\mathcal{G}$ was defined as the negative divergence of the drift ($\mathcal{G} = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}) \rangle$). In our linear system, this evaluates exactly to $\mathcal{G} = \mathrm{Tr}(\mathbf{A})$. Adding this to $4\mathcal{T}$, we find:

$$4\mathcal{T} + \mathcal{G} = \mathrm{Tr}\left[\mathbf{A}^T \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma} - 2\mathbf{A} \right] + \mathrm{Tr}(\mathbf{A}) = \mathrm{Tr}\left[\mathbf{A}^T \mathbf{D}^{-1} \mathbf{A} \boldsymbol{\Sigma} \right] - \mathrm{Tr}\left[\mathbf{A} \right] \ ,$$

which perfectly recovers the total entropy production rate derived in Eq. (2.24).

Traffic as the Energetic Cost of Transduction

While defining the energetic cost of transduction through the relative entropy production rate is thermodynamically rigorous, this choice presents significant practical challenges, particularly when attempting to verify results through numerical simulations or time-series data from real experiments.

First, evaluating the entropy production rate requires precise knowledge of the underlying microscopic force fields acting on the system. As recently highlighted [26], inferring these forces from experimental data is notoriously difficult. This limitation also extends to analytically intractable theoretical models (such as those featuring non-linear couplings) where the exact force fields cannot be explicitly derived.

Second, estimating the relative entropy rate $\dot{S}_{rel}$ from time series inherently requires two distinct datasets: one tracking the fully coupled system and another observing the uncoupled baseline. In most biophysical or experimental setups, obtaining such counterfactual data by deliberately "turning off" the sensory coupling is simply unfeasible.

Consequently, relying on the relative entropy production limits the applicability of our framework to a restricted class of exactly solvable linear models. To overcome

these drawbacks, recent advances in stochastic thermodynamics have demonstrated that the traffic $\mathcal{T}$ serves as an effective alternative objective function for this specific class of transduction problems.

A fundamental advantage of adopting the traffic $\mathcal{T}$ is that it can be numerically evaluated without any explicit knowledge of the underlying microscopic forces and thus it can be robustly used to bound and estimate the thermodynamic dissipation of the system.

The base of this force-free estimation is the Variance-Sum Rule (VSR) [29, 26, 5]. For our multidimensional Langevin system, the VSR expresses the time-symmetric dynamical activity entirely in terms of the system's stationary fluctuations:

$$\mathcal{T} = -\frac{1}{4}\sum_{i,j}(\mathbf{D}^{-1})_{ij}\ddot{C}^{(\mathbf{z})}_{ij}(0) \ , \tag{2.30}$$

where $C^{(\mathbf{z})}_{ij}(t)$ is the stationary two-point cross-correlation function of the state vector $\mathbf{z}(t)$. In its most general and complete form, it is defined as the steady-state ensemble average of the fluctuations around the mean values:

$$C^{(\mathbf{z})}_{ij}(t) = \langle (z_i(t) - \langle z_i \rangle)\,(z_j(0) - \langle z_j \rangle) \rangle = \langle z_i(t) z_j(0) \rangle - \langle z_i \rangle \langle z_j \rangle \ . \tag{2.31}$$

Recalling that the deterministic drift of our linear system has the form $\mathbf{F}(\mathbf{z}) = -\mathbf{A}\mathbf{z}$ (with no constant additive terms), the steady-state ensemble average of all our variables is strictly zero, meaning $\langle z_i \rangle = \langle z_j \rangle = 0$. The general definition simplifies exactly to the uncentered mixed moment:

$$C^{(\mathbf{z})}_{ij}(t) = \langle z_i(t) z_j(0) \rangle \ . \tag{2.32}$$

The estimation of $\mathcal{T}$ requires the diffusion matrix $\mathbf{D}$, which itself can be extracted directly from the time series without prior knowledge of the noise amplitude. By evaluating the right-derivative (at $t \to 0^+$) of the symmetrized correlation matrix, the diffusion coefficients are recovered as:

$$D_{ij} = -\dot{C}^{(\mathbf{z})}_{(i,j)}(0^+) \ , \quad \text{where} \quad C^{(\mathbf{z})}_{(i,j)}(t) = \frac{C^{(\mathbf{z})}_{ij}(t) + C^{(\mathbf{z})}_{ji}(t)}{2} \ . \tag{2.33}$$

By combining these relations, one can infer the traffic entirely from experimental or simulated trajectories. As detailed in [26], this is achieved by computing the empirical correlation functions and evaluating their first and second time derivatives at the origin, thus completely bypassing the need for a closed-form analytical drift.

The theoretical framework established so far provides all the tools required to rigorously simplify the thermodynamic cost function used in the optimization problem we will define in the next section. We start by expanding the relative entropy production rate $\dot{S}_{rel}$ using the traffic-inflow decomposition ($\dot{S} = 4\mathcal{T} + \mathcal{G}$):

$$\dot{S}_{rel} = \dot{S}_a - \dot{S}_{a=0} = (4\mathcal{T}_a + \mathcal{G}_a) - (4\mathcal{T}_{a=0} + \mathcal{G}_{a=0}) \ . \tag{2.34}$$

Recall that the inflow rate for a linear system is given by the trace of the drift matrix, $\mathcal{G} = \mathrm{Tr}(\mathbf{A})$. Looking at the explicit definition of $\mathbf{A}$ for our three-state model (Eq. 2.3), the trace is simply the sum of the inverse relaxation times:

$$\mathcal{G} = \frac{1}{\tau_y} + \frac{1}{\tau_x} + \frac{1}{\tau_\eta} \ . \tag{2.35}$$

Because the diagonal elements do not depend on the coupling parameter a, the inflow rate is an absolute constant of the system ($\mathcal{G}_a = \mathcal{G}_{a=0}$). Because of that, the geometric contribution cancels out completely, leaving only the difference in dynamical activity:

$$\dot{S}_{rel} = 4\left(\mathcal{T}_a - \mathcal{T}_{a=0}\right) = 4\mathcal{T}_{rel} \ . \tag{2.36}$$

We can simplify this further by examining the structure of the traffic itself. Since the diffusion matrix $\mathbf{D}$ in our model is strictly diagonal (representing independent thermal noise sources for each degree of freedom), the total traffic mathematically decouples into the sum of the individual traffics of each variable:

$$\mathcal{T} = \mathcal{T}_\eta + \mathcal{T}_x + \mathcal{T}_y \ . \tag{2.37}$$

At this point, we can exploit the purely feed-forward topology of our network ($\eta \to x \to y$). Because there is no feedback loop from the readout y back to the membrane x or the bath η, the stationary dynamics of the upstream variables are completely blind to the value of the coupling a. Their correlation functions, and therefore their individual traffics $\mathcal{T}_\eta$ and $\mathcal{T}_x$, are identical whether the probe is actively listening ($a \neq 0$) or disconnected ($a = 0$).

When substituting this into Eq. (2.36), all the upstream contributions exactly cancel out:

$$\begin{aligned}\dot{S}_{rel} &= 4\Big[\big(\mathcal{T}_\eta(a) + \mathcal{T}_x(a) + \mathcal{T}_y(a)\big) - \big(\mathcal{T}_\eta(0) + \mathcal{T}_x(0) + \mathcal{T}_y(0)\big)\Big] \\ &= 4\big(\mathcal{T}_y(a) - \mathcal{T}_y(0)\big) \ .\end{aligned} \tag{2.38}$$

This result has a profound implication for our objective. In any mathematical optimization problem, such as maximizing the mutual information against a thermodynamic penalty, subtracting a constant offset that does not depend on the optimization variable a (in this case, $\mathcal{T}_y(0)$) is strictly irrelevant, as it does not alter the location of the optimal point.

Therefore, minimizing the full relative entropy production rate of the network is mathematically and physically equivalent to minimizing just the absolute traffic of the probe, and this establishes $4\mathcal{T}_y(a)$ as the simplified objective function for our thermodynamic cost. However, we must make a mathematical remark regarding the matrix arrangement. While the total traffic of the system can be equivalently written using different matrix permutations under a full trace, the individual diagonal components depend on the chosen arrangement. Mathematically, the form $\mathbf{A}^T\mathbf{D}^{-1}\mathbf{A}\boldsymbol{\Sigma}$ mixes the rows during the multiplication with the transposed drift matrix $\mathbf{A}^T$. This algebraically spreads the coupling parameter a (which physically belongs only to y) into the upstream diagonal elements, such as the xx component. In contrast, the alternative arrangement $\mathbf{D}^{-1}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$ strictly respects the physical feed-forward topology of the system. Since a appears solely in the row of $\mathbf{A}$ corresponding to the probe y, the xx and $\eta\eta$ diagonal elements of this second matrix remain completely independent of a. The reason our optimization scheme is robust to this ambiguity is precisely because only the yy component of the traffic physically depends on the coupling a, meaning any variation of the total trace is perfectly localized in $\mathcal{T}_y$. Specifically, to maintain a strict mathematical and numerical correspondence

with the Variance-Sum Rule (VSR) estimators evaluated on the time-series data, we define the cost for the probe y using the precise matrix order that isolates a:

$$4\mathcal{T}_y(a) = \left[\mathbf{D}^{-1}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T - 2\mathbf{A}\right]_{yy} \ , \tag{2.39}$$

where the subscript yy indicates the diagonal element corresponding to the probe variable y. For the sake of completeness, we recall that the sum over all diagonal elements (the trace) yields the total system traffic:

$$4\mathcal{T} = \mathrm{Tr}\left[\mathbf{D}^{-1}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T - 2\mathbf{A}\right] \ , \tag{2.40}$$

and due to the cyclic property of the trace, this total sum is algebraically equivalent to the alternative quadratic forms, but focusing on the single diagonal element requires this specific arrangement to match the physics we are describing.

2.4 Coupling Optimization

Having defined the macroscopic quantities of our model, we now explain the optimization strategy used to tune the coupling parameter a. In particular, we introduce a Pareto functional that balances the mutual information $I(x;y)$ and the thermodynamic cost of the sensory mechanism, given by the traffic $4\mathcal{T}_y(a)$:

$$\mathcal{L}(a) = \lambda I(x;y) - (1-\lambda)4\mathcal{T}_y(a) \ , \tag{2.41}$$

where the parameter $\lambda \in [0,1]$ sets the relative weight of the two terms. Formally, to maintain dimensional consistency between the mutual information and the traffic, the latter should be multiplied by an observation time T. Since this detail is not crucial in these optimization frameworks, we set $T = 1$ without loss of generality.

The physical meaning of this parameter is straightforward. A value of λ close to 1 describes a system with a large energy budget, which can focus on extracting the maximum amount of information. On the other hand, a value of λ close to 0 represents a system with a limited energy budget, which must minimize its thermodynamic cost over the sensory precision.

The optimal coupling a^* is defined as the value that maximizes the functional in Eq. (2.41):

$$a^* = \underset{a}{\mathrm{argmax}} \left[\lambda I(x;y) - (1-\lambda)\ 4\mathcal{T}_y(a)\right] \ , \tag{2.42}$$

where it is clear that both terms explicitly depend on a.

In the following, we present our results by plotting the three fundamental quantities for different values of σ, which we remind is the active bath-membrane coupling (see Figure 2.2):

- Mutual Information ($I(y;\eta)$): evaluated at the optimal coupling a^*. Notice that we are now plotting the mutual information between the probe and the hidden active bath. This is the true goal of the transduction process: understanding how much information the external probe can capture about the inaccessible dynamics.

- Traffic cost ($4\mathcal{T}_y(a^*)$): the relative thermodynamic dissipation paid by the system operating at the optimal point.

- Optimal coupling values (a^*): the exact value of the parameter that maximizes the Pareto functional for a given setup.

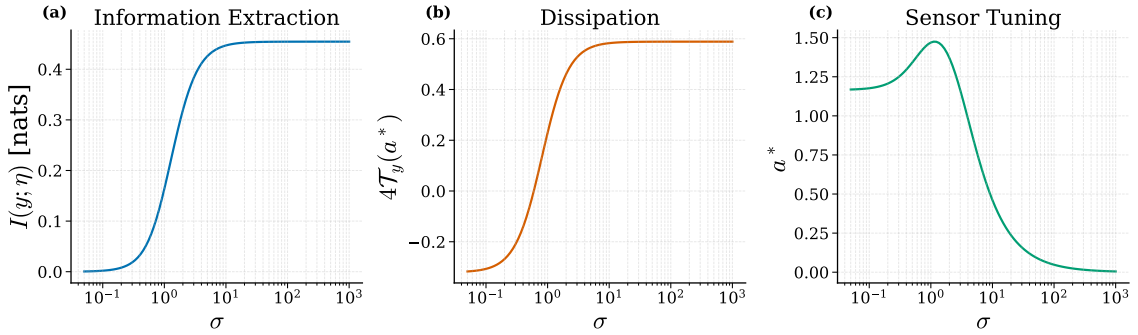


Figure 2.2: Optimization results for different values of σ. The plots show the mutual information $I(y;\eta)$ (a), the traffic cost $4\mathcal{T}_y$ (b), and the optimal coupling parameter a^* (c). In these plots, we used $\lambda = 0.9$; $\tau_y = \tau_x = 1$, $\tau_\eta = 5$.

From Figure 2.2, we can draw several key observations that will serve as a baseline for the multidimensional model. Since the parameter σ represents the coupling strength between the active bath and the membrane, it can be physically interpreted as the amount of signal transmitted to the system.

Specifically, we observe the following distinct behaviors:

- Information saturation: A low value of σ corresponds to a weak signal, resulting in near-zero mutual information. As the signal strength increases, the extracted information increases accordingly, until it eventually reaches a plateau. Beyond this point, an even stronger signal does not provide any additional useful information to the probe.

- Optimal sensor tuning: The signal intensity directly dictates the shape of the optimal coupling a^*. For weak and intermediate values of σ, the coupling must increase to effectively capture the poor signal. However, after reaching a maximum, a^* decreases monotonically. The physical reason is intuitive: if the incoming signal is already very strong, the probe does not need a thermodynamically expensive coupling to extract it, so the system lowers a^* to save energy.

- Thermodynamic cost and local traffic: $4\mathcal{T}_y(a^*)$ closely tracks the amount of information extracted by the probe. At low signal strengths, the thermodynamic cost is minimal, and in this regime the observable can take negative values. This is not in contradiction with the second law: only the total entropy production rate $\dot{S} = 4\mathcal{T} + \mathcal{G}$ is guaranteed to be non-negative, while the traffic $\mathcal{T}$ alone contains a divergence term (Sec. 1.3.6) that is not sign-definite, so both the global dynamical activity and the localized contribution of the probe, $\mathcal{T}_y$, can in principle fall below zero. As σ grows, the probe is forced

into higher dynamical activity to track the incoming signal, causing a sharp rise in the thermodynamic cost, which eventually plateaus in tandem with the mutual information.

In the following chapters, we will use these results as a baseline to make comparisons with more complex topologies.

Chapter 3

Multiple-state model

In this chapter, we extend our model to a generalized network with arbitrary M and N nodes. To first clarify the setup, we report the equations that govern this extended system, which are based on Eq. (2.1).

$$\begin{cases} \dot{y} = -\frac{y}{\tau_y} + \frac{1}{\tau_y}\sum_{i=1}^{M} a_i x_i + \sqrt{\frac{2D_y}{\tau_y}}\xi_y(t) \\ \dot{x}_i = -\frac{x_i}{\tau_{x_i}} + \sum_{\substack{l=1 \\ l\neq i}}^{M} \frac{k_{il}^{(x)}}{\tau_{x_i}} x_l + \sum_{j=1}^{N} \frac{\sigma_{ij}}{\tau_{x_i}}\eta_j + \sqrt{\frac{2D_{x_i}}{\tau_{x_i}}}\xi_{x_i}(t) \quad i = 1, \dots, M \\ \dot{\eta}_j = -\frac{\eta_j}{\tau_{\eta_j}} + \sum_{\substack{l=1 \\ l\neq j}}^{N} \frac{k_{jl}^{(\eta)}}{\tau_{\eta_j}} \eta_l + \sqrt{\frac{2D_{\eta_j}}{\tau_{\eta_j}}}\xi_{\eta_j}(t) \quad j = 1, \dots, N \end{cases} \tag{3.1}$$

From Eq. (3.1), we can see that the overall structure of the equations remains unchanged, except that a single node in the network can now be influenced by multiple entities instead of just one. In particular, the parameters a_i, σ_{ij}, τ_{η_j}, τ_{x_i}, and τ_y maintain the same meaning as in the three-state model. The new addition is the parameter $k_{ij}^{(\dots)}$. This parameter acts as an internal structural coupling within the membrane layer and the active bath layer. It is needed to model mechanisms such as inhibition or co-activation between states of the same group, which was obviously not possible in the simpler three-state model.

In Fig. 3.1 we sketch a diagram of the extended system.

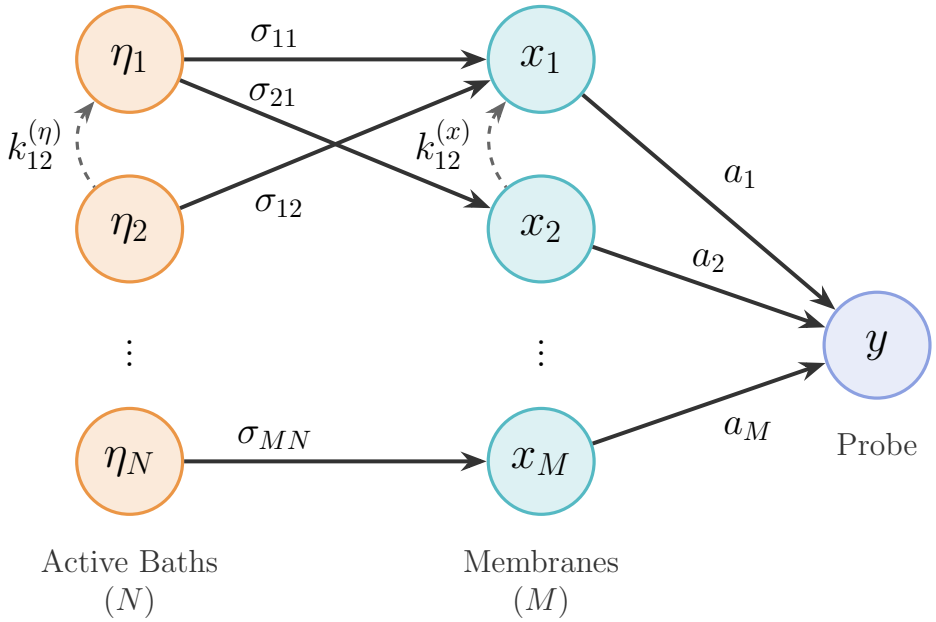


Figure 3.1: General multidimensional transduction network with N active baths, M membrane states, and a single external probe y. The parameter σ_{ij} represents the feed-forward coupling from the j-th bath to the i-th membrane. Similarly, the parameters $k_{12}^{(x)}$ and $k_{12}^{(\eta)}$ denote the directed internal structural couplings (from state 2 to state 1), effectively modeling mechanisms such as inhibition ($k < 0$) or co-activation ($k > 0$) within the sensory and environmental layers.

Adopting the vectorial representation for the $(1 + M + N)$-dimensional state, we can rewrite Eq. (3.1) in a compact matrix form:

$$\frac{d}{dt}\begin{pmatrix} y \\ \mathbf{x} \\ \boldsymbol{\eta} \end{pmatrix} = -A\begin{pmatrix} y \\ \mathbf{x} \\ \boldsymbol{\eta} \end{pmatrix} + \tilde{D}\begin{pmatrix} \xi_y(t) \\ \boldsymbol{\xi}_x(t) \\ \boldsymbol{\xi}_\eta(t) \end{pmatrix} \tag{3.2}$$

To maintain mathematical rigor and avoid ambiguous element-wise array divisions, we introduce the diagonal timescale matrices $T_x = \mathrm{diag}(\tau_{x_1}, \ldots, \tau_{x_M})$ and $T_\eta = \mathrm{diag}(\tau_{\eta_1}, \ldots, \tau_{\eta_N})$. The block drift matrix A can then be rigorously defined as:

$$A = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{1}{\tau_y}\mathbf{a}^T & O_{1\times N} \\ -T_x^{-1}F_{xy} & T_x^{-1}A_x & -T_x^{-1}\boldsymbol{\sigma} \\ O_{N\times 1} & -T_\eta^{-1}F_{\eta x} & T_\eta^{-1}A_\eta \end{pmatrix} \tag{3.3}$$

The individual blocks of the generalized drift matrix A have the following dimensions and physical meanings:

- Probe dynamics (Row 1): $A_{11} = \frac{1}{\tau_y} \in \mathbb{R}$ is the intrinsic relaxation rate of the probe. $A_{12} = -\frac{1}{\tau_y}\mathbf{a}^T \in \mathbb{R}^{1\times M}$ contains the row vector $\mathbf{a}^T$. These elements represent the tunable gains of the probe, dictating how strongly the probe itself chooses to couple to (or "listen to") the incoming feed-forward signals from the membrane layer.

- Membrane dynamics (Block row 2): $A_{22} = T_x^{-1} A_x \in \mathbb{R}^{M\times M}$ describes the internal dynamics of the membrane layer. The matrix A_x contains 1s on the diagonal (self-decay) and the structural couplings $-k_{ij}^{(x)}$ on the off-diagonals. $A_{23} = -T_x^{-1}\boldsymbol{\sigma} \in \mathbb{R}^{M\times N}$ encodes the active forces exerted by the baths, where $\boldsymbol{\sigma}$ is the matrix containing the σ_{ij} elements.

- Bath dynamics (Block row 3): $A_{33} = T_\eta^{-1} A_\eta \in \mathbb{R}^{N\times N}$ dictates the internal dynamics of the active baths, similar to the membrane layer. If we assume independence between the baths (no internal correlations), A_η becomes simply the identity matrix.

- Topological constraints and back-actions: $A_{13} = O_{1\times N}$ and $A_{31} = O_{N\times 1}$ are null matrices, enforcing the topological constraint that the external probe and the active baths do not interact directly. Furthermore, $F_{xy} \in \mathbb{R}^{M\times 1}$ and $F_{\eta x} \in \mathbb{R}^{N\times M}$ represent the mechanical back-actions (from the probe to the membranes, and from the membranes to the baths, respectively), which are assumed to be null in our feed-forward model.

Finally, the fluctuation matrix $\tilde{D}$ takes the block-diagonal form:

$$\tilde{D} = \sqrt{2}\begin{pmatrix} \sqrt{\frac{D_y}{\tau_y}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \bar{D}_x & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \bar{D}_\eta \end{pmatrix} \tag{3.4}$$

where the noise amplitudes for the multidimensional layers are explicitly defined by the diagonal sub-matrices:

$$\bar{D}_x = \begin{pmatrix} \sqrt{\frac{D_{x_1}}{\tau_{x_1}}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\frac{D_{x_M}}{\tau_{x_M}}} \end{pmatrix}, \quad \bar{D}_\eta = \begin{pmatrix} \sqrt{\frac{D_{\eta_1}}{\tau_{\eta_1}}} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\frac{D_{\eta_N}}{\tau_{\eta_N}}} \end{pmatrix}$$

In the subsequent sections, we generalize the analytical framework developed for the three-state model to incorporate this multidimensional block structure.

3.1 Mutual Information

The mutual information between two scalar random variables was introduced in Eq. (2.16). In our generalized framework, however, the source variable is a d-dimensional vector (where d corresponds to the number of membranes M or baths N, depending on the specific layer of interest), while the probe y remains a scalar.

Let $y \in \mathbb{R}$ denote the scalar random variable and $\mathbf{x} \in \mathbb{R}^d$ the vectorial one. As demonstrated in Sec. 2.2, the unforced system relaxes to a steady state characterized

by a zero-mean Gaussian distribution. Consequently, their respective probability density functions are given by:

$$p(y) = \frac{1}{\sqrt{2\pi\Sigma_{yy}}} \exp\left(-\frac{y^2}{2\Sigma_{yy}}\right) \tag{3.5}$$

and

$$p(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^d \det(\mathbf{\Sigma_{xx}})}} \exp\left(-\frac{1}{2}\mathbf{x}^T\mathbf{\Sigma_{xx}^{-1}}\mathbf{x}\right) \tag{3.6}$$

where Σ_{yy} is the variance of the probe y, and $\mathbf{\Sigma_{xx}}$ represents the $d \times d$ covariance matrix of the state vector $\mathbf{x}$.

By substituting the expressions for the differential entropy of the d-dimensional vector in Eq. (2.12), the entropy of the scalar variable in Eq. (2.11), and the joint entropy of the $(d+1)$-dimensional multivariate Gaussian distribution $(y, \mathbf{x})$ into the fundamental definition of mutual information (Eq. (2.9)), we obtain the final expression:

$$I(\mathbf{x}; y) = \frac{1}{2} \ln\left(\frac{\Sigma_{yy} \det(\mathbf{\Sigma_{xx}})}{\det(\mathbf{\Sigma}_{\text{joint}})}\right) \tag{3.7}$$

where $\mathbf{\Sigma}_{\text{joint}}$ is the $(d+1) \times (d+1)$ joint covariance matrix of the coupled system, defined in block form as:

$$\mathbf{\Sigma}_{\text{joint}} = \begin{pmatrix} \Sigma_{yy} & \mathbf{\Sigma}_{y\mathbf{x}} \\ \mathbf{\Sigma}_{\mathbf{x}y} & \mathbf{\Sigma_{xx}} \end{pmatrix} . \tag{3.8}$$

Here, $\mathbf{\Sigma}_{\mathbf{x}y} \in \mathbb{R}^{d\times 1}$ represents the column vector of cross-covariances between the elements of the multidimensional layer $\mathbf{x}$ and the scalar probe y, while $\mathbf{\Sigma}_{y\mathbf{x}} = \mathbf{\Sigma}_{\mathbf{x}y}^T \in \mathbb{R}^{1\times d}$ is the corresponding row vector, satisfying the intrinsic symmetry of the joint covariance matrix.

3.2 Relative Entropy Rate and Traffic

The generalization of the relative entropy rate, and consequently of the traffic, is straightforward within the new block-matrix formalism. By leveraging the exact same procedure detailed in Sec. 2.3 and defining the $(1+M+N)$-dimensional state vector as $\mathbf{z} = (y, \mathbf{x}^T, \boldsymbol{\eta}^T)^T$, with the drift matrix $\mathbf{A}$ given by Eq. (3.3) and the diffusion matrix $\mathbf{D} = \frac{1}{2}\tilde{\mathbf{D}}\tilde{\mathbf{D}}^T$ derived from Eq. (3.4), we recover the exact same structure for the total entropy production rate:

$$\dot{S} = \text{Tr}\left[\mathbf{A}^T\mathbf{D}^{-1}\mathbf{A}\mathbf{\Sigma}\right] - \text{Tr}[\mathbf{A}] , \tag{3.9}$$

where $\mathbf{\Sigma} = \langle \mathbf{z}\mathbf{z}^T \rangle$ is the full $(1+M+N) \times (1+M+N)$ covariance matrix of the entire coupled system.

The relative entropy production rate $\dot{S}_{rel}$ is then obtained by evaluating the difference between the fully coupled and the uncoupled system:

$$\dot{S}_{rel} = \dot{S}_{\mathbf{a}} - \dot{S}_{\mathbf{a}=\mathbf{0}} = \text{Tr}\left[\mathbf{A}_{\mathbf{a}}^T\mathbf{D}^{-1}\mathbf{A}_{\mathbf{a}}\mathbf{\Sigma}_{\mathbf{a}}\right] - \text{Tr}\left[\mathbf{A}_{\mathbf{0}}^T\mathbf{D}^{-1}\mathbf{A}_{\mathbf{0}}\mathbf{\Sigma}_{\mathbf{0}}\right] . \tag{3.10}$$

This is completely in accordance with our previous findings for the three-state model, confirming that the topological scaling of the network does not alter the fundamental thermodynamic structure of the transduction cost.

Regarding the traffic, applying the exact same procedure to the estimation of $4\mathcal{T}_y(\mathbf{a})$ (see Subsec. 2.3) yields:

$$4\mathcal{T}_y(\mathbf{a}) = \left[\mathbf{D}^{-1}\mathbf{A}\mathbf{\Sigma}\mathbf{A}^T - 2\mathbf{A}\right]_{yy} \ , \tag{3.11}$$

where the subscript yy indicates again the diagonal element corresponding to the probe variable y. Indeed, while taking the full trace of the matrix would give the global dynamical activity (or total traffic) of the entire $(1 + M + N)$-dimensional system, isolating the yy component is conceptually crucial. It ensures that the traffic is measured exclusively at the probe level, matching the thermodynamic cost evaluated from the local perspective.

In summary, the central conclusion of these last two sections is that the block-matrix framework easily extends the physical quantities established in Chapter 2 to a multidimensional setting.

3.3 Multi-Source Optimization Scheme

Extending our analysis to multidimensional topologies provides the opportunity to explore emergent phenomena that cannot be captured by the minimal three-state model. Specifically, we seek to determine whether synergistic and redundant information components facilitate or impede the overall transduction process. To investigate this, we must adapt the optimization procedure outlined in Sec. 2.4, although its core principles and overall structure remain invariant.

We consider again the same Pareto functional, parameterized by $\lambda \in [0, 1]$, which balances an information gain term against a thermodynamic dissipation penalty. The key novelty here is that multidimensional topologies allow us to exploit the Partial Information Decomposition (PID) framework (see Sec. 1.2.3 for details).

Since there is no unique way to evaluate the distinct contributions of these topological behaviors, we explicitly parameterize them within the information gain term.

However, as highlighted in our theoretical introduction in Subsec. 1.2.3, expanding the PID framework for an arbitrary number of $M > 2$ membranes leads to a combinatorial explosion of partial interaction terms. In such expanded systems, macroscopic synergy and redundancy fragment into localized sub-components (e.g., partial redundancies shared only by subsets of membranes), making a precise decomposition intractable. Therefore, to explicitly control the synergy-redundancy trade-off without information loss or double-counting, from this point forward we restrict our multi-source analysis to exactly $M = 2$ membranes.

To systematically assess the respective roles of synergy and redundancy in the optimal transduction process, we introduce a continuous tuning parameter $\gamma \in [0, 1]$ that weights these two contributions. For our two-sensor architecture (x_1, x_2), our multi-source optimization scheme evaluates the following functional:

$$\mathcal{L}(\mathbf{a}; \lambda, \gamma) = \lambda\Big[U(x_1 \setminus x_2; y) + U(x_2 \setminus x_1; y) + 2\gamma S(x_1, x_2; y) + 2(1-\gamma)R(x_1, x_2; y)\Big] - (1-\lambda)4\mathcal{T}_y(\mathbf{a}) \tag{3.12}$$

where $\mathbf{a} = (a_1, a_2)^T$ is the vector of coupling parameters to be optimized, $U(x_1 \setminus x_2; y)$ and $U(x_2 \setminus x_1; y)$ represent the unique information provided solely by each respective membrane, and S and R denote the total synergistic and redundant information components of the sensory layer. Note that Eq. (3.12) is exact only for $M = 2$.

To explicitly demonstrate the mathematical consistency of this parameterization, we can evaluate the objective functional for the symmetric case $\gamma = 1/2$. In this regime, the multiplicative weights for synergy and redundancy both evaluate to unity, and the term in square brackets naturally and exactly recovers the standard total joint mutual information (Eq. (1.15)):

$$\mathcal{L}(\mathbf{a}; \lambda, \tfrac{1}{2}) = \lambda \underbrace{\Big[U(x_1 \setminus x_2; y) + U(x_2 \setminus x_1; y) + S(x_1, x_2; y) + R(x_1, x_2; y)\Big]}_{I(x_1, x_2; y)} - (1-\lambda)\, 4\mathcal{T}_y(\mathbf{a})\,. \quad (3.13)$$

By tuning the parameter γ across its defined domain, we can identify three distinct operational strategies regarding the information gain:

- Synergy-driven ($\gamma = 1$): The system prioritizes the emergent information that is accessible only through the joint observation of all the membranes.
- Balanced ($\gamma = 0.5$): The system treats all information components with equal statistical weight. As previously noted, this specific tuning exactly recovers the standard total mutual information $I(x_1, x_2; y)$ (see Eq. (3.13)).
- Redundancy-driven ($\gamma = 0$): The probe favors a robust representation where the same information is duplicated and shared across multiple membranes.

Finally, for any given choice of the trade-off parameter λ and the strategy selector γ, the optimal vector of coupling parameters $\mathbf{a}^*$ is determined by maximizing the multi-source objective functional:

$$\mathbf{a}^*(\lambda, \gamma) = \arg\max_{\mathbf{a}} \mathcal{L}(\mathbf{a}; \lambda, \gamma)\ . \quad (3.14)$$

This optimal configuration $\mathbf{a}^*$ dictates the specific network tuning that yields the best possible compromise between the selected informational target (whether synergy, redundancy, or a balance of both) and the intrinsic thermodynamic dissipation required to sustain the sensory process. Once this optimal configuration is established, we can evaluate the relevant physical observables and directly compare the multidimensional network's performance against the baseline three-state model discussed in Chapter 2. Specifically, this comparison aims to determine whether prioritizing a particular component of the joint mutual information (synergy or redundancy) enhances the overall amount of information the probe can extract from the underlying active baths.

3.3.1 Comparing Systems at Constant Dissipation

Having defined our generalized multidimensional model, a natural question arises: does an increase in architectural complexity inherently enhance the extraction of mutual information from the active bath(s)? However, comparing different network architectures without imposing physical constraints may be biased. Since information processing is thermodynamically bounded, an unconstrained topology might appear to perform better simply because it dissipates more energy. To ensure a fair comparison, we must evaluate the models strictly at parity of thermodynamic cost.

To eliminate this thermodynamic bias, we perform the optimization procedure introduced in Sec. 3.3 under a strict relative equal-dissipation constraint. Specifically, we equate the relative entropy production rate of the system (M membranes and N active baths) to that of the baseline three-state model:

$$\dot{S}^{rel}_{M,N}(a_1, \dots, a_M) = \dot{S}^{rel}_{M=1,N=1}(a^*) \ , \tag{3.15}$$

Where the general expression $\dot{S}^{rel}_{M,N}(a_1, \dots, a_M)$ is given by Eq. (3.10) and a^* is the optimal coupling value that maximizes the functional in Eq. (2.41) for the baseline architecture.

This equality translates into an algebraic constraint on the coupling parameters, setting a function of the couplings to a constant target value, $f(a_1, \dots, a_M) = \mathcal{C}$. By exploiting this relation, we can isolate one of the parameters. For instance, expressing a_1 explicitly as a function of the remaining ones:

$$a_1 = g(a_2, \dots, a_M) \ . \tag{3.16}$$

By substituting this relation back into the system, we effectively restrict our search to an $(M-1)$-dimensional manifold that preserves the total energy consumption, and the optimization problem is recast as a constrained maximization over the reduced parameter space:

$$\mathbf{a}^* = \underset{a_2,\dots,a_M}{\operatorname{argmax}} \ \mathcal{L}\big(g(a_2, \dots, a_M), a_2, \dots, a_M\big) \ . \tag{3.17}$$

Note that, in general, a physically meaningful solution to this problem is not guaranteed to exist. For the optimization to be physically meaningful, two conditions must be satisfied. First, the overall network must be dynamically stable, meaning that the eigenvalues of the drift matrix $\mathbf{A}$ must have strictly positive real parts to ensure the existence of a NESS. Second, the constraint equation must yield a real and non-negative solution for the parameter $a_1(a_2, \dots, a_M)$. Therefore, if this condition is violated, or if the resulting point falls outside the valid $(M-1)$-dimensional relative equal-dissipation manifold, the configuration is invalid and must be discarded.

Having established this theoretical foundation, we now proceed to investigate two specific architectures, directly comparing their optimized results with those of the three-state model.

Chapter 4

$M = 2$, $N = 1$ Topology: Two Membranes and One Active Bath

The first particular topology we are going to study in detail is composed by two membranes ($M = 2$) and one active bath (N=1). According to Eq. (3.1), in this specific case the system of equations becomes:

$$\begin{cases} \dot{y} = -\frac{y}{\tau_y} + \frac{a_1}{\tau_y} x_1 + \frac{a_2}{\tau_y} x_2 + \sqrt{\frac{2D_y}{\tau_y}} \xi_y(t) \\ \dot{x}_1 = -\frac{x_1}{\tau_{x_1}} + \frac{k_{12}^{(x)}}{\tau_{x_1}} x_2 + \frac{\sigma_{11}}{\tau_{x_1}} \eta + \sqrt{\frac{2D_{x_1}}{\tau_{x_1}}} \xi_{x_1}(t) \\ \dot{x}_2 = -\frac{x_2}{\tau_{x_2}} + \frac{k_{21}^{(x)}}{\tau_{x_2}} x_1 + \frac{\sigma_{21}}{\tau_{x_2}} \eta + \sqrt{\frac{2D_{x_2}}{\tau_{x_2}}} \xi_{x_2}(t) \\ \dot{\eta} = -\frac{\eta}{\tau_\eta} + \sqrt{\frac{2D_\eta}{\tau_\eta}} \xi_\eta(t) \end{cases} \tag{4.1}$$

Here, a_1 and a_2 are the informational couplings to be optimized. The parameters σ_{11} and σ_{21} quantify how strongly each sensory membrane is driven by the active bath, while $k_{12}^{(x)}$ and $k_{21}^{(x)}$ represent the intrinsic lateral couplings between the two membranes.

Defining the four-dimensional state vector $\mathbf{z} = (y, x_1, x_2, \eta)^T$, we can express the dynamics in the compact vectorial form:

$$\dot{\mathbf{z}} = -A\mathbf{z} + \tilde{D}\boldsymbol{\xi}(t) \ , \tag{4.2}$$

where $\boldsymbol{\xi}(t) = (\xi_y(t), \xi_{x_1}(t), \xi_{x_2}(t), \xi_\eta(t))^T$ is a vector of independent standard white noises.

By applying the block-matrix definition introduced in Eq. (3.3), the explicit 4×4 drift matrix A for this topology reads:

$$A = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{a_1}{\tau_y} & -\frac{a_2}{\tau_y} & 0 \\ 0 & \frac{1}{\tau_{x_1}} & -\frac{k_{12}^{(x)}}{\tau_{x_1}} & -\frac{\sigma_{11}}{\tau_{x_1}} \\ 0 & -\frac{k_{21}^{(x)}}{\tau_{x_2}} & \frac{1}{\tau_{x_2}} & -\frac{\sigma_{21}}{\tau_{x_2}} \\ 0 & 0 & 0 & \frac{1}{\tau_\eta} \end{pmatrix} . \tag{4.3}$$

Notice how the strict feed-forward nature of the model is reflected in the upper block-triangular structure drift matrix, preventing any back-action from the probe to the membranes or from the membranes to the bath.

Similarly, the diagonal noise amplitude matrix $\tilde{D}$ is explicitly given by:

$$\tilde{D} = \sqrt{2} \begin{pmatrix} \sqrt{\frac{D_y}{\tau_y}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{D_{x_1}}{\tau_{x_1}}} & 0 & 0 \\ 0 & 0 & \sqrt{\frac{D_{x_2}}{\tau_{x_2}}} & 0 \\ 0 & 0 & 0 & \sqrt{\frac{D_\eta}{\tau_\eta}} \end{pmatrix} . \tag{4.4}$$

For the sake of simplicity and to ensure a rigorous comparison with the baseline three-state model, in the following we assume a symmetric configuration for the membrane layer. Specifically, we set the lateral interactions to be identical, $k_{12}^{(x)} = k_{21}^{(x)} \equiv k$, equal characteristic timescales $\tau_{x_1} = \tau_{x_2} \equiv \tau_x$ and we assume also that both sensors are equally driven by the single active bath, setting $\sigma_{11} = \sigma_{21} \equiv \sigma$. By maintaining this structural symmetry, we ensure that the overall setup remains as close as possible to the reference model, guaranteeing that any emergent behaviour is driven strictly by the optimization of the informational couplings.

In Fig. 4.1 we sketch a schematic of the system:

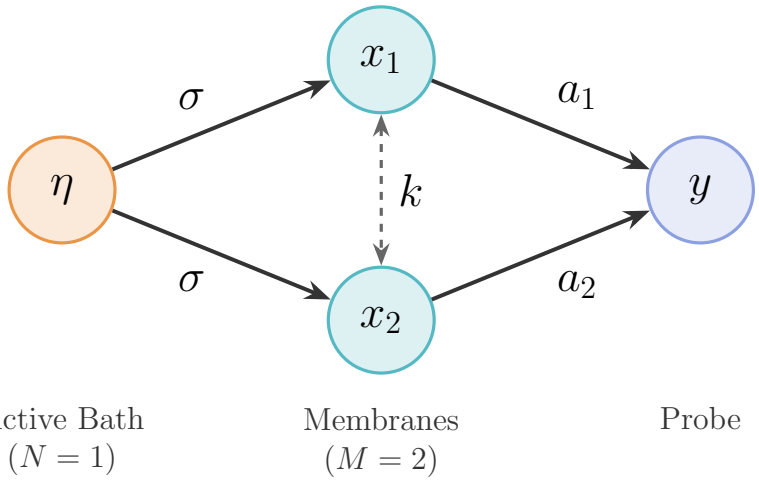


Figure 4.1: Network topology for $M = 2$ membranes and $N = 1$ active bath. Inter-membrane structural correlation is denoted by k.

Stability Conditions

Before proceeding with the optimization of the informational parameters, we must ensure that the dynamical system admits a well-defined Non-Equilibrium Steady State. Mathematically, since the deterministic part of our Langevin dynamics is governed by $\dot{\mathbf{z}} = -A\mathbf{z}$, the system is asymptotically stable if and only if all the eigenvalues of the drift matrix A have strictly positive real parts.

Thanks to the strictly feed-forward topology of our network (i.e., the absence of structural back-actions from the probe to the membranes and from the membranes to the bath), the matrix A is block-triangular. Consequently, its eigenvalues are simply the union of the eigenvalues of its diagonal blocks.

The characteristic timescales of the probe ($1/\tau_y$) and the active bath ($1/\tau_\eta$) are positive, granting unconditional stability to these individual layers. Thus, the overall stability of the network is entirely dictated by the 2×2 sub-matrix representing the internal dynamics of the membrane layer. Under our symmetric assumptions, this block takes the form:

$$A_{xx} = \frac{1}{\tau_x} \begin{pmatrix} 1 & -k \\ -k & 1 \end{pmatrix} . \tag{4.5}$$

By solving the characteristic polynomial $\det(A_{xx} - \lambda \mathbb{I}) = 0$, we find the two eigenvalues associated with the membrane layer:

$$\lambda_\pm = \frac{1 \pm k}{\tau_x} . \tag{4.6}$$

Imposing the stability requirement $\lambda_\pm > 0$, we immediately obtain the fundamental structural constraint:

$$|k| < 1 . \tag{4.7}$$

4.1 Equal-Dissipation Constraint

As stated in Subsec. 3.3.1, to perform a proper comparison with the baseline three-state model, we must fix the thermodynamic energy budget the system can access. To do so, according to the outlined procedure, we impose:

$$\dot{S}^{rel}_{M=2,N=1}(a_1, a_2) = \dot{S}^{rel}_{M=1,N=1}(a^*) , \tag{4.8}$$

where $\dot{S}^{rel}_{M=1,N=1}(a^*)$ is the total relative entropy production rate of the baseline three-state model evaluated at its optimum. During the optimization of the extended topology, this term acts as a strictly positive constant. Here, a_1 and a_2 are the target parameters we need to optimize. Importantly, the baseline dissipation $\dot{S}^{rel}_{M=1,N=1}(a^*)$ depends parametrically on the trade-off parameter λ of the Pareto functional, as λ dictates the position of the optimal reference coupling a^*.

We can then solve this equation to express one coupling parameter as a function of the other (substituting the corresponding expressions into each side of the equation):

$$\mathcal{A}(k, \sigma, \tau) a_1^2 + \mathcal{B}(k, \sigma, \tau) a_1 a_2 + \mathcal{C}(k, \sigma, \tau) a_2^2 = \dot{S}^{rel}_{M=1,N=1}(a^*) \mathcal{Q}(k, \tau) . \tag{4.9}$$

Here, the symbol τ encapsulates a generic dependence on all the characteristic timescales of the system. Furthermore, the coefficients explicitly reflect the symmetric baseline assumed earlier, incorporating the uniform environmental drive (σ) and the reciprocal structural interactions (k). For the sake of compactness, we decided to omit the explicit expressions of the conic coefficients due to their extensive algebraic length.

From a geometric perspective, Eq. (4.9) represents a conic section in the (a_1, a_2) parameter space. By evaluating the discriminant of this quadratic form, $\Delta = \mathcal{B}^2 - 4\mathcal{A}\mathcal{C}$, we can classify the nature of the relative equal-dissipation manifold. For all physically meaningful parameter regimes, we numerically verify that $\Delta < 0$, which strictly classifies the manifold as an ellipse. A visual representation of the first quadrant of these relative equal-dissipation ellipses is provided in Fig. 4.2.

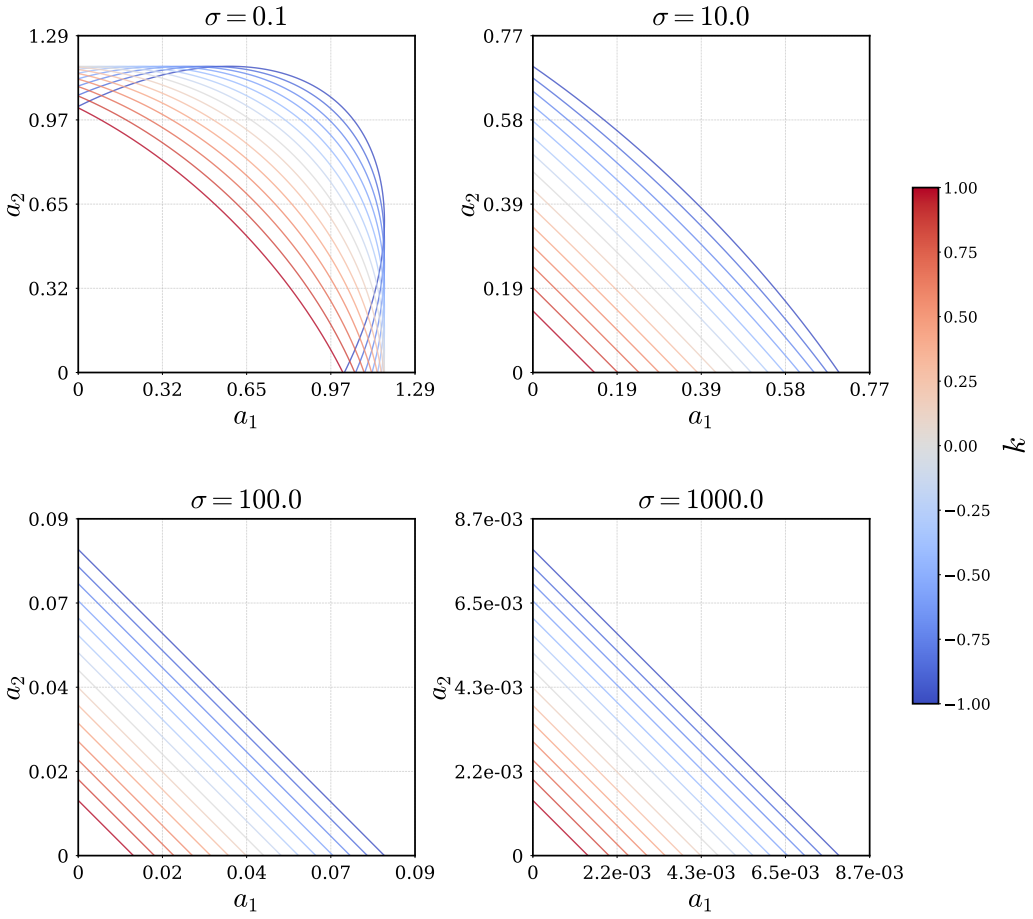


Figure 4.2: equal-dissipation manifolds in the (a_1, a_2) parameter space. The elliptical curves represent the set of points satisfying the constant relative entropy production rate constraint, $\dot{S}^{rel}_{M=1,N=1}(a^*)$. In accordance with the physical assumptions, the search space is restricted to the first quadrant $(a_1, a_2 \geq 0)$. The structural and thermodynamic parameters used for the numerical evaluation are set as follows: Diffusion coefficients $D_{x_1} = D_{x_2} = D_\eta = D_y = 1$; relaxation timescales $\tau_{x_1} = \tau_{x_2} = \tau_y = 1$ and $\tau_\eta = 5$. The parameter space explored spans the stable structural coupling regime $k \in (-1, 1)$ and the environmental drive $\sigma \in [0.1, 1000]$.

As justified previously, we restrict our analysis to non-negative sensory couplings. Geometrically, this confines our parameter search space to the first quadrant of the (a_1, a_2) plane (i.e., $a_1, a_2 \geq 0$). The closed topology of the ellipse bounds the search space, defining finite ranges for the permissible values of both parameters.

Isolating a_1 via the quadratic formula, we extract the physical branch(es) lying

within the first quadrant:

$$a_1(a_2) = \frac{-\mathcal{B}a_2 \pm \sqrt{\mathcal{B}^2 a_2^2 - 4\mathcal{A}\left(\mathcal{C}a_2^2 - \dot{S}^{rel}_{M=1,N=1}(a^*)\mathcal{Q}\right)}}{2\mathcal{A}}\,, \tag{4.10}$$

where the valid search domain for a_2 is constrained by both the non-negativity of the discriminant and the physical requirement $a_1 \geq 0$. Note that in some regions of the parameter space both roots yield valid non-negative solutions, thus we need to consider both solutions in general in the optimization.

4.2 Functional Optimization and Observables Calculation

After having outlined the stability conditions and the relative equal-dissipation constraint, we can proceed with the multi-source functional optimization. By decomposing the mutual information using PID, the functional for our specific $M = 2$ architecture assumes the explicit form:

$$\mathcal{L}(a_1, a_2; \lambda, \gamma) = \lambda\Big[U(x_1 \setminus x_2; y) + U(x_2 \setminus x_1; y) + 2\gamma S(x_1, x_2; y) + 2(1-\gamma)R(x_1, x_2; y)\Big] - (1-\lambda)4\mathcal{T}_y(a_1, a_2) \tag{4.11}$$

To practically evaluate this functional, the individual PID components must be explicitly computed. For our two-sensor architecture, this procedure relies on the standard information-theoretic relationships, executed in the following order (adapting to this specific case the procedure outlined in Sec. 1.2.3):

1. Joint and Marginal Mutual Information: we first compute the marginal mutual information for each individual sensor, $I(x_1; y)$ and $I(x_2; y)$, along with the joint mutual information between the probe and the full sensory layer, $I(x_1, x_2; y)$. As we have already derived in the previous chapters, they assume the following form:

   $$I(x_i; y) = \frac{1}{2}\ln\left(\frac{\Sigma_{x_i x_i}\Sigma_{yy}}{\Sigma_{x_i x_i}\Sigma_{yy} - \Sigma^2_{x_i y}}\right) \quad \text{for } i \in \{1, 2\},$$

   and:

   $$I(x_1, x_2; y) = \frac{1}{2}\ln\left(\frac{\Sigma_{yy}\det(\mathbf{\Sigma_{xx}})}{\det(\mathbf{\Sigma}_{\text{joint}})}\right),$$

   where Σ_{yy} is the variance of the probe, $\mathbf{\Sigma_{xx}}$ is the 2×2 sub-covariance matrix of the sensory layer, and $\mathbf{\Sigma}_{\text{joint}}$ is the full 3×3 joint covariance matrix of the subsystem $(y, x_1, x_2)^T$.

2. Redundancy: we evaluate the redundant information shared by the two sensors using the Minimum Mutual Information (MMI) estimator, defined as the minimum of the individual marginal mutual information:

   $$R(x_1, x_2; y) = \min\{I(x_1; y), I(x_2; y)\} \tag{4.12}$$

3. Unique Information: the unique information provided by each sensor is obtained by subtracting the redundant contribution from its respective marginal mutual information:

$$U(x_1 \setminus x_2; y) = I(x_1; y) - R(x_1, x_2; y) \tag{4.13}$$
$$U(x_2 \setminus x_1; y) = I(x_2; y) - R(x_1, x_2; y) \tag{4.14}$$

4. Synergy: Finally, the synergistic information is derived from the global information balance, subtracting all the other contributions from the joint mutual information:

$$S(x_1, x_2; y) = I(x_1, x_2; y) - I(x_1; y) - I(x_2; y) + R(x_1, x_2; y) \tag{4.15}$$

Having established this pipeline and fixed the specific information-harvesting strategy by setting the value of γ, we can then determine the optimum by maximizing the functional with respect to our independent variable:

$$a_2^* = \underset{a_2}{\operatorname{argmax}}\ \mathcal{L}\big(a_1(a_2), a_2; \lambda, \gamma\big)\ , \tag{4.16}$$

where the full optimal coupling vector is subsequently reconstructed as $\mathbf{a}^* = (a_1(a_2^*), a_2^*)^T$. This procedure effectively reduces the search to a one-dimensional optimization problem along the equal-dissipation manifold, as outlined in the previous section.

Once these optimal values are determined, we can evaluate several key macroscopic quantities to assess the system's performance. Specifically, we investigate:

- The Whole Minus Sum (WMS) evaluated at optimality (see Sec. 1.2.3 for details). This metric reveals whether the system, under a specific information-harvesting strategy (fixed γ), naturally gravitates toward synergistic or redundant signal integration. Mathematically, this is because in a two-source architecture, the WMS reduces exactly to the difference between synergy and redundancy ($S - R$).

- The mutual information between the probe and the active bath ($I(y; \eta)$), representing the actual information successfully extracted from the environment, consistent with the analysis performed on the baseline three-state model.

- The absolute optimal coupling difference, $|a_1^* - a_2^*|$. This observable is interesting for identifying potential sensory specialization patterns. Given the structural symmetry of the network, the trivial thermodynamic solution would suggest a uniform resource allocation ($a_1^* = a_2^*$); however, under a constrained energy budget, asymmetric topologies may emerge as the optimal strategy.

Note that we do not explicitly analyze the traffic in this context. Because the entropy production rate is constrained to match the reference model, the traffic behaves identically to that of the baseline three-state architecture, offering no additional discriminative insights.

The subsequent figures present various phase diagrams mapped across the (k, σ) parameter space, evaluated at explicitly fixed values of γ. Rather than plotting

the raw mutual information, we visualize the information gain, $\Delta I(y;\eta)$, which quantifies the net performance difference between our optimized $M = 2$ architecture and the baseline three-state model.

To provide a comprehensive synthesis of the system's behaviour, the final figure employs a multi-layered visualization strategy. Specifically, we represent the coupling asymmetry $|a_1^* - a_2^*|$ as the primary color map, upon which we superimpose the contour lines of both $\Delta I(y;\eta)$ and the WMS. This superimposed representation allows for a direct visual correlation between topological specialization and information gain.

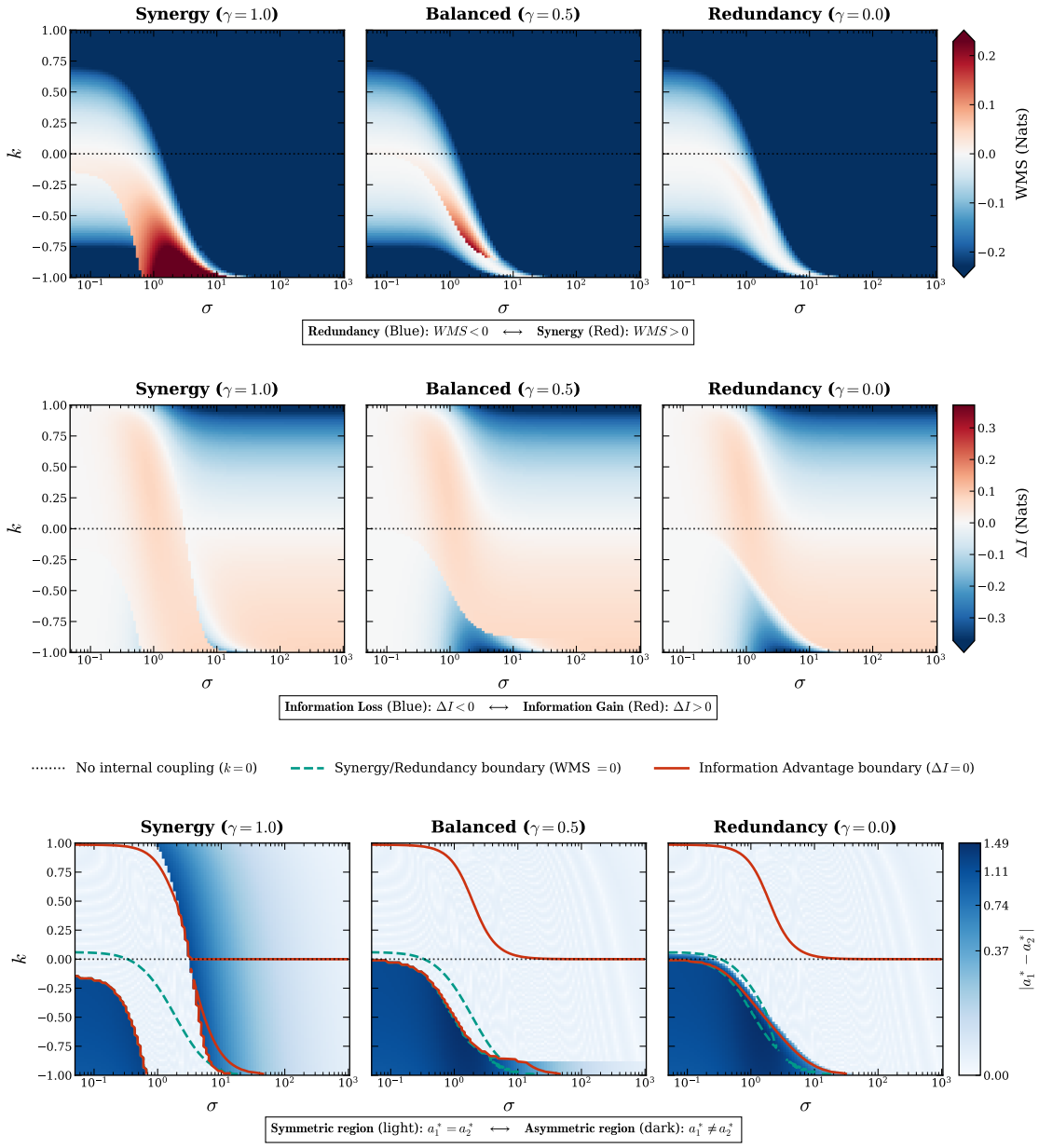


Figure 4.3: Phase diagrams of the $M = 2$, $N = 1$ topology. The three panels illustrate the network configuration at optimality and their informational performance across the (σ, k) parameter space, evaluated for three distinct optimization strategies ($\gamma = 1.0, 0.5, 0.0$). **(Top)** Whole-Minus-Sum (WMS) informational landscape. A saturated divergent scale highlights the transition from a redundancy-dominated regime (blue, $WMS < 0$) to a synergy-dominated regime (red, $WMS > 0$). **(Middle)** Information Gain (ΔI) compared to the single-membrane baseline ($M = 1$). The divergent scale (extended downwards) emphasizes the parameter region where the multidimensional topology yields a net informational advantage (red) against the broad domain of thermodynamic information loss (blue). **(Bottom)** Coupling difference magnitude $|a_1^* - a_2^*|$. The system transitions from a symmetric configuration (light) to an asymmetric one (dark) that maximizes the functional. The dashed green line and solid red line represent the $WMS = 0$ and $\Delta I = 0$ boundaries, respectively.

There are multiple ways to understand Fig. 4.3. Here, we will describe the behaviour of the system in each heatmap according to the different regions of the (σ, k) parameter space. First of all, we recall the physical intuition behind these parameters:

1. As already described in the introduction, the structural coupling k models the co-activation or the inhibition between the states of the same layer. This means that a negative k induces a signal reduction between the membranes (similar to a negative feedback), while a positive k indicates that the signal is mutually reinforced by the membranes.

2. σ_{11} and σ_{21} indicate how much signal is transmitted from the active bath to the respective membrane. This means that a lower σ corresponds to a weaker signal that can be read from the membranes, whereas a higher σ means a stronger signal reaching each membrane.

Another preliminary consideration is that this topology is inherently redundant by construction: two sensors are used to measure the same active bath. Thus, we generally expect redundancy to dominate, as both membranes naturally capture a shared portion of the incoming signal.

This is confirmed by the top plot in Fig. 4.3, which displays the WMS. Redundancy is generally prevalent across the parameter space, except in a specific region: for negative values of k and intermediate values of σ, the red band indicates that, at optimality, synergistic interaction prevails (while some synergy may also be observed for small positive values of k, this regime is not central to our present analysis). This feature is present across all strategies, although it is most pronounced in the synergy-driven one. Indeed, optimizing for redundancy does not strictly prevent synergistic behaviour, which can emerge regardless of the specific optimization target. The converse is also true: redundant behaviour can emerge at optimality even when explicitly optimizing for synergy. A physical hypothesis for why synergy emerges in this specific region lies in the interplay between σ and k. Since a negative k induces an inhibition between the membranes, it may attenuate the overall macroscopic signal. An intermediate σ provides a driving force that is strong enough to be captured, yet not strong enough to be trivially resolved given the internal inhibition. In contrast, at low σ, the signal is simply too weak to be useful. Conversely, at high σ, the external signal is so intense that it overwhelms the negative structural feedback, making specialized informational strategies unnecessary. We will return to this point in the following section.

In the information gain plot (middle panel), all three strategies show very similar behaviour when k is positive. The system performs better than the baseline model when σ is small or medium. However, when σ becomes large, the performance drops. This drop is associated with the structural coupling: a larger k makes the region of advantage smaller, while a smaller positive k makes it wider. We can get an intuition of it mathematically by looking at the double effect of the coupling. If we add the two membrane equations in Eq. (4.1) assuming symmetry, the sum of the signals $x_1 + x_2$ follows this equation for its homogeneous part:

$$\tau_x \frac{d(x_1 + x_2)}{dt} = -(1 - k)(x_1 + x_2) + \dots, \tag{4.17}$$

This means that the system relaxes at an effective rate of $(1-k)/\tau_x$. A positive k helps the membranes share the signal, but it also makes this relaxation rate smaller. Because the relaxation is slower, the system's response is delayed. As a result, the membranes accumulate random fluctuations that do not match the current state of the bath. We believe that, at high σ, this delay is associated with the break of the useful correlation between the membranes and the bath, resulting in the final drop in information gain. In contrast, in the low-to-intermediate σ regime, this co-activation is connected with the incoming signal without introducing significant noise, thereby providing a net performance advantage.

Turning to the negative k regime, we observe an interesting behaviour: the system's performance varies depending on the chosen strategy. For intermediate-to-high values of σ, we find an overall performance gain across all three strategies. As discussed above, k controls the effective relaxation rate of the combined membrane response; a negative k increases this rate, making the combined response of the two membranes faster. This allows the system to build stronger correlations with the bath's fluctuations, increasing the overall information available. The most significant differences emerge in the low-to-intermediate σ regime. Here, the synergy-driven strategy yields a broader parameter region of performance gain compared to the balanced and redundancy-driven approaches. It is worth noting that while regions exhibiting a performance advantage generally feature a dominant synergistic component, this is not strictly necessary. Indeed, there are also areas of performance gain characterized by a prevalence of redundancy, a point we will examine further when discussing the bottom panel. Overall, we can conclude from this panel that the combination of structural inhibition and a synergy-driven strategy provides a broader and more robust performance advantage for low-to-intermediate values of σ. This is likely because exploiting the joint observation of both membranes within that specific signal-strength range allows the system to extract complementary information, thereby outperforming the other strategies in terms of parameter space coverage.

Finally, the bottom panel presents a heatmap that combines the features of the first two panels with the coupling difference. As previously stated, this latter observable indicates a specific adaptation of the system to the environment, a behaviour likely driven by the relative equal-dissipation constraint. Indeed, given the perfect structural symmetry of our system, one may expect that the optimal solution would always dictate an equal distribution of the coupling strength between the membranes, resulting in a zero coupling difference. In our constrained system, however, it is evident that distinct specialization pathways emerge across different parameter ranges. In several cases, these regions of asymmetry closely correlate with the WMS and ΔI boundaries discussed above. For instance, consider again the $k > 0$ regime: for the balanced and redundancy-driven strategies, we observe no specialization. Across the entire parameter range, the coupling strength remains equally distributed between the two sensors. In contrast, the synergy-driven strategy exhibits a spontaneous coupling tuning. While the coupling is initially equally distributed for low σ values, the system transitions into a regime where one of the couplings is effectively turned off (visible as the blue region). The strength of the single active coupling then progressively decreases as σ increases. This is likely because

a stronger external signal does not require an extremely high coupling to be accurately captured. Since the relative dissipation $\dot{S}^{\mathrm{rel}}$, and hence the cost term $4\mathcal{T}_y$, is constant along the equal-dissipation manifold, We hypothesize that the membranes specialization arises from the informational term of the functional, whose maximum on the manifold need not lie at the symmetric point $a_1 = a_2$ in this structure. Finally, it is worth noting that the boundary of the information advantage (the red line) in this scenario coincides almost perfectly with the onset of this blue asymmetric region. Relying on only one active membrane state may therefore be an additional factor contributing to the observed performance degradation, adding to the detrimental effects of the membranes' slow collective relaxation rate.

Regarding the $k < 0$ regime, several distinct observations can be made. For both the balanced and redundancy-driven strategies, the system typically starts with one coupling turned off at low σ values. In the balanced strategy, for strong inhibitory couplings ($k \to -1$), the optimal solution is to never activate the second coupling across the entire range. As discussed previously, this symmetry breaking is likely enforced by the relative equal-dissipation constraint. The synergy-driven strategy, however, exhibits a different behaviour. For values of k close to zero, both couplings start activated, whereas for $k \to -1$, the system initiates with one coupling turned off. As σ increases, the system traverses an intermediate region where both couplings are activated and symmetric; however, this phase is temporary and eventually gives way to the same asymmetric behaviour observed in the $k > 0$ regime at higher σ. A remarkable feature across all three strategies, and almost perfectly delineated in the synergy-driven one, is that the exact moment both couplings become active coincides with both the information gain boundary ($\Delta I = 0$) and the transition into the synergistic regime. This is consistent with our previous physical deductions. Furthermore, there is a region of negative information gain around $\sigma \approx 10$: in this specific area, one coupling is turned off, meaning that a performance gain can only be achieved if the external signal becomes strong enough to efficiently drive the probe through a single channel.

Finally, the relationship between k and the energetic cost strengthens our earlier arguments. A stronger structural inhibition (a more negative k) demands more energy to activate the second coupling. Consequently, to achieve an information advantage, the system must rely on the synergistic component over a broader range of σ, a feature that distinguishes it from the other two strategies. It is important to note that having both couplings activated does not imply synergy. For instance, in certain regions of the synergy-driven strategy and, more prominently, in the balanced and redundancy-driven ones, we observe a positive ΔI characterized by a prevalence of redundancy even when both couplings are active.

4.3 ΔI vs WMS

Following the general considerations discussed in the previous section, we now investigate more closely the relationship between ΔI and the WMS. To achieve this, for each strategy, we plot ΔI as a function of σ at fixed values of k. Each data point in the plots is color-coded according to its corresponding WMS value (see the caption of Fig. 4.4 for details). We begin our analysis by examining the $k \geq 0$ regime:

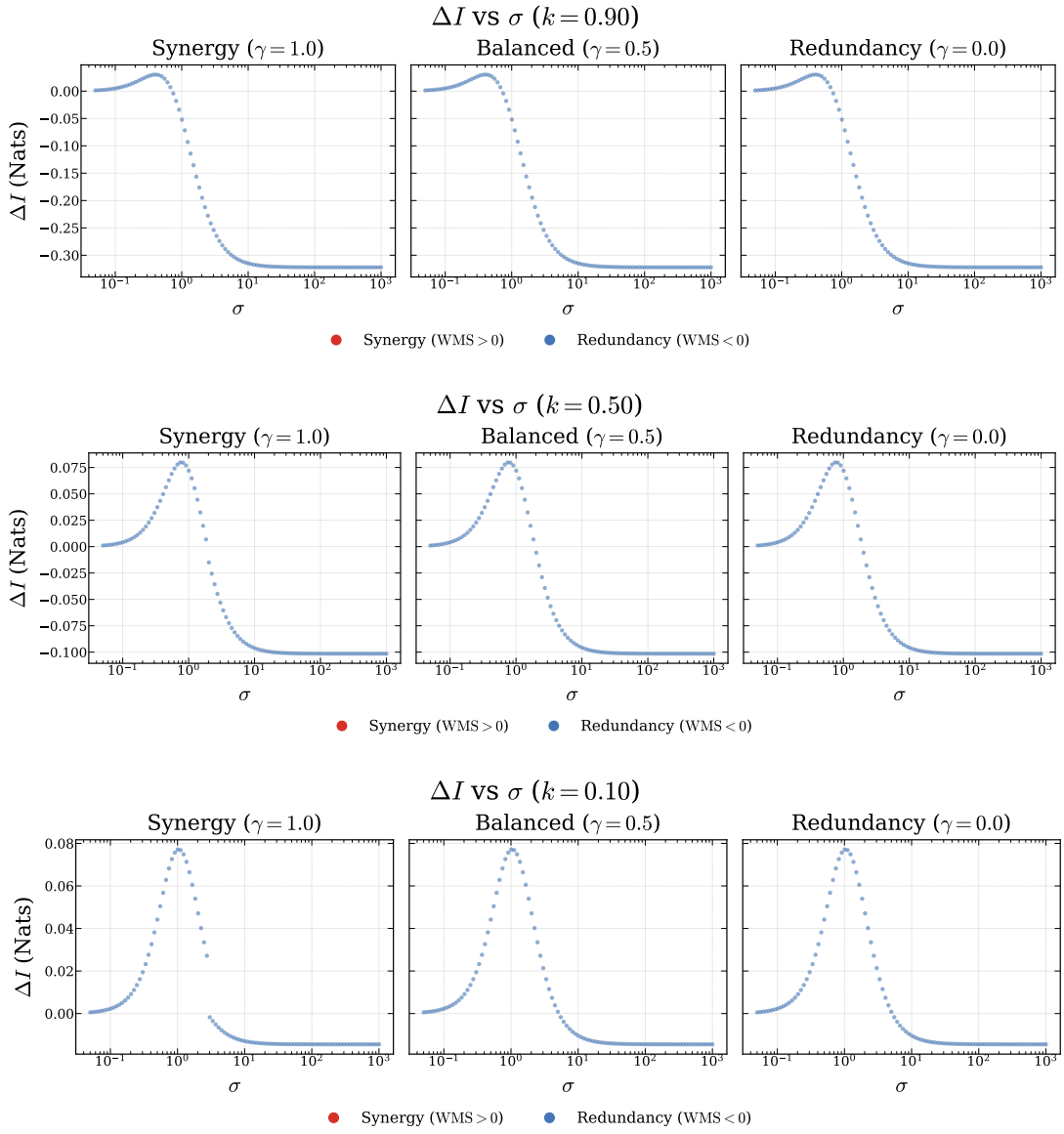


Figure 4.4: Information gain ΔI as a function of the signal strength σ for selected positive structural couplings: $k = 0.90$ (top), $k = 0.50$ (middle), and $k = 0.10$ (bottom). Each panel compares the synergy-driven ($\gamma = 1.0$), balanced ($\gamma = 0.5$), and redundancy-driven ($\gamma = 0.0$) strategies. The scatter points are color-coded according to the Whole Minus Sum (WMS) sign, highlighting regions dominated by synergy (WMS > 0) or redundancy (WMS < 0).

Fig. 4.4 provides further confirmation of the behaviours discussed previously. Across all three strategies, the system exhibits a predominantly redundant behaviour over the entire explored range of σ. An actual performance gain ($\Delta I > 0$) is exclusively localized in the low-to-intermediate σ regime. For intermediate-to-high values of σ instead we observe a steep performance degradation, the severity of which scales directly with the magnitude of k. Furthermore, these plots reveal that even within the region of positive information gain, there are diminishing returns: the maximum achievable advantage decreases as the structural coupling k grows. This reduction in the peak performance is entirely consistent with the relaxation-rate mismatch mechanism discussed above, since a larger k corresponds to a larger delay between the collective membrane response and the bath dynamics.

A richer set of behaviours emerges when analyzing the $k < 0$ regime, as depicted in Fig. 4.5. As noted earlier, the synergy-driven strategy consistently achieves a positive information gain ($\Delta I > 0$) and significantly outperforms the balanced and redundancy-driven approaches in the low-to-intermediate σ range. Indeed, across all three values of k, the performance peak for the synergy strategy is robustly characterized by synergistic interactions (indicated by the red data points in the upper half-plane), whereas the other two strategies display predominantly redundant states (blue points) in the corresponding regions. Consistent with the observations from the previous section, we also note that as the structural inhibition becomes stronger (for instance, comparing $k = -0.90$ against $k = -0.10$), the region of positive information gain relies entirely on a WMS > 0. It becomes evident that exploiting complementary information is strictly vital to maintain an advantage when k approaches -1. A particularly insightful detail is the appearance of synergistic states (red points) within the redundancy-driven strategy, even when located in the negative ΔI half-plane. Rather than contradicting our hypothesis, this reinforces our argument: although the objective function in this case is explicitly tuned to maximize redundancy ($\gamma = 0$), the system is physically forced into a synergistic configuration as its optimal state under constraints (see the top and middle panels of Fig. 4.5).

Furthermore, the discontinuities visible in the curves illustrate the symmetry breaking of the couplings discussed earlier, tracking the points where one of the couplings switches on or off to satisfy the equal-dissipation bound.

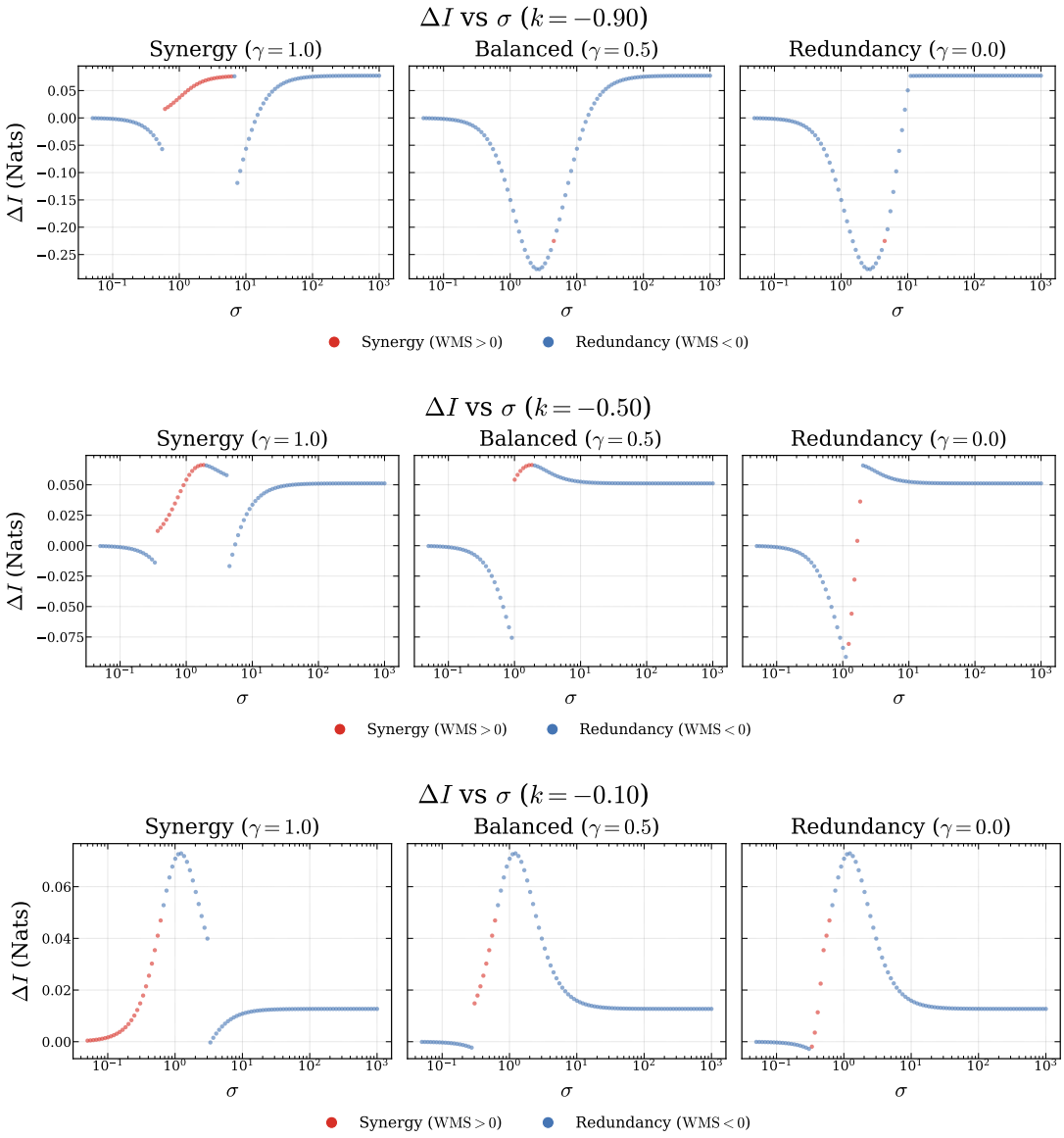


Figure 4.5: Information gain ΔI as a function of the signal strength σ for selected negative structural couplings: $k = -0.90$ (top), $k = -0.50$ (middle), and $k = -0.10$ (bottom). Each panel compares the synergy-driven ($\gamma = 1.0$), balanced ($\gamma = 0.5$), and redundancy-driven ($\gamma = 0.0$) strategies.

Chapter 5

$M = 1$, $N = 2$ Topology: One Membrane and Two Active Baths

In this chapter, we investigate the dual topology to the one analyzed previously, focusing on a system composed of a single membrane interacting with two distinct active baths ($M = 1, N = 2$). As before, we begin by defining the most general set of Langevin equations governing the dynamics of this system:

$$\begin{cases} \dot{y} = -\frac{y}{\tau_y} + \frac{a}{\tau_y}x + \sqrt{\frac{2D_y}{\tau_y}}\xi_y(t) \\ \dot{x} = -\frac{x}{\tau_x} + \frac{\sigma_1}{\tau_x}\eta_1 + \frac{\sigma_2}{\tau_x}\eta_2 + \sqrt{\frac{2D_x}{\tau_x}}\xi_x(t) \\ \dot{\eta}_1 = -\frac{\eta_1}{\tau_{\eta_1}} + \frac{k_{12}^{(\eta)}}{\tau_{\eta_1}}\eta_2 + \sqrt{\frac{2D_{\eta_1}}{\tau_{\eta_1}}}\xi_{\eta_1}(t) \\ \dot{\eta}_2 = -\frac{\eta_2}{\tau_{\eta_2}} + \frac{k_{21}^{(\eta)}}{\tau_{\eta_2}}\eta_1 + \sqrt{\frac{2D_{\eta_2}}{\tau_{\eta_2}}}\xi_{\eta_2}(t) \end{cases} \tag{5.1}$$

Here, for the sake of notational simplicity, we omit the first subscript in the signal strength parameters. We thus define $\sigma_1 \equiv \sigma_{11}$ and $\sigma_2 \equiv \sigma_{12}$, which respectively quantify the incoming connections from active baths 1 and 2 to the single membrane. Defining also in this case the four-dimensional state vector $\mathbf{z} = (y, x, \eta_1, \eta_2)^T$, we express the dynamics in vectorial form:

$$\dot{\mathbf{z}} = -A\mathbf{z} + \tilde{D}\boldsymbol{\xi}(t) \ , \tag{5.2}$$

The block drift matrix A is:

$$A = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{a}{\tau_y} & 0 & 0 \\ 0 & \frac{1}{\tau_x} & -\frac{\sigma_1}{\tau_x} & -\frac{\sigma_2}{\tau_x} \\ 0 & 0 & \frac{1}{\tau_{\eta_1}} & -\frac{k_{12}^{(\eta)}}{\tau_{\eta_1}} \\ 0 & 0 & -\frac{k_{21}^{(\eta)}}{\tau_{\eta_2}} & \frac{1}{\tau_{\eta_2}} \end{pmatrix} \ . \tag{5.3}$$

The diagonal noise amplitude matrix $\tilde{D}$ is explicitly given by:

$$\tilde{D} = \sqrt{2} \begin{pmatrix} \sqrt{\frac{D_y}{\tau_y}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{D_x}{\tau_x}} & 0 & 0 \\ 0 & 0 & \sqrt{\frac{D_{\eta_1}}{\tau_{\eta_1}}} & 0 \\ 0 & 0 & 0 & \sqrt{\frac{D_{\eta_2}}{\tau_{\eta_2}}} \end{pmatrix} . \tag{5.4}$$

Following the approach established in the previous chapter, we impose a certain degree of symmetry on our system parameters. Specifically, we set the characteristic timescales as $\tau_{\eta_1} = \tau_{\eta_2}$. Additionally, we assume a symmetric environmental cross-talk by assigning a uniform value k to the coupling parameters between the two active baths (i.e., $k_{12}^{(\eta)} = k_{21}^{(\eta)} \equiv k$).

In Fig. 5.1 we show a schematic of this simplified system.

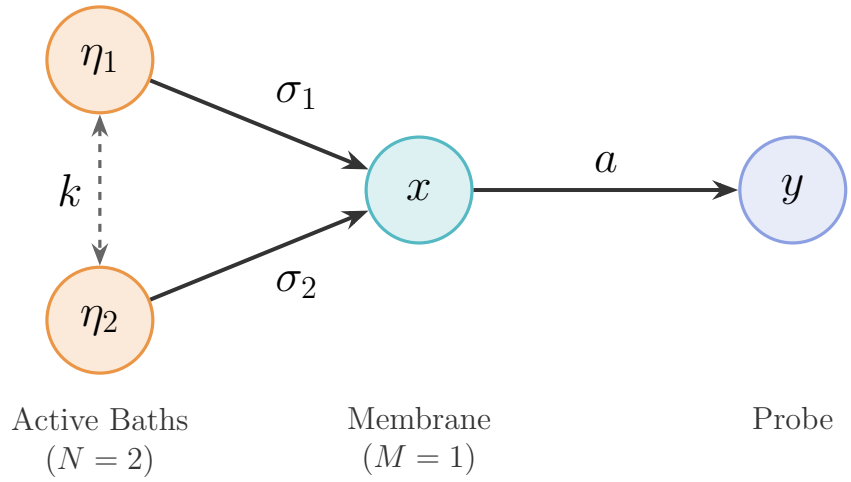


Figure 5.1: Network topology for $M = 1$ membrane and $N = 2$ active baths. Bath structural correlation is denoted by k.

Stability Conditions

To assess the stability of the system, we can leverage the same arguments presented in the previous chapter. In particular, we recall that since the drift matrix A is block-triangular, its eigenvalues are again simply the union of the eigenvalues of its diagonal blocks.

Since the characteristic relaxation rates of the probe $(1/\tau_y)$ and the single sensory membrane $(1/\tau_x)$ are positive, the overall stability of the network is entirely dictated by the 2×2 sub-matrix representing the internal dynamics of the active baths. Under our symmetric assumptions, this environmental block takes the form:

$$A_{\eta\eta} = \frac{1}{\tau_\eta} \begin{pmatrix} 1 & -k \\ -k & 1 \end{pmatrix} . \tag{5.5}$$

By solving the characteristic equation $\det(A_{\eta\eta} - \lambda\mathbb{I}) = 0$, we easily recover the same result found for the $M = 2, N = 1$ topology:

$$\lambda_{\pm} = \frac{1 \pm k}{\tau_\eta} \ . \tag{5.6}$$

Imposing the stability requirement $\lambda_{\pm} > 0$, we immediately obtain the same fundamental constraint:

$$|k| < 1 \ . \tag{5.7}$$

5.1 Equal-Dissipation Constraint

As established in the previous chapter we express also for this topology the equal-dissipation constraint. Following the same methodology, we impose the equality of the relative entropy production rates between the two topologies:

$$\dot{S}^{rel}_{M=1,N=2}(a^*, \sigma_1, \sigma_2) = \dot{S}^{rel}_{M=1,N=1}(a^*, \sigma) \ , \tag{5.8}$$

where $\dot{S}^{rel}_{M=1,N=1}(a^*, \sigma)$ acts as a positive constant target, parameterized by the optimal configuration of the baseline model. Note that in both topologies we assume the same optimal probe-membrane coupling a^*, effectively reducing the constraint only to the membrane-baths space. A crucial distinction emerges in this topology: upon explicitly computing the relative entropy rates and equating them, the squared probe coupling $(a^*)^2$ exactly cancels out from both sides. Consequently, the relative equal-dissipation manifold is independent of the sensor's informational allocation. By factoring out this coupling, we can define a reduced baseline dissipation $\tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau) = \dot{S}^{rel}_{M=1,N=1}/(a^*)^2$. The fixed thermodynamic budget constrains the environmental drives, leaving σ_1 and σ_2 as the free parameters defining the space. By algebraically rearranging the constraint, we can express the relative equal-dissipation manifold as a general conic section in the (σ_1, σ_2) parameter space:

$$\mathcal{A}(k, \tau)\sigma_1^2 + \mathcal{B}(k, \tau)\sigma_1\sigma_2 + \mathcal{C}(k, \tau)\sigma_2^2 = \tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau)\delta(k, \tau) - \Phi(k, \tau) \ . \tag{5.9}$$

Here, the symbol τ encapsulates a generic dependence on all the characteristic timescales of the system. Also in this case, for the sake of compactness we omit the explicit expressions of these coefficients due to their extensive algebraic length. As in the previous topology, evaluating the discriminant of this conic section ($\Delta = \mathcal{B}^2 - 4\mathcal{A}\mathcal{C}$) yields $\Delta < 0$ across all physically meaningful parameter regimes. This classifies the conic as an ellipse.

In accordance with our physical assumptions, we restrict our analysis to non-negative drives. Geometrically, this confines our parameter search space to the first quadrant of the (σ_1, σ_2) plane. By isolating σ_1 via the quadratic formula, we extract the two possible solutions:

$$\sigma_1(\sigma_2) = \frac{-\mathcal{B}\sigma_2 \pm \sqrt{\mathcal{B}^2\sigma_2^2 - 4\mathcal{A}\left[\mathcal{C}\sigma_2^2 - \left(\tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau)\delta(k, \tau) - \Phi(k, \tau)\right)\right]}}{2\mathcal{A}} \ , \tag{5.10}$$

where the valid search domain for σ_2 is bounded by both the non-negativity of the discriminant and the physical requirement $\sigma_1 \geq 0$. Note that also in this scenario, certain regions of the parameter space yield two strictly positive roots, both of which are explicitly taken into account during the optimization.

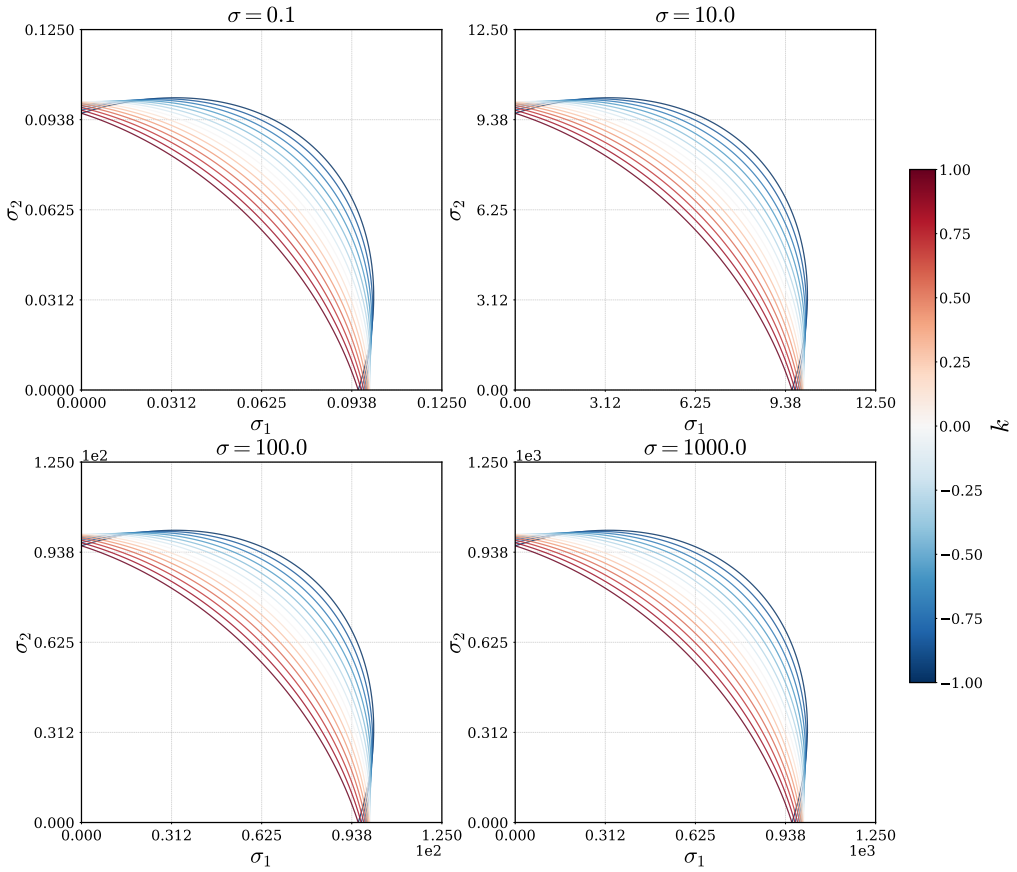


Figure 5.2: equal-dissipation manifolds in the (σ_1, σ_2) parameter space. The elliptical curves represent the set of points satisfying the reduced constant relative entropy production rate constraint, $\tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau)$. In accordance with the physical assumptions, the search space is restricted to the first quadrant $(\sigma_1, \sigma_2 \geq 0)$. The structural and thermodynamic parameters used for the numerical evaluation are set as follows: Diffusion coefficients $D_x = D_y = D_{\eta_1} = D_{\eta_2} = 1$; relaxation timescales $\tau_x = \tau_y = 1$ and $\tau_{\eta_1} = \tau_{\eta_2} = 5$. The parameter space explored spans the stable structural coupling regime $k \in (-1, 1)$ (endpoints excluded), with the baseline environmental drive evaluated at $\sigma \in \{0.1, 10.0, 100.0, 1000.0\}$.

Before proceeding to the numerical optimization, a brief clarifying note on the notation used in the upcoming results is necessary. In the subsequent plots analyzing the extended $M = 1, N = 2$ architecture, the independent variable denoted simply as σ refers to the environmental drive of the baseline $M = 1, N = 1$ reference model. For the new topology, this σ value does not represent a single physical bath coupling; rather, it acts as a parameter that fixes the thermodynamic target $\tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau)$. Consequently, each value of σ on the axes dictates a specific relative equal-dissipation manifold, and the performance of the $M = 1, N = 2$ system is evaluated by optimizing over the corresponding valid (σ_1, σ_2) pairs that lie on this curve.

Conceptual Shift: Agency and Traffic

Before delving into the functional optimization for this specific topology, we need a brief preamble to specify the conceptual difference with respect to the $M = 2, N = 1$ architecture. Up to now, the agent taking decisions according to its accessible degrees of freedom was the probe, which could tune its couplings a_1 and a_2. In that context, it was physically consistent to use the MMI PID framework to study the trade-offs between synergy and redundancy directly between the membranes and the probe. However, for the $M = 1, N = 2$ topology, this approach is not valid: since the probe is not directly coupled with the baths, applying the PID framework at the probe level would not be a fair comparison. In order to properly study this topology, we need to invert our perspective. As anticipated in the previous section, the probe coupling is fixed to the optimal baseline value a^*. The free parameters over which we perform the optimization are now the environmental drives, σ_1 and σ_2. This effectively shifts the "agency" from the probe to the membrane, which is now the entity tuning its couplings given the environment. Because there is a direct physical coupling between the baths and the membrane, the PID framework can now be applied coherently at this level.

Note that, since the parameters being optimized now control the membrane's interaction with the baths, the relevant thermodynamic cost is associated with the membrane dynamics rather than with the probe's coupling optimization. Following again the procedure outlined in Subsec. 2.3, we conclude that the proper thermodynamic cost is the localized traffic of the membrane, $4\mathcal{T}_x$, which is given by:

$$4\mathcal{T}_x(a^*) = \left[\mathbf{D}^{-1}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T - 2\mathbf{A}\right]_{xx} , \tag{5.11}$$

where the dependence on a^* indicates that the downstream probe is active (otherwise we would have used $a = 0$, consistently with the three-state model) and fixed at its optimal baseline value, and the subscript xx isolates the diagonal element of the traffic matrix corresponding to the membrane.

5.2 Functional Optimization and Observables Calculation

After this preamble, we express the functional used for the optimization process:

$$\mathcal{L}(\sigma_1, \sigma_2; \lambda, \gamma) = \lambda\Big[U(\eta_1 \setminus \eta_2; x) + U(\eta_2 \setminus \eta_1; x) + 2\gamma S(\eta_1, \eta_2; x) + 2(1-\gamma)R(\eta_1, \eta_2; x)\Big] - (1-\lambda)4\mathcal{T}_x(\sigma_1, \sigma_2) \quad (5.12)$$

Following the logical steps already introduced for the previous architecture, we must adapt the explicit computation of the PID components to this membrane-centric framework:

1. Mutual Information terms: The marginal dependencies for each individual active bath, $I(\eta_1; x)$ and $I(\eta_2; x)$, alongside the joint mutual information $I(\eta_1, \eta_2; x)$, are computed via standard Gaussian formulas:

$$I(\eta_i; x) = \frac{1}{2} \ln \left(\frac{\Sigma_{\eta_i \eta_i} \Sigma_{xx}}{\Sigma_{\eta_i \eta_i} \Sigma_{xx} - \Sigma_{\eta_i x}^2} \right) \quad \text{for } i \in \{1, 2\},$$

and:

$$I(\eta_1, \eta_2; x) = \frac{1}{2} \ln \left(\frac{\Sigma_{xx} \det(\boldsymbol{\Sigma}_{\boldsymbol{\eta\eta}})}{\det(\boldsymbol{\Sigma}_{\text{joint}})} \right),$$

where Σ_{xx} represents the membrane's variance, $\boldsymbol{\Sigma}_{\boldsymbol{\eta\eta}}$ denotes the 2×2 covariance block of the active environment, and $\boldsymbol{\Sigma}_{\text{joint}}$ is the complete 3×3 covariance matrix for the relevant $(x, \eta_1, \eta_2)^T$ subspace.

2. Redundancy: Applying the Minimum Mutual Information (MMI) prescription, the overlapping information shared by the two environments is extracted as:

$$R(\eta_1, \eta_2; x) = \min\{I(\eta_1; x), I(\eta_2; x)\} \ . \quad (5.13)$$

3. Unique Information: The exclusive contribution from each environmental source is isolated by removing the redundant portion from its corresponding marginal information:

$$U(\eta_1 \setminus \eta_2; x) = I(\eta_1; x) - R(\eta_1, \eta_2; x) \quad (5.14)$$
$$U(\eta_2 \setminus \eta_1; x) = I(\eta_2; x) - R(\eta_1, \eta_2; x) \ . \quad (5.15)$$

4. Synergy: The cooperative information emerging from the joint interaction is finally recovered through the fundamental PID balance equation:

$$S(\eta_1, \eta_2; x) = I(\eta_1, \eta_2; x) - I(\eta_1; x) - I(\eta_2; x) + R(\eta_1, \eta_2; x) \ . \quad (5.16)$$

With the functional explicitly defined for a chosen information-harvesting strategy (parameterized by γ), the optimization task reduces to finding the optimal drive along the equal-dissipation curve:

$$\sigma_2^* = \underset{\sigma_2}{\text{argmax}} \ \mathcal{L}\big(\sigma_1(\sigma_2), \sigma_2; \lambda, \gamma\big) \ . \quad (5.17)$$

Once σ_2^* is identified, the complete optimal environmental coupling vector is naturally reconstructed as $\boldsymbol{\sigma}^* = (\sigma_1(\sigma_2^*), \sigma_2^*)^T$. This approach effectively projects what would be a two-dimensional search down to a targeted one-dimensional optimization, strictly bounded by the fixed thermodynamic budget.

Once the optimal environmental parameters are determined, we can evaluate the same key macroscopic quantities introduced in the previous chapter to assess the system's performance. For this specific topology, we investigate:

- The Whole Minus Sum (WMS) evaluated at optimality under a given information-harvesting strategy (fixed γ).

- The joint mutual information between the downstream probe and the active environment, $I(\eta_1, \eta_2; y)$. This represents the actual macroscopic information successfully extracted from the dual-bath setup, serving as the direct analogue to the performance metric evaluated in the baseline three-state model.

- The absolute optimal drive difference, $|\sigma_1^* - \sigma_2^*|$. Similar to the sensory coupling difference analyzed earlier, this observable tracks potential symmetry breaking. Given the structural equivalence of the two active baths, the trivial thermodynamic solution would suggest a uniform drive allocation ($\sigma_1^* = \sigma_2^*$); however, it is interesting to investigate whether asymmetric environmental couplings emerge as an optimal strategy under a constrained energy budget.

In this section, we will focus exclusively on the first two observables, as the analysis of the optimal drive differences will be addressed separately in the following one. The corresponding results for the WMS and the mutual information are presented in Fig. 5.3:

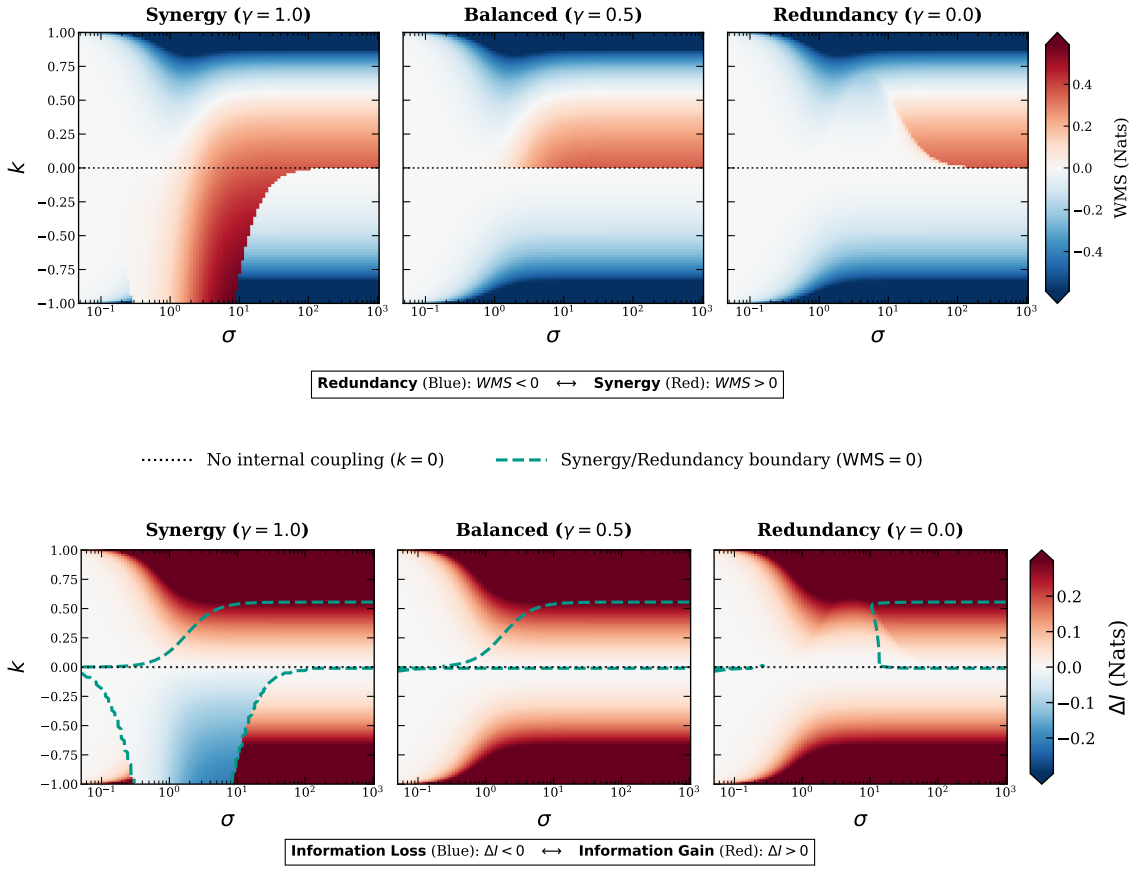


Figure 5.3: Observables heatmaps for the optimal $M = 1, N = 2$ architecture. The three columns correspond to different information-harvesting strategies: Synergy-driven ($\gamma = 1.0$), Balanced ($\gamma = 0.5$), and Redundancy-driven ($\gamma = 0.0$). The independent variable σ parametrizes the fixed equal-dissipation budget $\tilde{S}^{rel}_{M=1,N=1}(\sigma, \tau)$. **(Top)** Whole Minus Sum (WMS) evaluated at the optimal environmental coupling configuration (σ_1^*, σ_2^*). The diverging colormap visually separates the parameter space into distinct informational regimes: red areas indicate a naturally synergistic integration (WMS > 0), while blue areas denote a redundant overlap (WMS < 0). **(Bottom)** Macroscopic performance (ΔI) relative to the single-bath baseline. To correlate the performance with the intrinsic information partitioning, the zero-WMS boundary (dashed blue line) is superimposed on the heatmaps, marking the topological transitions between synergistic and redundant regimes. The horizontal dotted black line ($k = 0$) serves as a reference for structurally independent baths.

We begin our analysis by examining the WMS heatmaps for positive internal couplings ($k > 0$). In this regime, the environmental baths are structurally correlated, meaning they tend to co-activate and share overlapping fluctuations. As observed in the Synergy-driven ($\gamma = 1.0$) and Balanced ($\gamma = 0.5$) strategies, a naturally synergistic regime (indicated by the red areas where WMS > 0) emerges only for intermediate-to-high values of the environmental drive σ, and for weak internal couplings ($k \lesssim 0.5$). The transition between these informational regions illustrates the physical competition between the signal intensity (σ) and the structural correlation (k). As k increases, the environment becomes intrinsically more redundant, meaning that progressively higher σ values are required to extract any synergistic information. More importantly, the WMS maps exhibit a clear structural limit: above a critical coupling threshold ($k \approx 0.5$), the synergistic region vanishes entirely. The high degree of co-activation likely renders the two baths essentially indistinguishable from an information-theoretic standpoint, trapping the system in a redundant regime (solid blue areas) regardless of the available thermodynamic budget σ. A similar topological boundary is observed in the Redundancy-driven strategy ($\gamma = 0.0$). While the critical k threshold remains essentially unchanged, the emergence of the synergistic region is significantly shifted towards higher σ values.

Turning our attention to the anti-correlated regime ($k < 0$), we observe a radically different informational landscape. As shown in the WMS heatmaps, both the Balanced ($\gamma = 0.5$) and Redundancy-driven ($\gamma = 0.0$) strategies are dominated by a globally redundant behaviour (WMS < 0). However, the Synergy strategy ($\gamma = 1.0$) exhibits a distinct topology. For structurally independent or weakly anti-correlated baths ($k \approx 0$), the naturally synergistic regime (WMS > 0) spans almost the entire range of σ. Yet, as the internal anti-correlation strengthens towards $k = -1$, this synergistic area becomes progressively constrained, shrinking towards intermediate-to-low values of the signal drive. This specific topological shape likely arises from the interference between the opposed environmental fluctuations. Outside this region, the behaviour is overall redundant. To understand exactly how the membrane structurally adapts to these anti-correlated drives and why this informational partitioning occurs, we must analyze its optimal coupling configurations, which will be the focus of the following section.

Focusing now on the macroscopic performance shown in the bottom panels (ΔI), a clear connection between the informational regions and the system's performance emerges. For structurally correlated environments ($k > 0$), the two-bath architecture always performs better than the single-bath baseline ($\Delta I > 0$). Notably, as the system moves away from the synergistic region and the WMS becomes more negative, we observe a continuous increase in ΔI across all strategies. In this domain, a higher degree of redundancy is associated with a larger macroscopic informational advantage. However, the most evident physical behavior is present in the anti-correlated regime ($k < 0$). Here, the boundaries of the WMS closely match the macroscopic performance, showing a clear relation: the emergence of synergy corresponds exactly to a loss in information ($\Delta I < 0$), whereas the redundant regions are strictly associated with an informational gain ($\Delta I > 0$).

According to these two panels, the data shows a unified principle for this architecture: regardless of the internal coupling, maximizing redundancy is structurally

connected to maximizing the net information gain.

5.3 Sensory Specialization and Coupling Asymmetry

Further structural insights emerge when we link the phenomena discussed in the previous section with the absolute coupling difference, $|\sigma_1^* - \sigma_2^*|$. To investigate how the system physically allocates its thermodynamic budget in response to the environment, Fig. 5.4 illustrates this optimal sensory specialization across the different strategies, separated into structurally correlated ($k > 0$) and anti-correlated ($k < 0$) regimes:

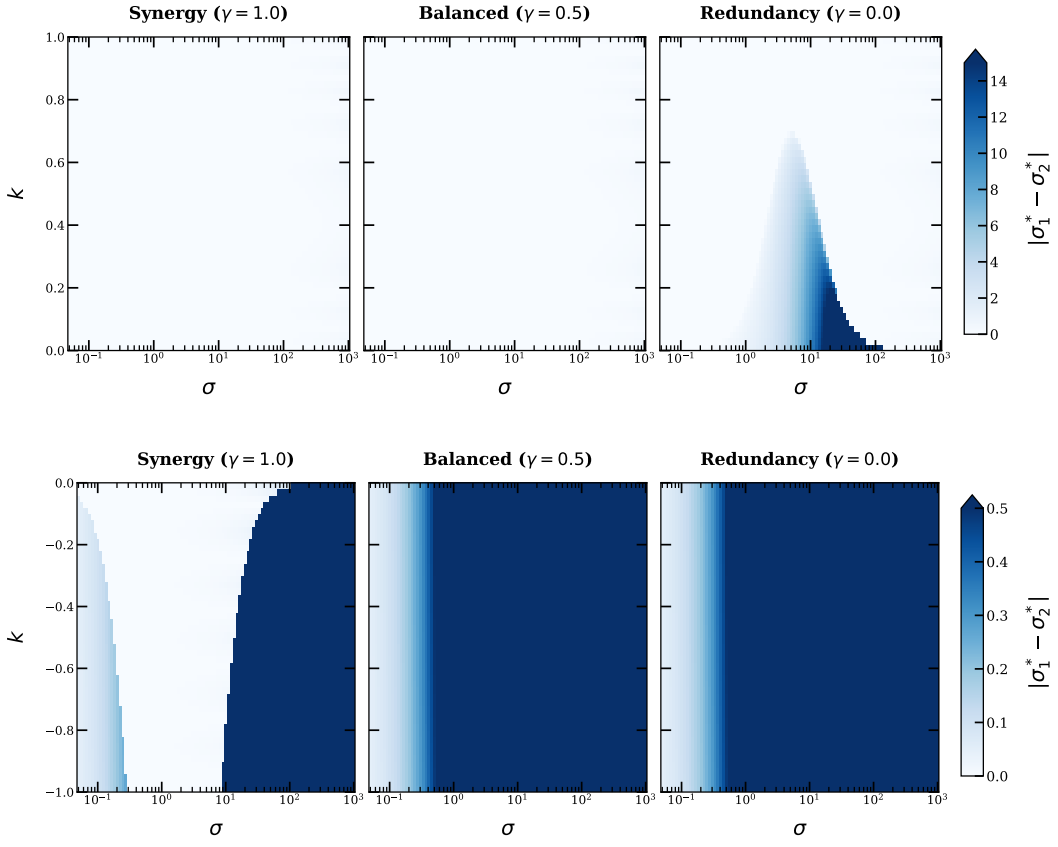


Figure 5.4: Optimal absolute coupling difference $|\sigma_1^* - \sigma_2^*|$ as a function of the signal drive σ and internal coupling k. **(Top)** Structurally correlated regime ($k \geq 0$). **(Bottom)** Anti-correlated regime ($k < 0$, saturated colormap to highlight the symmetry-breaking onset). The heatmaps demonstrate the emergence of sensory specialization across the Synergy-driven, Balanced, and Redundancy-driven strategies.

We begin our analysis with the anti-correlated regime ($k < 0$), as it presents an interesting physical behaviour, perfectly contextualizing the macroscopic observations of the WMS and ΔI discussed above. As observed in the bottom panels of Fig. 5.4 (where the colormap is saturated at a low threshold to highlight the exact

onset of the structural transitions), the system's structural response depends on the imposed functional objective:

- When optimizing for redundancy ($\gamma = 0.0$) or for the total joint mutual information ($\gamma = 0.5$), we observe that the system immediately shuts down one of the two sensory couplings (σ_1^* or σ_2^*). This specialization allows the system to bypass destructive interference, establishing an overall redundant regime and securing a macroscopic information gain ($\Delta I > 0$) across the entire parameter space, consistent with our previous analysis.

- Conversely, when optimizing for synergy ($\gamma = 1.0$), we observe a structural progression similar to the $M = 2, N = 1$ topology. At extremely low signal drives, the system exhibits a slight asymmetry (light blue region). This is followed by a vast intermediate region where both couplings are strictly balanced (structural symmetry, represented by the white area). Finally, the third region, at high thermodynamic budgets, is characterized by a symmetry breaking where one coupling is switched off.

These structural transitions are driven by the interplay between the chosen optimization strategy and the relative equal-dissipation condition. In fact, while maintaining a single active coupling is intuitively the most efficient solution to avoid noise in an anti-correlated environment, the maximization of synergy requires using all available information sources. However, maintaining both couplings active across the entire range of signal drives is probably thermodynamically too expensive, ultimately leading the system to shut one down.

While the consequences of immediate symmetry breaking are straightforward to interpret for the Redundancy-driven and Balanced strategies, the structural evolution of the Synergy-driven strategy requires closer inspection. To determine how this forced symmetry and subsequent switch-off directly correlate with the macroscopic performance, we superimpose the isolines of the structural asymmetry onto the ΔI landscape in Fig. 5.5.

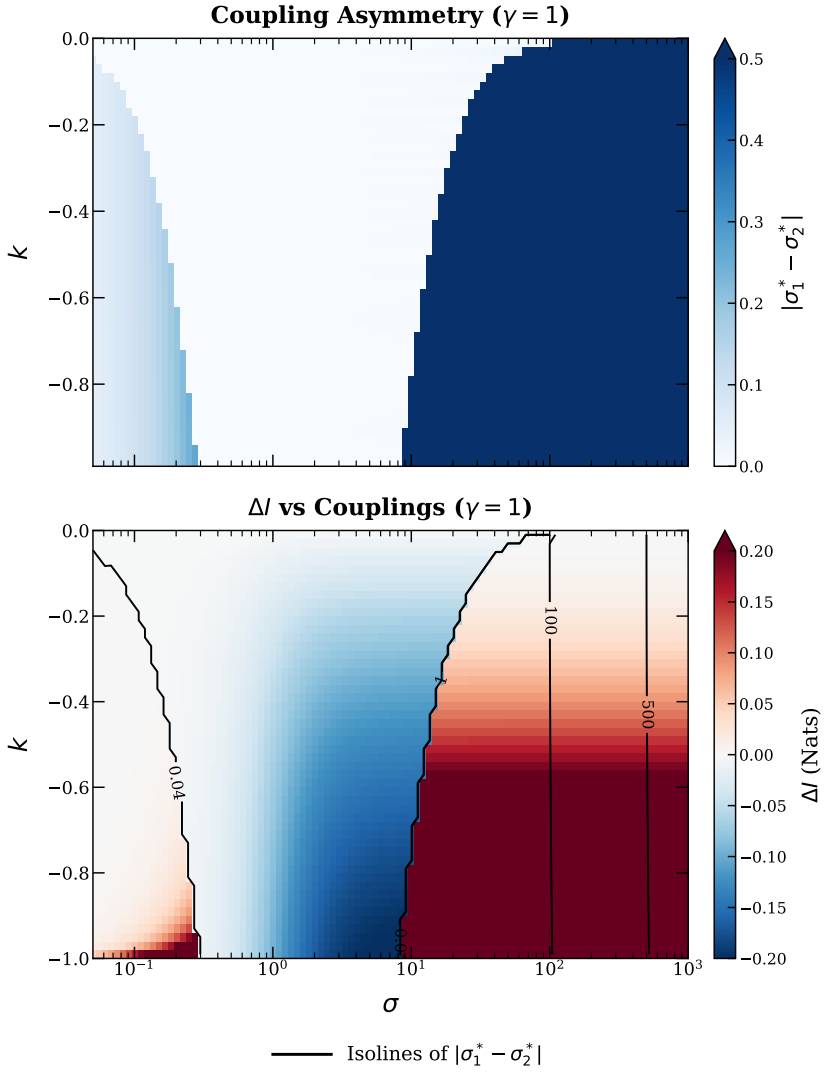


Figure 5.5: Structural origin of the topological frustration for the Synergy-driven strategy ($\gamma = 1.0$) in the anti-correlated regime ($k < 0$). **(Top)** Saturated heatmap of the optimal absolute coupling difference, $|\sigma_1^* - \sigma_2^*|$. The white central region highlights the region where the system maintains a symmetric coupling. **(Bottom)** Macroscopic information advantage (ΔI) with superimposed isolines of the structural asymmetry. The spatial matching demonstrates that the forced structural symmetry directly traps the system in a sub-optimal state ($\Delta I < 0$, blue region) due to destructive interference. Conversely, transitioning to a highly asymmetric state allows the system to recover its macroscopic informational advantage ($\Delta I > 0$, dark red region).

As is clearly evident from the figure, the synergy-dominated region, previously identified as the regime of macroscopic information loss ($\Delta I < 0$), maps exactly to the parameter space where both sensory couplings are active. Conversely, transitioning to an asymmetric configuration, where the system listens to only a single bath, shifts the informational partitioning towards redundancy producing a net information gain in this specific topology.

Before concluding this section, it is worth discussing a peculiar behavior observed for the Redundancy-driven strategy ($\gamma = 0.0$) in the structurally correlated regime (top panel of Fig. 5.4). We note that this does not contradict the unified principle stated above, namely that "maximizing redundancy is connected to maximizing the net information gain". Indeed, this was related to how the informational partitioning of a given optimal solution correlates with its resulting ΔI, not a claim that the Redundancy-driven strategy ($\gamma = 0$) necessarily outperforms the others. Specifically, there appears to be an intermediate range of the signal drive σ characterized by an asymmetric coupling configuration where neither of the two sensors is completely switched off. To clearly visualize this phenomenon, we plot the relative difference in macroscopic performance between the Redundancy strategy and its counterparts, superimposing the isolines of the structural coupling difference:

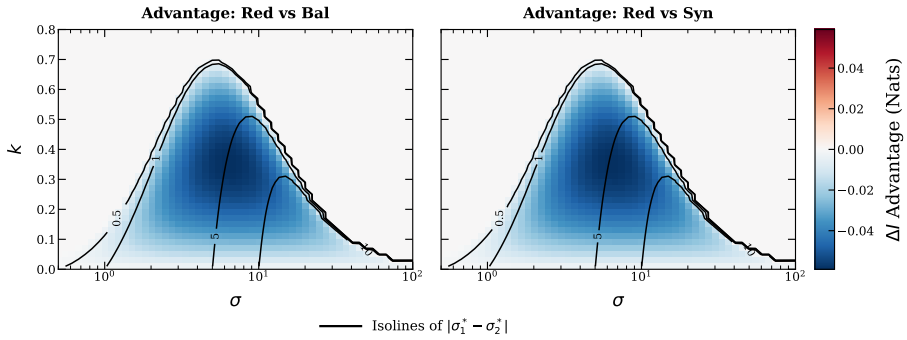


Figure 5.6: Relative macroscopic performance advantage (ΔI) of the Redundancy-driven strategy ($\gamma = 0.0$) compared to the Balanced (**Left panel**) and Synergy-driven (**Right panel**) strategies in the structurally correlated regime ($k > 0$). The colormap indicates the information advantage in Nats, where the prominent blue region signifies a relative performance deficit for the Redundancy strategy. Superimposed are the black isolines of the absolute coupling difference, $|\sigma_1^* - \sigma_2^*|$. The spatial matching shows that the performance drop coincides with an intermediate topological configuration where the system adopts a partially asymmetric coupling without fully shutting down either sensory channel.

The spatial matching shows that this partial asymmetry is associated with a relative performance deficit in the Redundancy-driven strategy. One may question whether such a deviation is a physical phenomenon or a numerical artifact of the optimization algorithm. To understand it, we can inspect the underlying landscape of the associated Pareto functional, L. By fixing the environmental cross-talk at $k = 0.3$, a representative slice through the center of this anomalous region, and plotting the functional shape across varying signal drives σ, we can visually track the exact evolution of the system's optimal structural state:

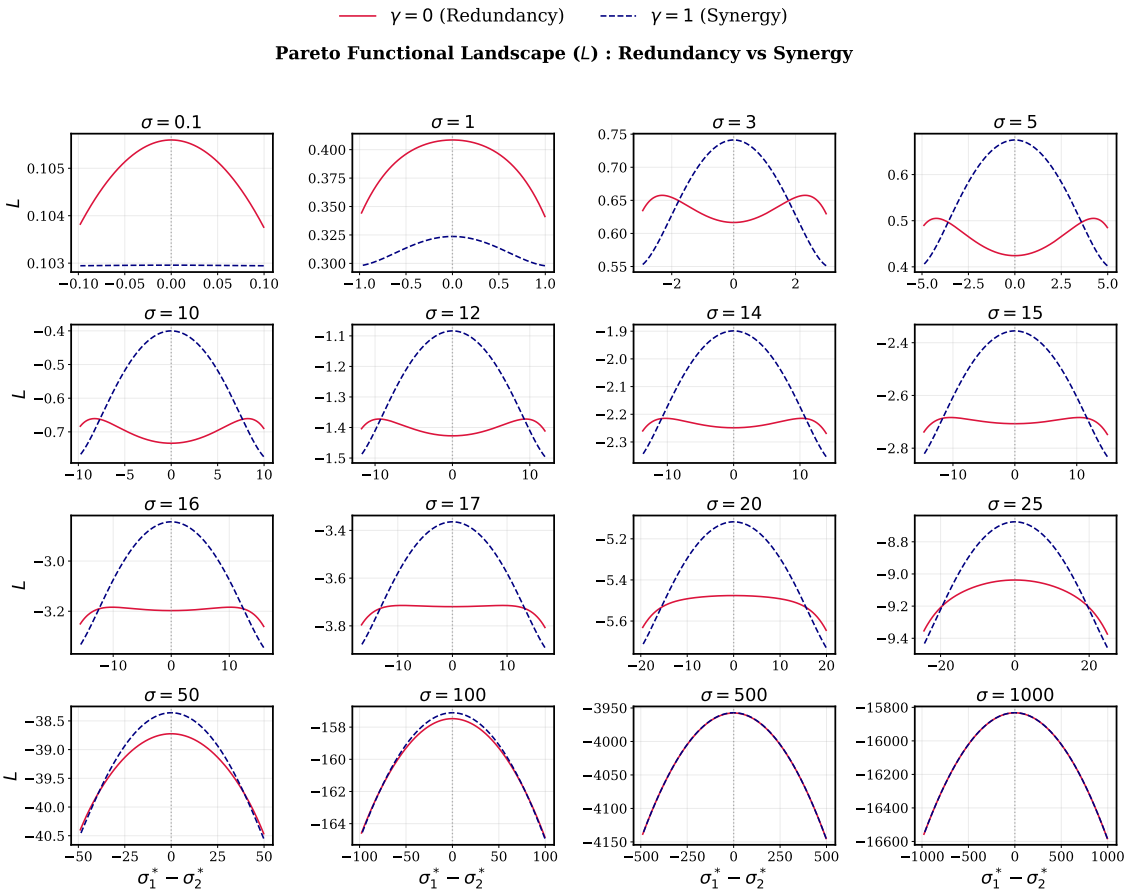


Figure 5.7: Evolution of the Pareto functional landscape (L) as a function of the structural coupling difference $\sigma_1^* - \sigma_2^*$, evaluated at a fixed environmental cross-talk $k = 0.3$. The panels compare the Redundancy-driven strategy ($\gamma = 0$, solid red line) against the Synergy-driven strategy ($\gamma = 1$, dashed navy line) across varying signal drives σ. While the Synergy functional consistently maintains a single global maximum at zero (symmetric coupling), the Redundancy functional exhibits a distinct structural phase transition. At intermediate thermodynamic budgets (e.g., $3 \leq \sigma \leq 17$), the optimal state bifurcates, shifting the maxima away from the center and creating two degenerate, partially asymmetric optimal configurations.

This analysis confirms that the relative performance deficit is a direct manifestation of the changing topology of the Pareto functional, which undergoes a bifurcation at intermediate signal drives. While this links the macroscopic performance drop to the emergence of partial asymmetry, the precise thermodynamic arguments dictating this specific functional shape remain an open question and require further investigation.

5.4 ΔI vs WMS

Having completed the analysis of the individual observables, we now shift our focus to the direct relationship between the information advantage ΔI and the WMS. Following the same approach used in Sec. 4.3, we present the corresponding figures. We begin by examining the case of positive structural coupling, $k > 0$, shown in Fig. 5.8:

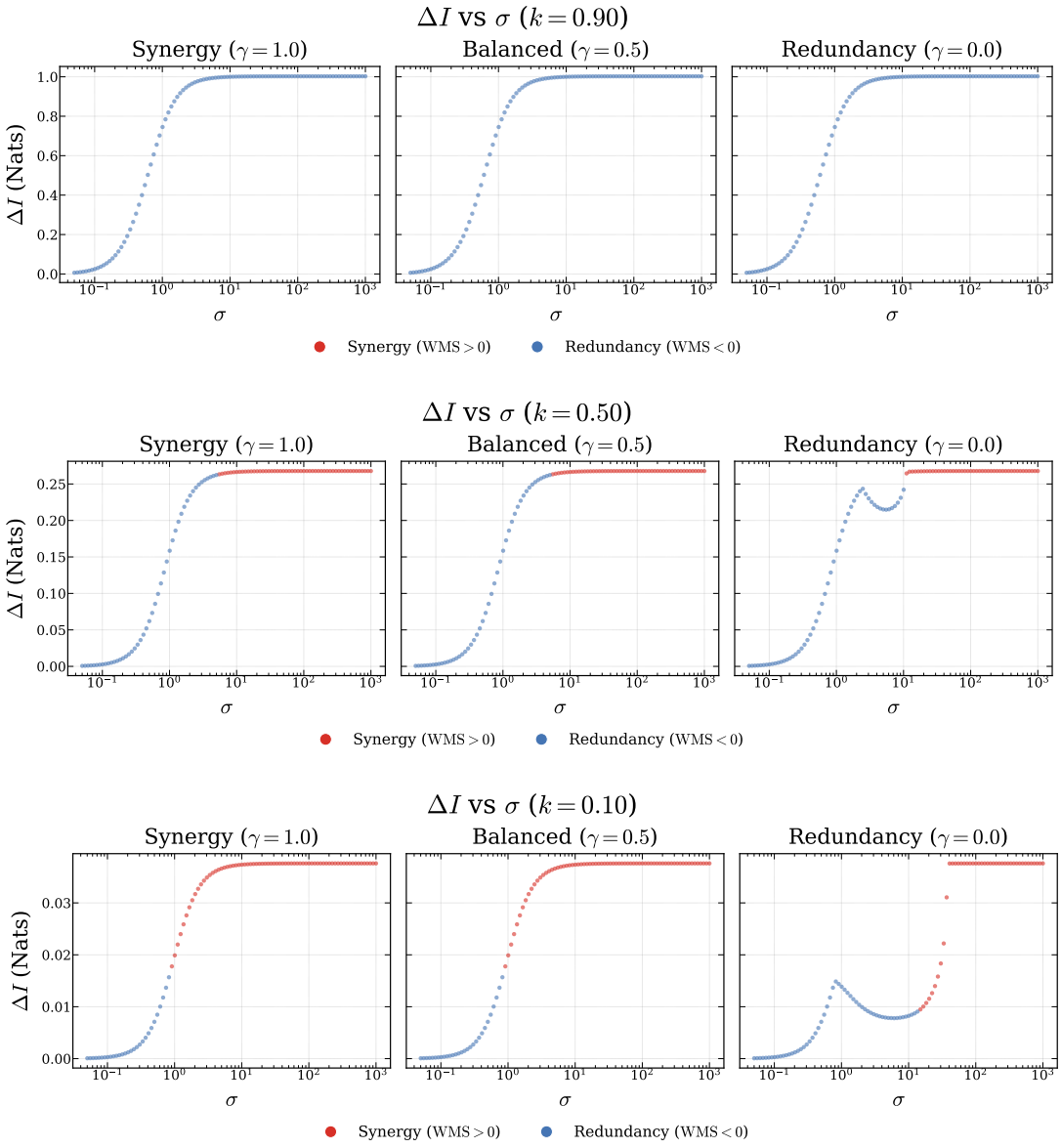


Figure 5.8: Information gain ΔI as a function of the signal strength σ for selected positive structural couplings: $k = 0.90$ (top), $k = 0.50$ (middle), and $k = 0.10$ (bottom).

Consistent with our previous analysis, we observe that $\Delta I > 0$ strictly holds regardless of the chosen optimization strategy. Furthermore, the effects of the partial asymmetry noted earlier become evident: the Redundancy-driven strategy experiences a sharp drop in information gain for values of k close to zero, which then progressively recovers as k approaches 1. Finally, we also note that in this regime, the synergistic contribution tends to vanish as $k \to 1$, as shown in the heatmaps.

Turning to the case of negative structural coupling ($k < 0$), the corresponding plots are presented in Fig. 5.9:

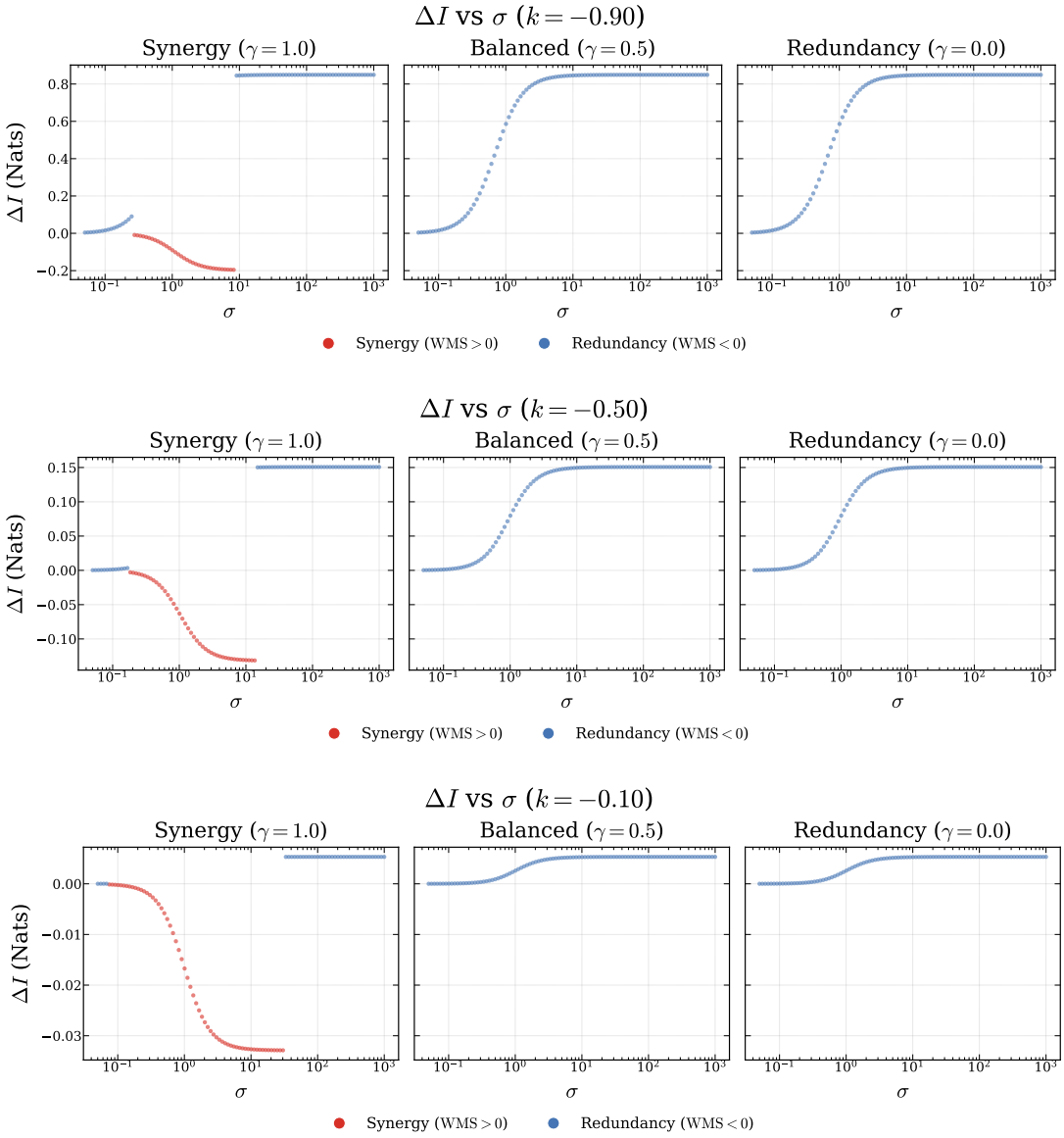


Figure 5.9: Information gain ΔI as a function of the signal strength σ for selected negative structural couplings: $k = -0.90$ (top), $k = -0.50$ (middle), and $k = -0.10$ (bottom).

In this regime, a clear physical trend emerges: the presence of synergy is associated with a negative information gain ($\Delta I < 0$). We also observe distinct discontinuities in the Synergy-driven strategy; these arise from the activation or

deactivation (switching on or off) of one of the two input couplings, σ_1 or σ_2, during the optimization process. Finally, as the intra-bath inhibition grows stronger ($k \to -1$), the parametric region capable of supporting synergistic behavior narrows significantly.

Chapter 6

Conclusions

In this work, we extended the framework used to study information transduction to include more complex topologies beyond the foundational three-state model. We showed that even a minimal increase in structural complexity, i.e. adding just a single state to either the sensory membrane ($M = 2$) or the active bath ($N = 2$), gives rise to non-trivial patterns in the system's capacity to extract information from a noisy environment.

To ensure a fair comparison between the multidimensional topologies and the three-state baseline, we imposed a relative equal-dissipation constraint matching that of the base case. Consequently, any observed information advantage (ΔI) is strictly attributable to the architectural properties of the specific topology rather than an increased energy expenditure.

In the following tables, we summarize the key findings of our study for both extended topologies. We recall that σ represents the signal strength (the coupling driving the membrane from the active bath), while k denotes the structural coupling parameter.

Topology $M = 2$, $N = 1$: Two Membranes and One Active Bath

Regime (k)	**Best Strategy**	**Couplings Specialization (a_1, a_2)**
$k \geq 0$	Indifferent	Specialization (e.g., deactivating one sensory coupling) occurs exclusively when the system is constrained to a synergy-driven strategy.
$k < 0$	Synergy-driven	Specialization occurs across all strategies.

Table 6.1: Summary of the optimal transduction strategies for the $M = 2, N = 1$ topology. The structural parameter k determines the correlation ($k \geq 0$) or anti-correlation ($k < 0$) between the two sensory membranes.

Topology $M = 1$, $N = 2$: One Membrane and Two Active Baths

Regime (k)	**Best Strategy**	**Couplings Specialization (σ_1, σ_2)**
$k \geq 0$	Indifferent	Both couplings remain active; however, a partial asymmetry between the inputs emerges specifically when adopting the redundancy-driven strategy.
$k < 0$	Balanced / Redundancy-driven	One input coupling is switched off in both redundancy-driven and balanced strategies; this structural specialization also extends to the synergy-driven strategy in specific parameter ranges.

Table 6.2: Summary of the optimal transduction strategies for the $M = 1, N = 2$ topology. The structural parameter k dictates the degree of intra-bath correlation ($k \geq 0$) or intra-bath inhibition ($k < 0$) between the two hidden sources.

As highlighted by our results across both topologies, the relationship between the information advantage ΔI and the Whole-Minus-Sum (WMS) metric reveals that synergy and redundancy play distinct roles depending on the environmental regime. An interesting insight of this work lies in the topological differences of information extraction: when multiple membranes monitor a single environment ($M = 2, N = 1$), the optimal performance is achieved by maximizing the synergistic information shared between the sensors. Conversely, when a single membrane is tasked with integrating signals from two distinct active baths ($M = 1, N = 2$), the system favors the maximization of redundancy.

Furthermore, we observed a clear operational requirement for the couplings: generating synergy strictly requires both connection pathways to remain active in either topology. Redundancy, on the other hand, exhibits more structural flexibility, as it can emerge both when the couplings are active and when one pathway is selectively switched off.

From a biophysical perspective, these findings lead to the consideration that there is no universal optimal architecture for information transduction. Biological sensors must dynamically adapt their topologies and coupling strategies based on the environmental statistics they are monitoring.

More broadly, the optimization perspective explored provides an intuitive physical interpretation of these distinct strategies. When a system needs to extract information from several identical environments, the most effective approach is to maximize redundancy. Intuitively, focusing on the overlapping signals between the baths allows the sensor to filter out independent local noise, thereby extracting the most reliable macroscopic information. In contrast, when multiple sensors monitor a single environment, synergy becomes the most beneficial strategy. In this scenario, the membranes can act as complementary probes, each capturing a different feature of the same external drive. By combining these distinct partial observations, the system can reconstruct a more detailed picture of the environment, achieving a

greater overall information gain than it would by simply measuring the same feature twice.

It is also important to acknowledge the theoretical boundaries of the current framework. Specifically, our analysis relies on the Minimum Mutual Information (MMI) approach to perform the Partial Information Decomposition (PID). While MMI provides a tractable measure for synergy and redundancy in our linear, Gaussian setting, the evaluation of PID for continuous variables remains an open challenge in information theory (see Sec. 1.2.3 for details).

Future extensions of this work could explore non-linear coupling functions or the inclusion of explicit feedback loops from the probe to the upstream variables. Furthermore, applying the Variance-Sum Rule (VSR) to real or simulated time-series data of these multidimensional networks would provide a force-free experimental validation of our theoretical predictions. Building upon these concepts, an ongoing extension focuses on developing a framework to simulate the dynamic adaptation of the coupling parameters in response to abrupt environmental shifts (e.g., sudden variations in the driving σ). Because this approach relies entirely on stochastic trajectories, it holds significant promise for the optimization of non-linear systems and represents a possible continuation of the present work.

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